TOPIC 04 · PHASE DIAGRAMS

Phase Diagrams and Solidification

Phases and microconstituents, binary diagrams, the Gibbs phase rule and the lever rule, eutectic systems, precipitation, the invariant reactions, intermetallic compounds, the Fe-C system with the microstructures of steels, and ceramic and ternary diagrams.

Source: Universidad Carlos III de Madrid · Session 7 T4 Phase diagrams I_2025.pdf, Session 8 T4 Phase diagrams II.pdf, Session 9 T4 Phase diagrams III.pdf and Session 10 T4 Phase diagrams IV.pdf

Central idea: a phase diagram is a map that tells which phases are stable in a material for each temperature, pressure and composition. Reading it gives three answers for any alloy at any temperature: which phases are present, what each phase is made of, and how much of each there is.

This page follows the official slides of Topic 4 in order. Part I (Session 7) covers the basic vocabulary, one-component diagrams, binary isomorphous systems, the Gibbs phase rule, the tie line and the lever rule, non-equilibrium solidification and binary eutectic systems. Part II (Session 8) covers solid-state precipitation, the invariant reactions, intermetallic compounds and congruent and incongruent melting. Part III (Session 9) covers the Fe-C system, the equilibrium transformations of steels and the influence of alloying elements. Part IV (Session 10) covers ceramic and ternary phase diagrams. Worked examples use only data printed on the slides.

Part I — Phase diagrams I (Session 7)

§ 1 Component, phase and microconstituent [Session 7 Slide 2]

Three words are used all through the topic, and they mean different things:

- **Component:** a chemical substance or compound with a fixed composition that can form part of a phase or of an alloy. Examples: aluminium (Al), alumina (Al_2O_3).
- **Phase:** a chemically and structurally homogeneous region that forms part of the microstructure. It can be made of one or of several components. Examples: ferrite (a solid solution of C in Fe with BCC structure), austenite (a solid solution of C in Fe with FCC structure), calcium carbonate ($CaCO_3$).
- **Microconstituent:** each of the different structures that can be seen on a polished surface of the material. It can be made of one phase or of several phases.

The slide illustrates the three words with food: the components of a potato are water, protein, fat, carbohydrates and so on (a table of grams per 100 g); the potato itself plays the role of a phase; and fries, a structure you can see and that is made from the potato, play the role of a microconstituent.

In a real material, the micrograph below shows a steel with two microconstituents. Region 1 is **ferrite**, a single phase. Regions 2 and 3 are **pearlite**, a microconstituent made of two phases (ferrite and cementite) arranged in thin alternating plates.

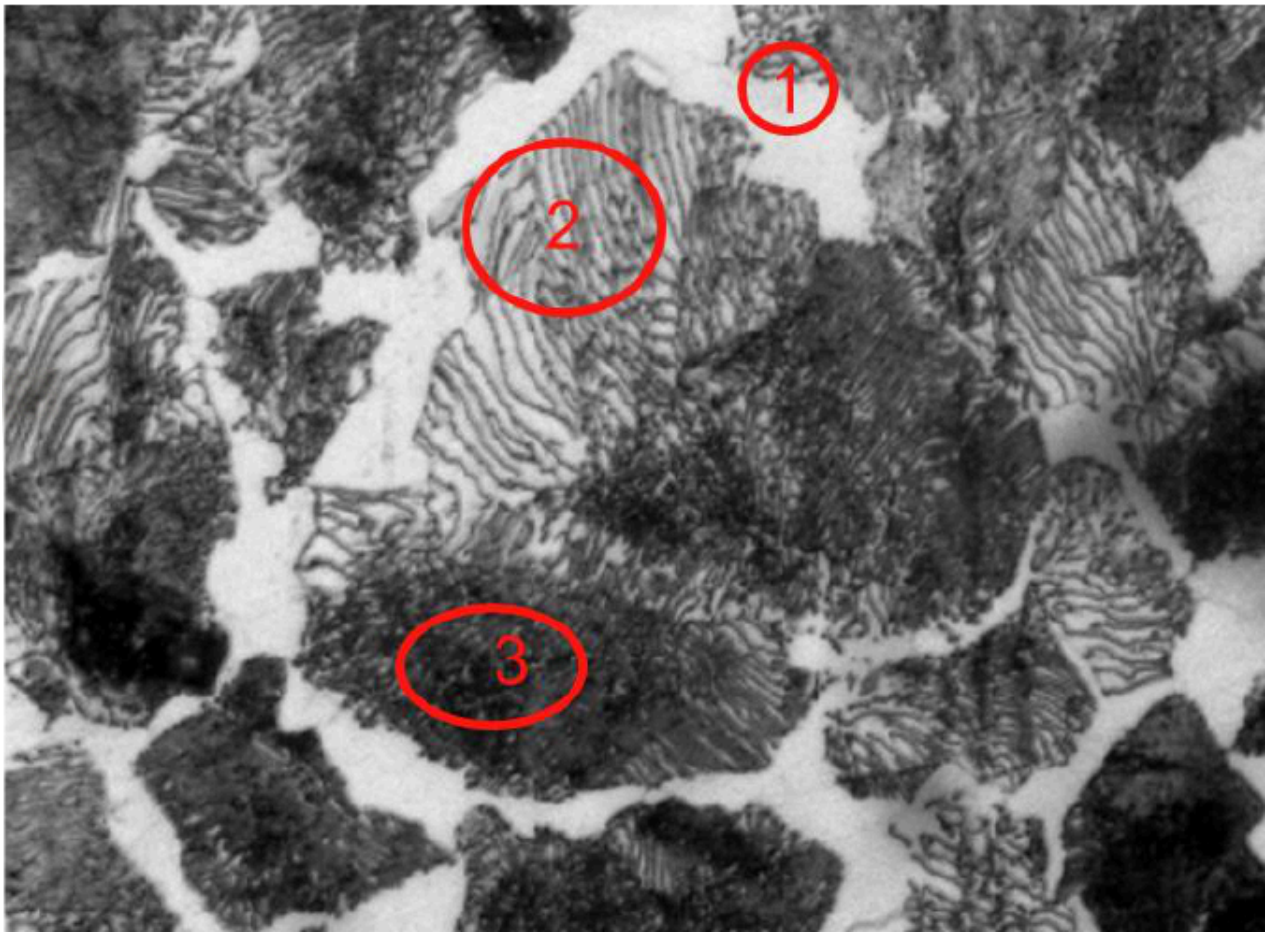


Figure 1. Microconstituents in a steel: 1 is ferrite, 2 and 3 are pearlite (ferrite + cementite). From Session 7 Slide 2.

§ 2 The equilibrium phase diagram [Session 7 Slide 3]

An **equilibrium phase diagram** (also called a **stability map**) is a map of the phases in thermodynamic equilibrium that are present in a system of materials at different pressures, temperatures and compositions.

The slide uses the phase diagram of CO_2 as an example:

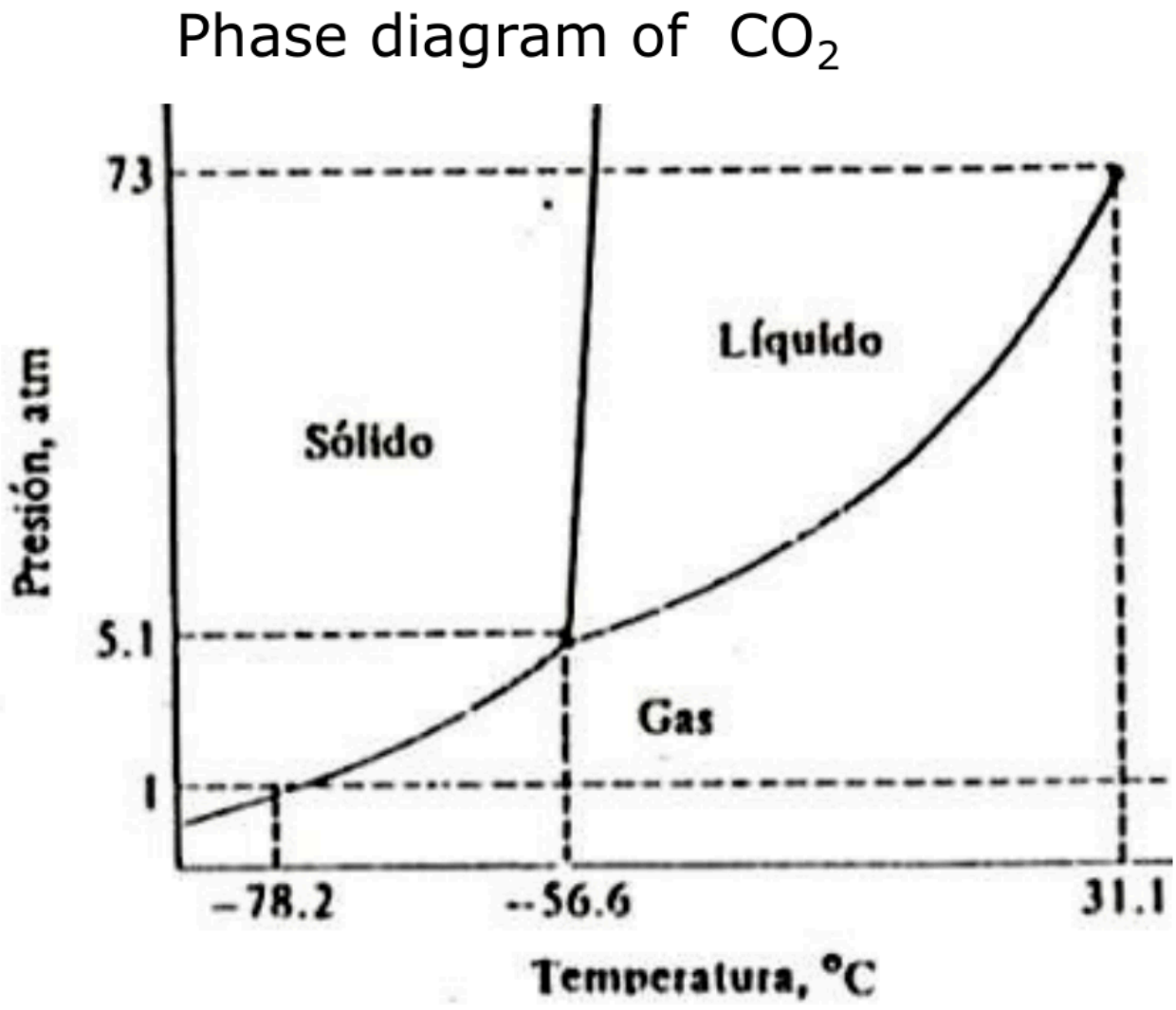


Figure 2. Phase diagram of CO2 (labels in Spanish: Sólido, Líquido, Gas; Presión, Temperatura): triple point at 5.1 atm and −56.6 °C, sublimation at 1 atm and −78.2 °C, critical point at 73 atm and 31.1 °C. Copied from Session 7 Slide 3.

- There is no liquid CO_2 at room temperature and ambient pressure.
- The slope of the solid-liquid line, dP/dT, is positive, as it is in almost all substances.
- At the **triple point** (5.1 atm, –56.6 °C) gas, liquid and solid are in equilibrium.
- At temperatures below the triple point CO_2 can sublime: gas → solid (dry ice) on cooling, and solid → gas on heating.
- Above the critical temperature T_c = 31.1 °C (the value printed on the diagram; the text of the slide writes 32 °C) and pressure P_c = 73 atm CO_2 is a **supercritical fluid** (used to extract caffeine from coffee).

WORKED EXAMPLE 1 — WHY DRY ICE DOES NOT MELT AT 1 ATM

Question. A block of solid CO_2 is heated at a constant pressure of 1 atm. What phases appear?

Reading the diagram. Follow the horizontal line P = 1 atm from left to right. Since 1 atm is below the pressure of the triple point (5.1 atm), the line never enters the liquid region: it goes from the solid region straight into the gas region, crossing the sublimation line at −78.2 °C.

Answer. The solid turns directly into gas at −78.2 °C (sublimation). Liquid CO_2 can exist only at pressures above 5.1 atm, which is why there is no liquid CO_2 at room conditions [Session 7 Slide 3].

§ 3 Unary (one-component) diagrams [Session 7 Slides 5-6]

With a single component the composition is fixed, so the diagram is drawn with **pressure and temperature** as axes (a P-T diagram).

3.1 Water [Session 7 Slide 5]

The P-T diagram of water has three lines that meet at the **triple point** P_T: the sublimation line (A to P_T), the melting line (P_T to B) and the vaporisation line (P_T to C). Cooling at constant pressure along the dashed line, the water vapour first crosses the vaporisation line at T_v and then the melting line at T_m.

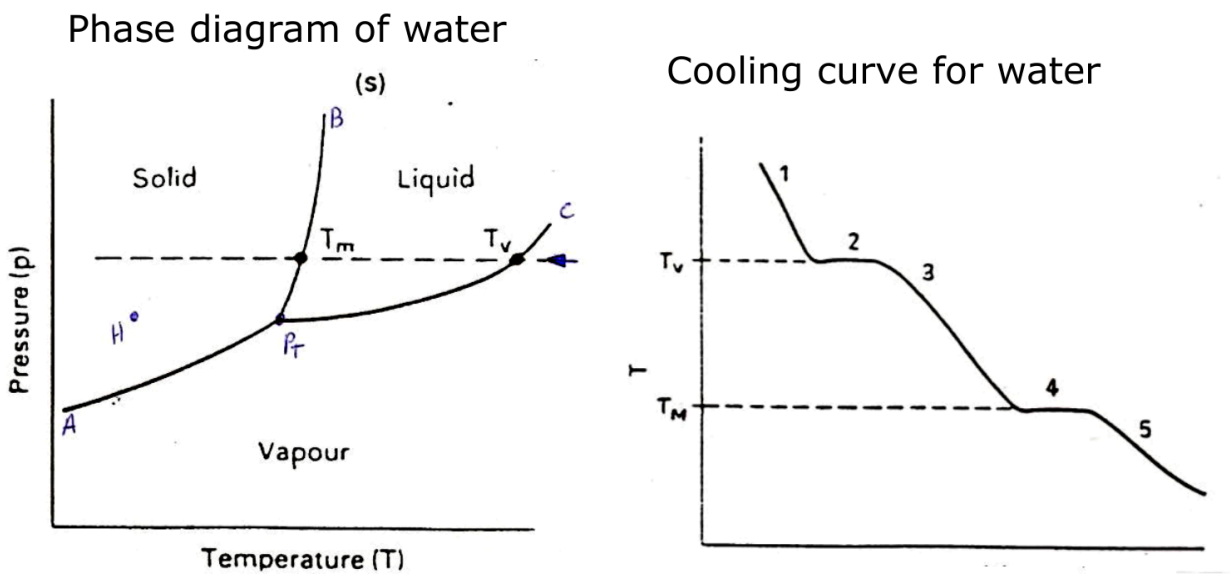


Figure 3. Left: P-T diagram of water with the sublimation (A to P_T), melting (P_T to B) and vaporisation (P_T to C) lines and the triple point P_T; the dashed line is the cooling path at constant pressure. Right: its cooling curve, with the plateaus 2 (condensation at T_v) and 4 (solidification at T_m). The blue marks are handwritten on the slide. Copied from Session 7 Slide 5.

The **cooling curve** on the right records the temperature against time along that path. In the stretches 1, 3 and 5 a single phase cools down. In the plateaus 2 (condensation at T_v) and 4 (solidification at T_m) the temperature stays constant while two phases coexist, because the heat released by the transformation compensates the heat that is lost.

On a diagram with pressure on a log scale, the vertical line at 1 atm meets the melting line at 0 °C (water/ice) and the vaporisation line at 100 °C (steam/water):

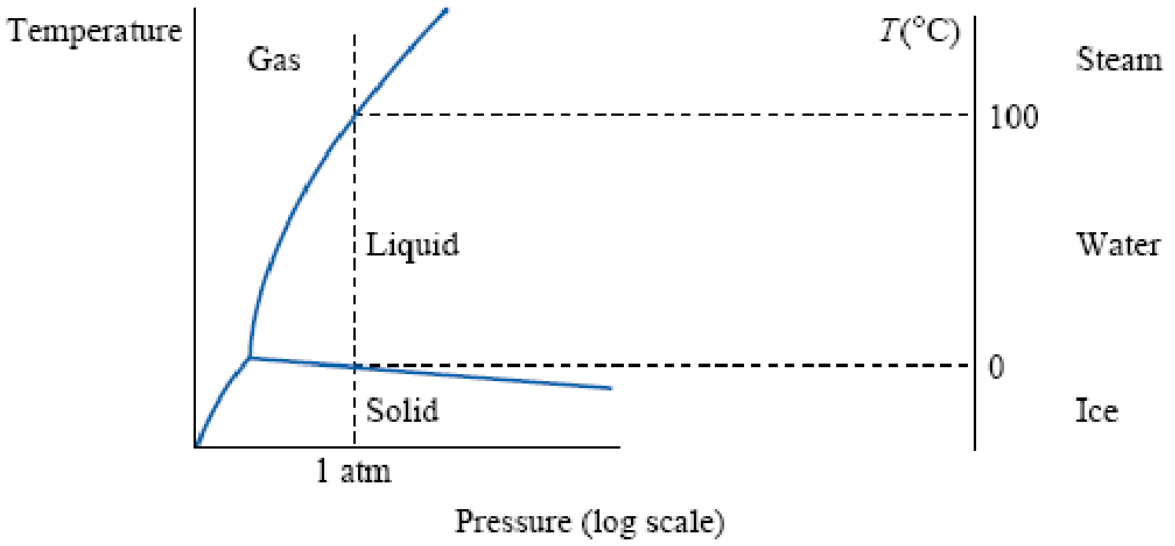


Figure 4. Temperature against pressure (log scale) for water: at 1 atm the vertical line meets the melting line at 0 °C and the vaporisation line at 100 °C. Copied from Session 7 Slide 5.

At very high pressures (up to several GPa) water has many different solid phases, the different crystal structures of ice:

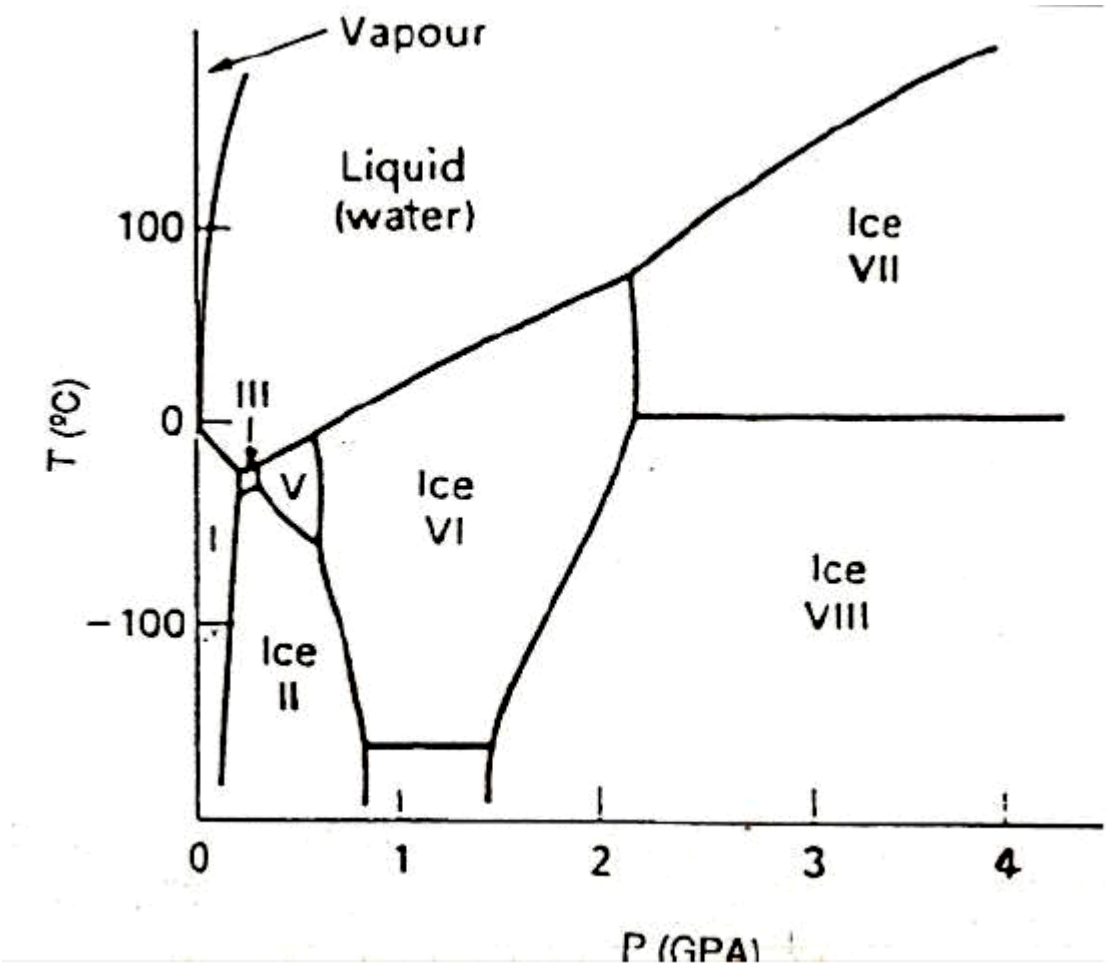


Figure 5. Water at high pressure (up to 4 GPa): several crystal structures of ice (I, II, III, V, VI, VII, VIII), each a different solid phase. From Session 7 Slide 5.

3.2 Pure iron [Session 7 Slide 6]

Solid iron has **three different crystal structures**, each a different phase: **α-Fe (BCC)** below 910 °C, **γ-Fe (FCC)** between 910 and 1394 °C, and **δ-Fe (BCC)** from 1394 °C up to the melting point, 1538 °C. Different phases coexist at the **triple points**. The slide compares the three solid structures with three varieties of potato (yellow, red and purple) and the liquid with potato purée: the same "substance" in different forms.

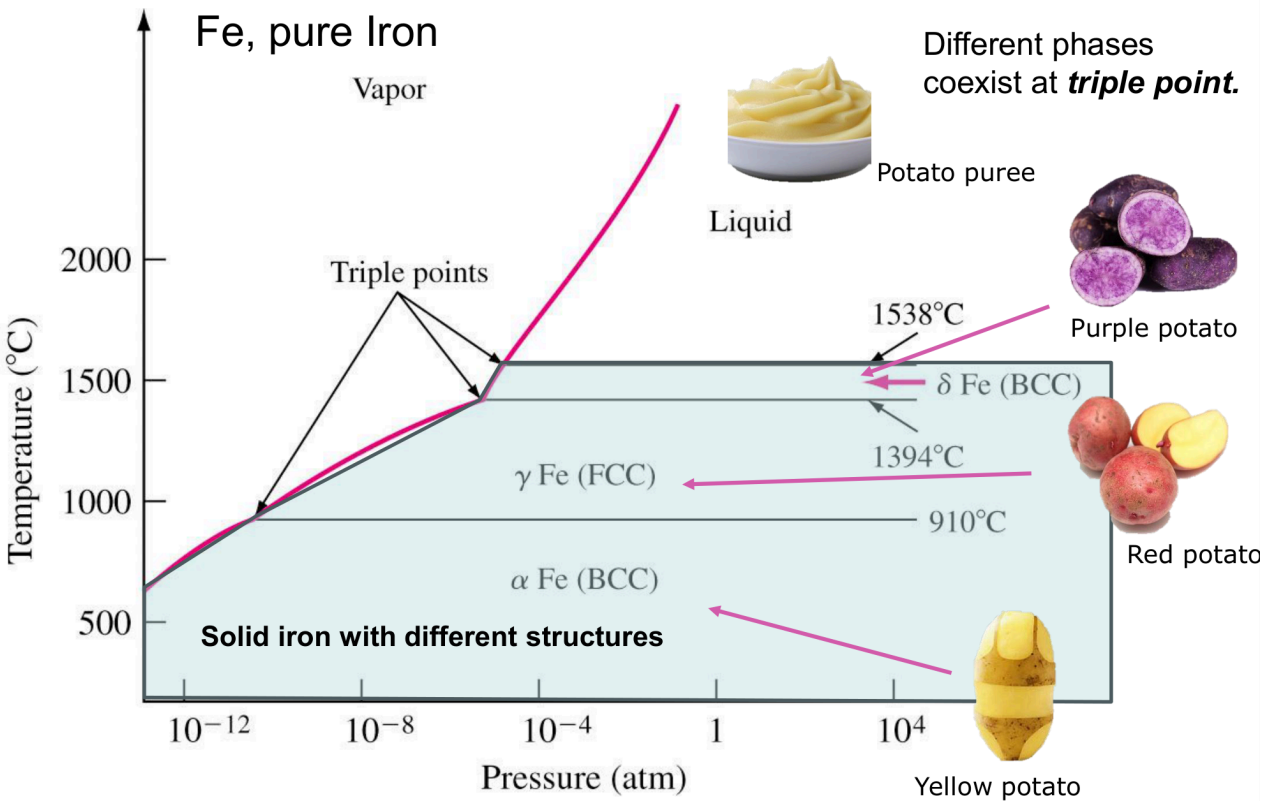


Figure 6. P-T diagram of pure iron (pressure on a log scale) with α-Fe (BCC), γ-Fe (FCC), δ-Fe (BCC), liquid and vapour, the triple points and the slide's potato analogy. Copied from Session 7 Slide 6.

§ 4 Binary isomorphous diagrams [Session 7 Slides 8-11]

4.1 Cooling curves [Session 7 Slide 8]

Cooling curves are used to determine the temperatures of the phase transitions: the temperature of a cooling molten metal is recorded against time and plotted.

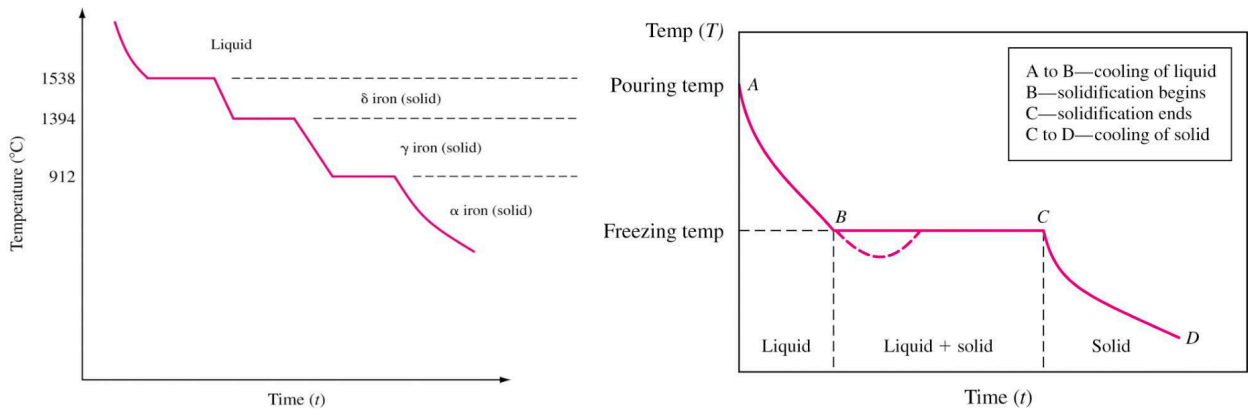


Figure 7. Left: cooling curve of pure iron, with a thermal arrest at 1538, 1394 and 912 °C. Right: cooling curve of a pure metal: A to B cooling of the liquid, B solidification begins, C solidification ends, C to D cooling of the solid. Copied from Session 7 Slide 8.

- **Thermal arrest:** while a pure metal solidifies, the heat lost to the surroundings equals the heat supplied by the solidifying metal, so the temperature stays constant (a plateau). Pure iron shows one plateau for solidification and one for each change of crystal structure. (This slide prints 912 °C for the γ → α change; Slide 6 prints 910 °C.)
- **Alloys** solidify over a range of temperatures: they show no thermal arrest, only changes of slope.

4.2 Building the diagram from cooling curves [Session 7 Slide 9]

A **binary isomorphous** system has two components with **complete solid solubility**. Its diagram is built from the cooling curves of several compositions: each curve gives the temperature where solidification starts and the one where it ends. Joining the start points gives the **liquidus**; joining the end points gives the **solidus**. Above the liquidus everything is liquid (L), below the solidus everything is a single solid solution (SS), and between them liquid and solid coexist (L + SS).

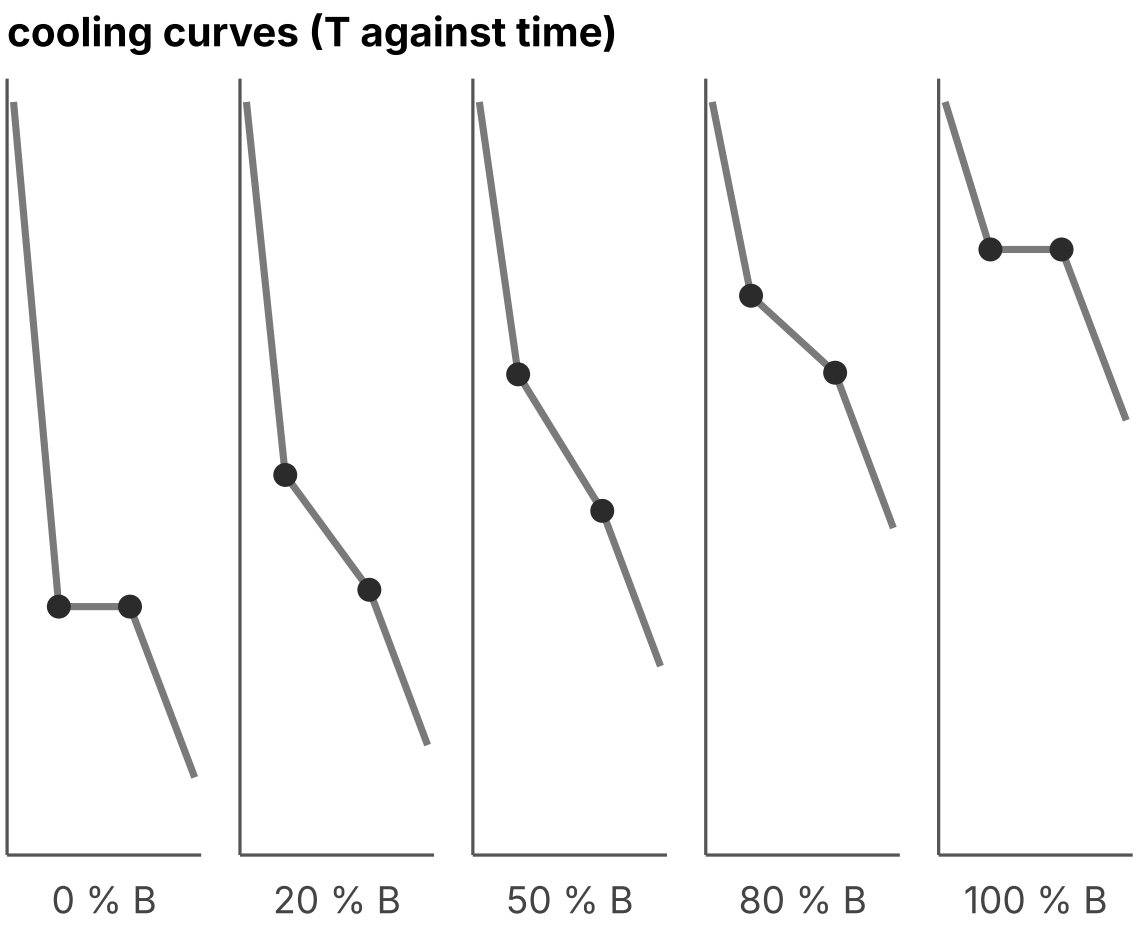


Figure 8. Cooling curves of alloys with 0, 20, 50, 80 and 100 % B: the pure components solidify at one temperature, the alloys between the start (liquidus) and the end (solidus) of solidification, the green points. Redrawn because on Session 7 Slide 9 the milk photographs cover the curves. Portal drawing, not in the slides (EXTRA)..

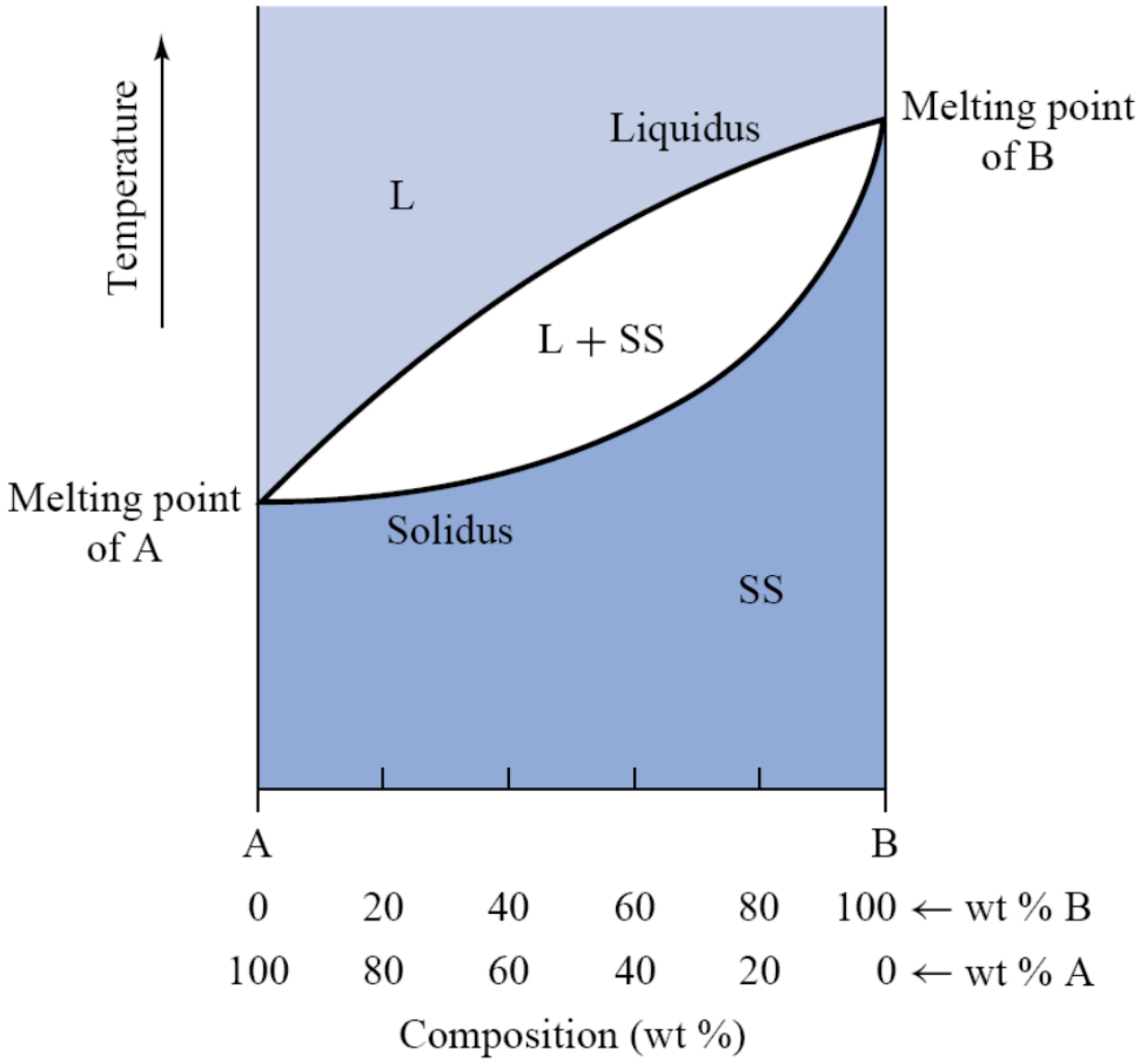


Figure 9. Binary isomorphous diagram: liquidus and solidus between the melting points of A and B, with the regions L, L + SS and SS. Copied from Session 7 Slide 9.

The slide compares the series of compositions with milk: whole, semi-skimmed and skimmed milk are mixtures of water and skim solids in different proportions, just as the alloys are mixtures of A and B.

4.3 Microstructure along the cooling [Session 7 Slide 10]

For an alloy of a given composition (the vertical line), the diagram tells what the microstructure is at each temperature:

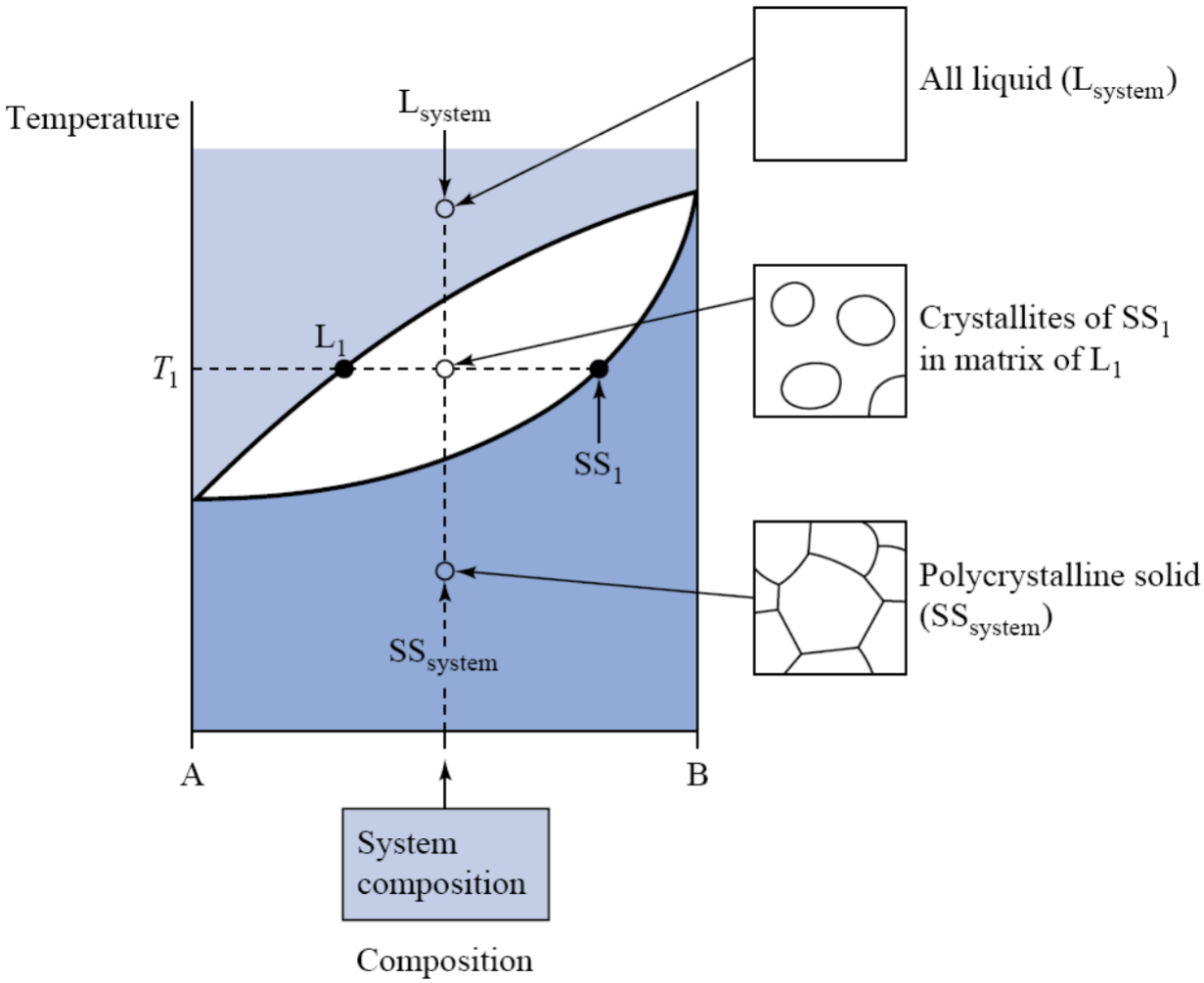


Figure 10. An alloy (system composition) cooled through the isomorphous diagram: all liquid L_system above the liquidus, crystallites of SS1 in a matrix of L1 at T1, and a polycrystalline solid SS_system below the solidus. Copied from Session 7 Slide 10.

- above the liquidus: **all liquid**;
- at a temperature T_1 between the liquidus and the solidus: **crystallites of the solid solution SS1 in a matrix of liquid L1**; the compositions of L1 and SS1 are read on the liquidus and the solidus at T_1;
- below the solidus: a **polycrystalline solid** of a single phase, the solid solution, with the composition of the alloy.

The slide's analogy: milk alone is the liquid; milk with pieces of Manchego cheese is the two-phase mixture; the cheese alone is the solid.

4.4 Complete solid solubility [Session 7 Slide 11]

With complete solubility in the solid state:

- the material is a **polycrystalline single-phase** material: its microstructure (FCC, BCC or any other) is not different from that of the pure metal, because the atoms of B simply replace atoms of A in the same lattice;
- when solute is added, the melting temperature T_m is modified;
- the alloy **does not solidify at a constant temperature** but over a range of temperatures, between the liquidus and the solidus.

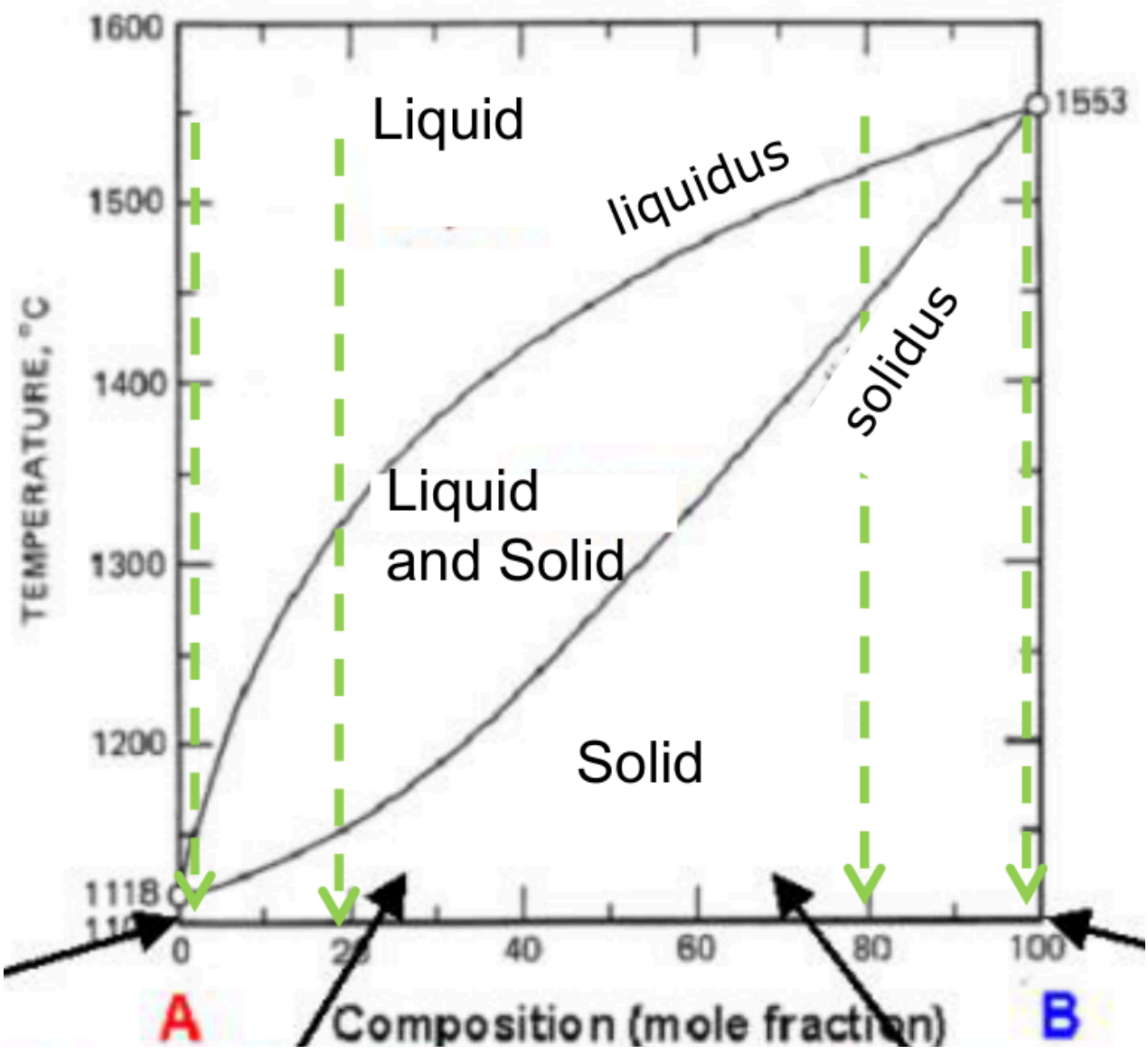


Figure 11. Isomorphous diagram with complete solid solubility from the melting point of A (1118 °C) to that of B (1553 °C), composition in mole fraction; the green dashed lines mark the alloys whose atom arrangements are shown below. Copied from Session 7 Slide 11.

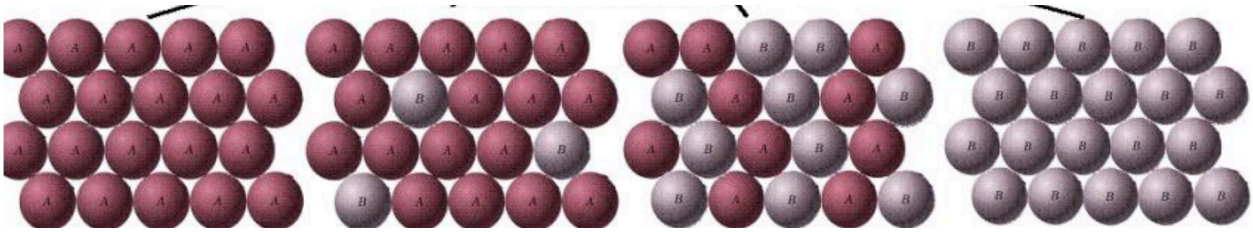


Figure 12. Arrangement of A and B atoms in the same lattice for increasing B content: B atoms replace A atoms at any proportion. Copied from Session 7 Slide 11.

§ 5 Gibbs phase rule [Session 7 Slide 13]

The **Gibbs phase rule** counts how many variables can be changed independently while keeping the same phases in equilibrium:

$$F = C - P + N$$

- F: degrees of freedom, the number of independent variables available to the system;
- C: number of components;
- P: number of phases;
- N: number of non-compositional variables (for example pressure and temperature).

With pressure and temperature both free, $N = 2$ and $F = C - P + 2$. For routine processing of materials the pressure can be considered fixed at 1 atm, so only temperature remains and

$$F = C - P + 1$$

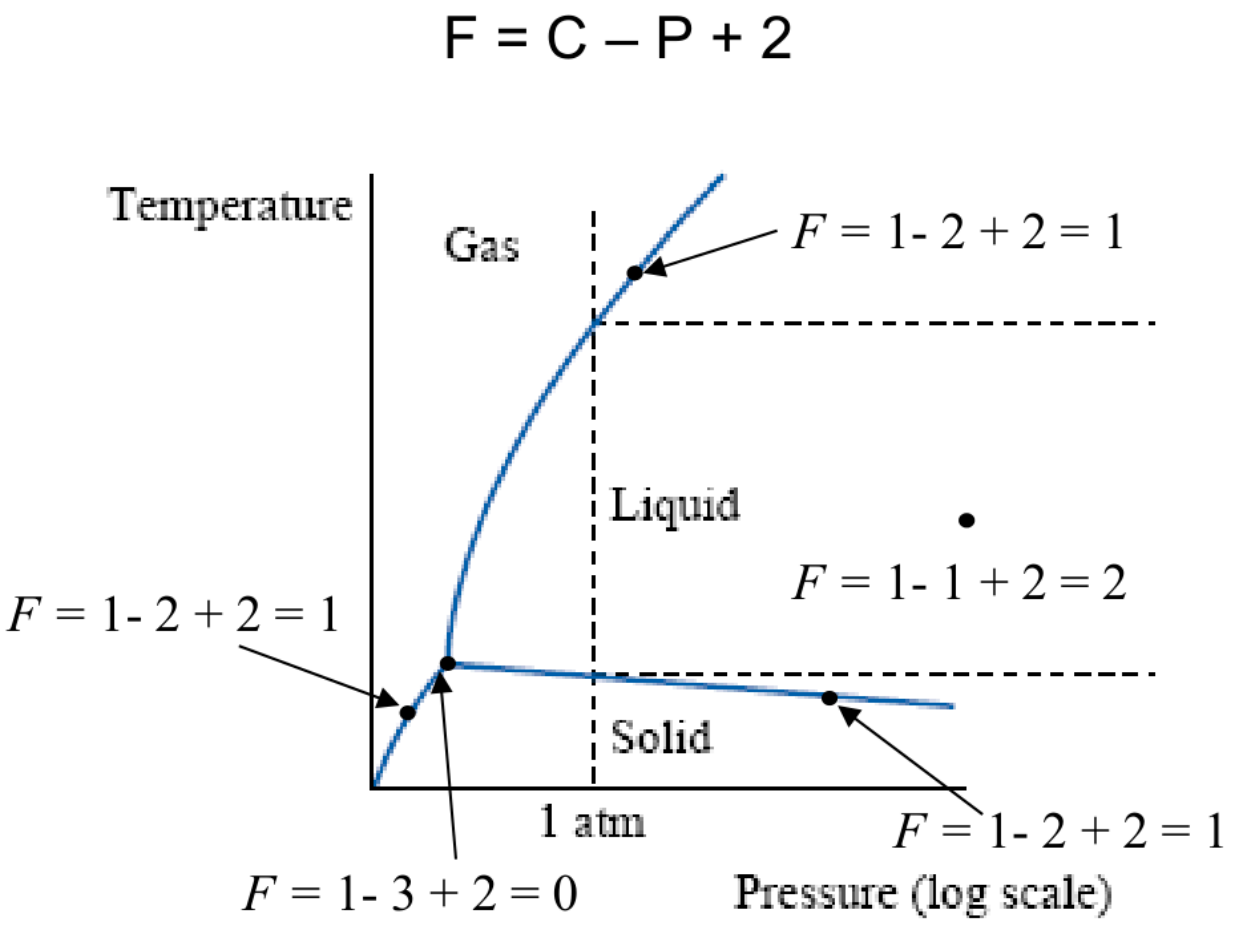


Figure 13. One component, F = C − P + 2: F = 2 in a single-phase region, 1 on a two-phase line and 0 at the triple point. Copied from Session 7 Slide 13.

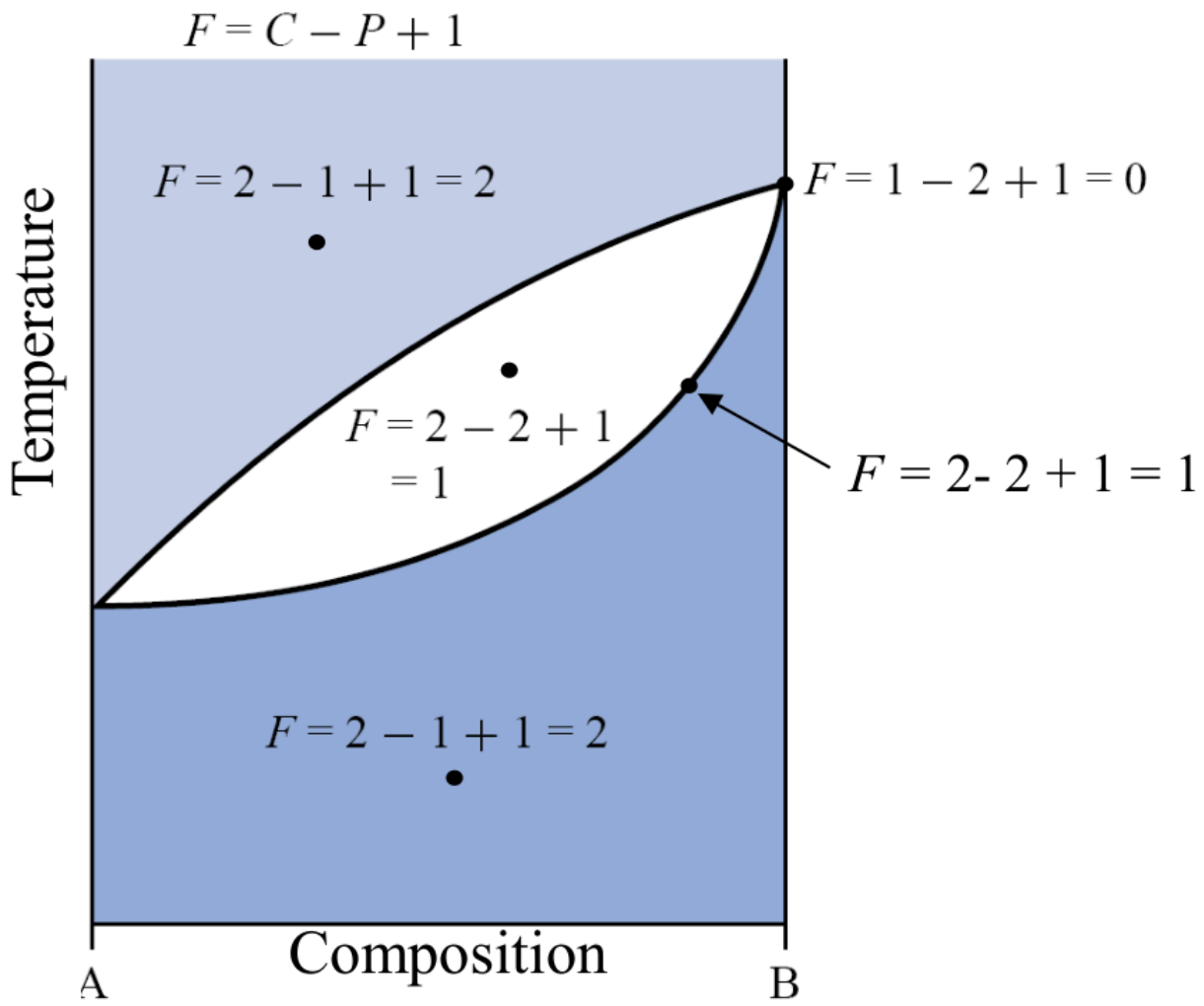


Figure 14. Two components at 1 atm, F = C − P + 1: F = 2 in L or in the solid, 1 in L + solid and on its boundaries, 0 at the melting point of a pure component. Copied from Session 7 Slide 13.

WORKED EXAMPLE 2 — COUNTING DEGREES OF FREEDOM ON THE SLIDE DIAGRAMS

One component (water), $F = 1 - P + 2$ [Session 7 Slide 13]:

- inside a single-phase region (liquid): $F = 1 - 1 + 2 = 2$; temperature and pressure can both change without leaving the region;
- on a two-phase line (for example solid-liquid): $F = 1 - 2 + 2 = 1$; if the temperature is chosen, the pressure is fixed by the line;
- at the triple point: $F = 1 - 3 + 2 = 0$; three phases coexist only at one temperature and one pressure.

Two components at 1 atm, $F = 2 - P + 1$:

- in L or in the solid solution: $F = 2 - 1 + 1 = 2$; temperature and composition can both change;
- in L + solid (and on its boundaries): $F = 2 - 2 + 1 = 1$;
- at the melting point of pure B, a single component with two phases: $F = 1 - 2 + 1 = 0$, which is why a pure metal melts at one fixed temperature.

Consequence for the tie line (§6). In the α + L region of the Cu-Ni diagram $F = 1$. Choosing the temperature uses that single degree of freedom, so the compositions of the two phases are no longer free: at about 1250 °C (a reading of the figure) they can only be C_L = 32 wt% Ni and C_α = 43 wt% Ni [Session 7 Slide 16]. Changing the overall composition of the alloy changes only the amounts of the two phases, not their compositions.

§ 6 Tie line and lever rule [Session 7 Slides 15-21]

In a two-phase region two questions are answered with two tools [Session 7 Slide 15]:

- **Phase composition: the tie line.** Draw a horizontal line (constant temperature) through the point of the alloy across the two-phase region. Its intersections with the phase boundaries give, on the composition axis, the composition of each phase.
- **Phase amounts: the lever rule.** The fraction of each phase is computed by taking the length of the tie line from the overall alloy composition to the phase boundary of the **other** phase, and dividing by the total length of the tie line [Session 7 Slide 16].

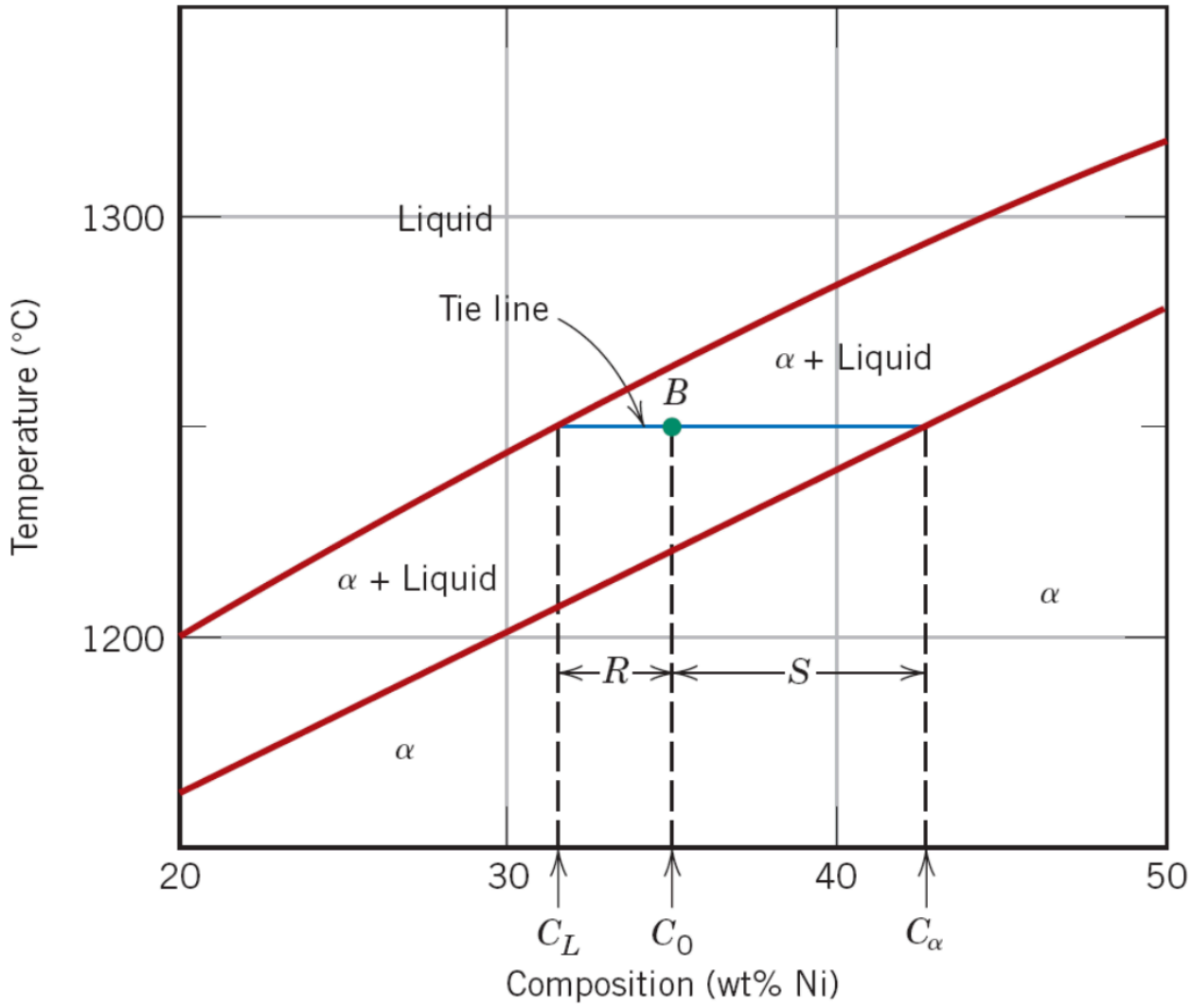


Figure 15. Tie line through the alloy B (C_0) in the α + liquid region: it ends at C_L on the liquidus and at C_α on the solidus, and R and S are its two arms; Slide 16 prints C_L = 32, C_0 = 35 and C_α = 43 wt% Ni. Copied from Session 7 Slide 15.

For the alloy B, with overall composition C_0 and the arms $R = C_0 - C_L$ and $S = C_\alpha - C_0$:

$$W_L = \frac{S}{R+S} = \frac{C_\alpha - C_0}{C_\alpha - C_L} \qquad W_\alpha = \frac{R}{R+S} = \frac{C_0 - C_L}{C_\alpha - C_L} \qquad W_L + W_\alpha = 1$$

where W_L is the mass fraction of liquid and W_α the mass fraction of α. They are **mass** fractions because the composition axis is in wt%; on an axis in atomic % the same construction gives atom (mole) fractions (extension, needed for the at.% diagrams of Problems 11-13 of the sheet).

WORKED EXAMPLE 3 — CU-35 WT% NI AT ABOUT 1250 °C [SESSION 7 SLIDES 16-21]

Data from the slide. $C_0 = 35$ wt% Ni at point B. The slide prints the three compositions but neither the temperature nor the name of the system: about 1250 °C is a reading of the figure (the tie line lies between the 1200 and 1300 °C grid lines), and "Cu-Ni" is assumed from the wt% Ni axis (extension). The tie line meets the liquidus at $C_L = 32$ wt% Ni and the solidus at $C_\alpha = 43$ wt% Ni.

Step 1, the arms of the lever. $R = C_0 - C_L = 35 - 32 = 3$ wt% Ni and $S = C_\alpha - C_0 = 43 - 35 = 8$ wt% Ni; the whole tie line is $R + S = C_\alpha - C_L = 43 - 32 = 11$ wt% Ni.

Step 2, the fractions. Each phase takes the arm on the side of the other phase:

$$W_L = \frac{S}{R+S} = \frac{43-35}{43-32} = \frac{8}{11} = 0.727 \qquad W_\alpha = \frac{R}{R+S} = \frac{35-32}{43-32} = \frac{3}{11} = 0.273$$

Step 3, checks. $W_L + W_\alpha = 8/11 + 3/11 = 1$. The nickel of the two phases adds up to the nickel of the alloy: $W_L C_L + W_\alpha C_\alpha = 0.727 \times 32 + 0.273 \times 43 = 23.27 + 11.73 = 35.0$ wt% Ni $= C_0$.

Answer. At about 1250 °C the alloy is 73 % liquid of 32 wt% Ni and 27 % solid α of 43 wt% Ni. The alloy point is closer to the liquidus, so there is more liquid.

ROUNDING ON THE SLIDE

Slide 21 writes $W_L = 0.72$ and $W_\alpha = 0.28$. The exact values are $8/11 = 0.727$ and $3/11 = 0.273$, so they round to 0.73 and 0.27, as the official solutions of the problem sheet write them (Solution problems Phase diagrams I.pdf, p. 5). The slide truncates W_L and then takes $W_\alpha = 1 - W_L$.

WORKED EXAMPLE 4 — THE SAME ALLOY DURING EQUILIBRIUM COOLING [SESSION 7 SLIDE 23]

Slide 23 prints the equilibrium compositions of the liquid and the solid at several temperatures for the same Cu-35 wt% Ni alloy. Applying the lever rule at each one:

Point	C_L (wt% Ni)	C_α (wt% Ni)	$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$	W_α
b′ (liquidus)	35	46	$\frac{46-35}{46-35} = 1$	0
c′	29	40	$\frac{40-35}{40-29} = \frac{5}{11} = 0.455$	0.545
d′ (solidus)	24	35	$\frac{35-35}{35-24} = 0$	1

Interpretation. At b′ the first solid appears, of 46 wt% Ni, in a liquid that still has the composition of the alloy. As the temperature falls both phases become poorer in Ni (the liquid follows the liquidus, the solid follows the solidus) and the amount of solid grows. At d′ the solid has reached the composition of the alloy, 35 wt% Ni, and the last liquid disappears: in equilibrium, solidification ends on the solidus.

§ 7 Non-equilibrium solidification [Session 7 Slides 23-24]

When the alloy goes down through the phase diagram too quickly (high cooling rates), solid-state diffusion has no time to keep the whole solid at the equilibrium composition. Only the surface of each growing crystal has the composition of the solidus; the inside keeps the composition it had when it formed. The result is a microstructure of **heterogeneous composition**, called **microsegregation** (or **cored structure**).

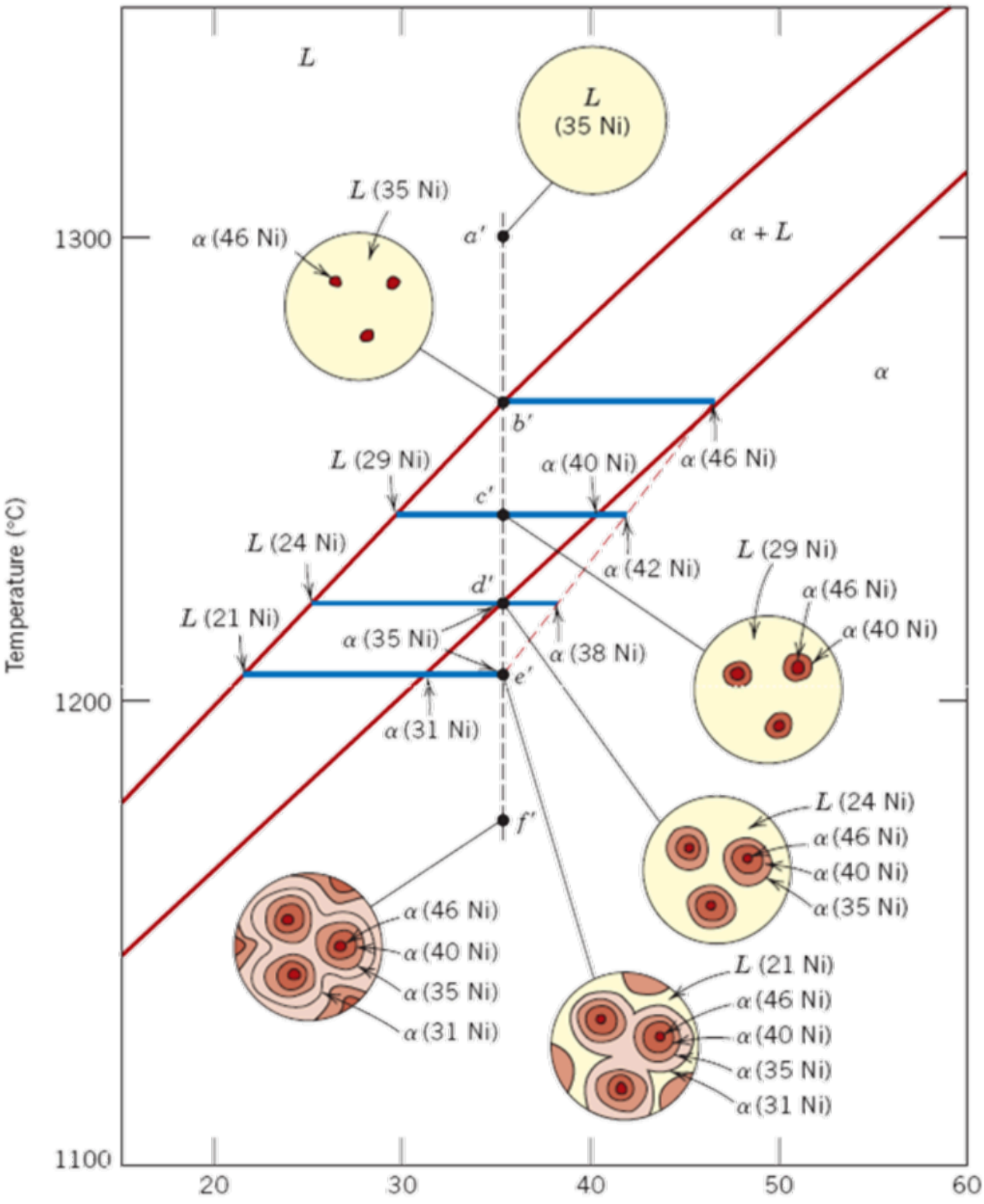


Figure 16. Cooling of a Cu-35 wt% Ni alloy too fast for equilibrium: tie lines at b′, c′, d′ and e′, the cored crystals (46, 40, 35 and 31 wt% Ni from the centre out) and the dashed non-equilibrium solidus. Temperature (°C) against composition (wt% Ni). Copied from Session 7 Slide 23.

In the Cu-35 wt% Ni alloy the first solid is α of 46 wt% Ni (b′). At c′ the new layer is α of 40 wt% Ni, but the crystal still has its 46 % core, so its **average** composition is 42 wt% Ni instead of 40; at d′ the average is 38 instead of 35; at e′ it is 35. The average compositions follow a **non-equilibrium solidus** (dashed) that lies to

the right of the equilibrium solidus, so liquid survives below the equilibrium solidus and solidification ends only at e′. The final grains are cored: 46 wt% Ni at the centre and 31 wt% Ni at the edge (f′).

The same behaviour in an alloy of Ni-30 wt% Cu (Slide 24): with fast cooling the solid follows the non-equilibrium solidus α′2 … α′7 and solidification ends at T_7 instead of T_5:

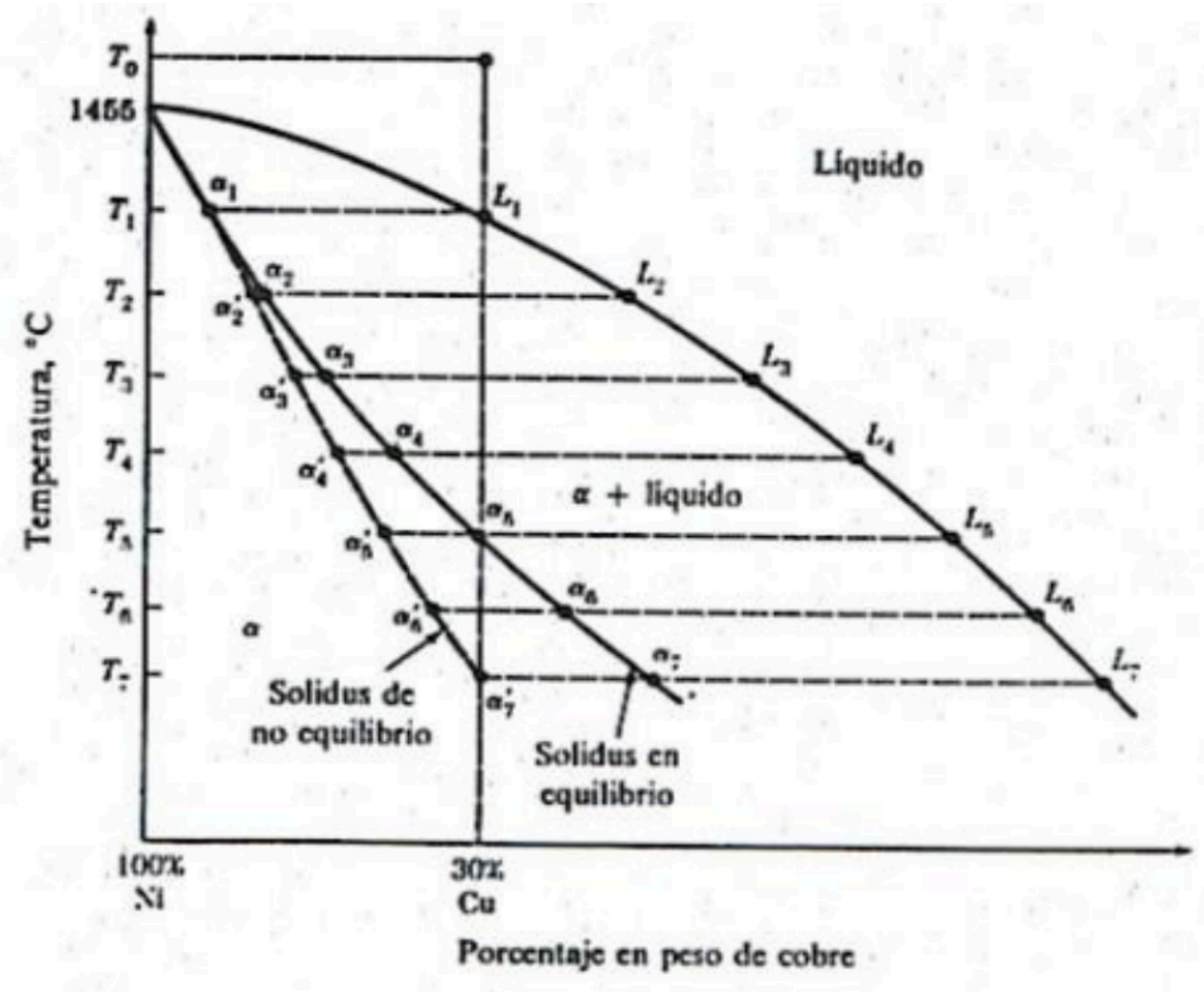


Figure 17. Ni-30 wt% Cu alloy (labels in Spanish): with equilibrium cooling the solid follows α1 … α5 and solidification ends at T5; with fast cooling the average composition follows the non-equilibrium solidus α′2 … α′7 and it ends at T7. Copied from Session 7 Slide 24.

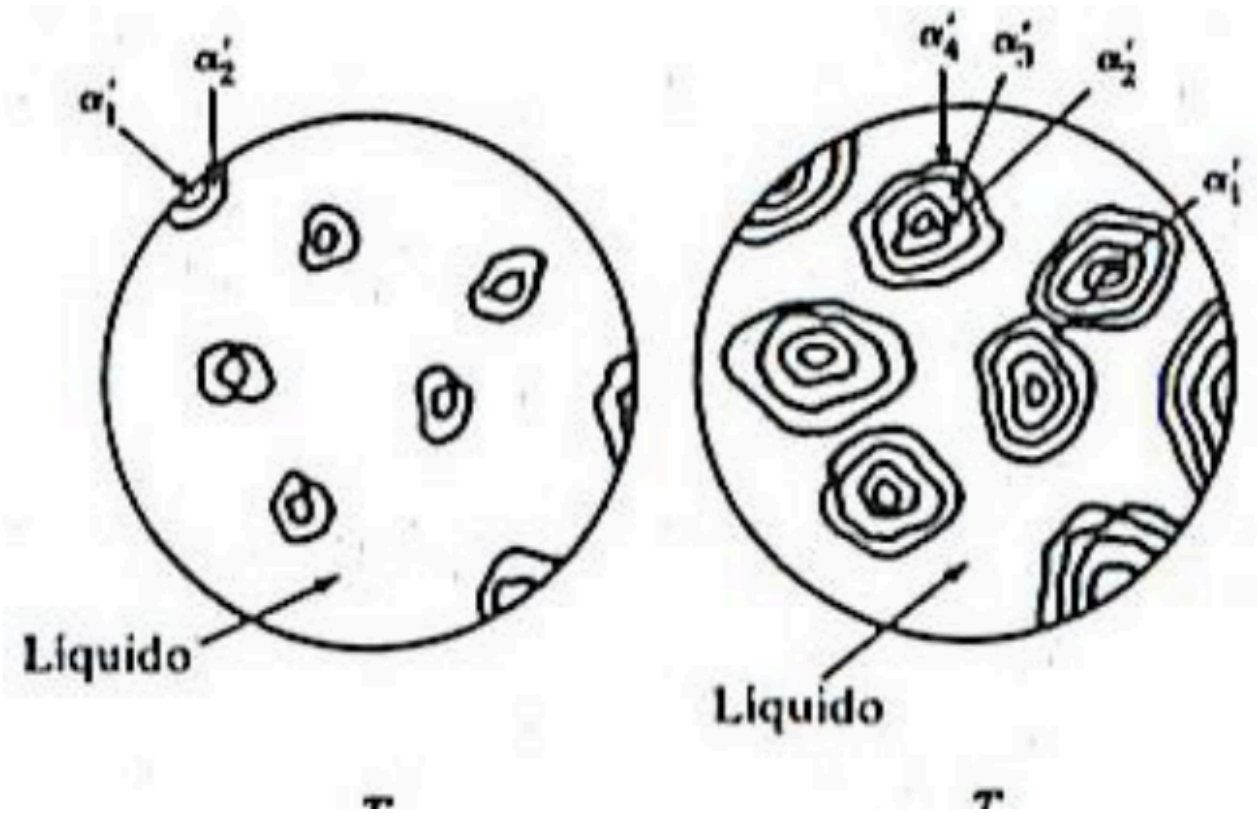


Figure 18. Cored crystals of the Ni-30 % Cu alloy growing in the liquid at T_2 and T_4: each new layer (α'_1, α'_2, ...) has a different composition. From Session 7 Slide 24.

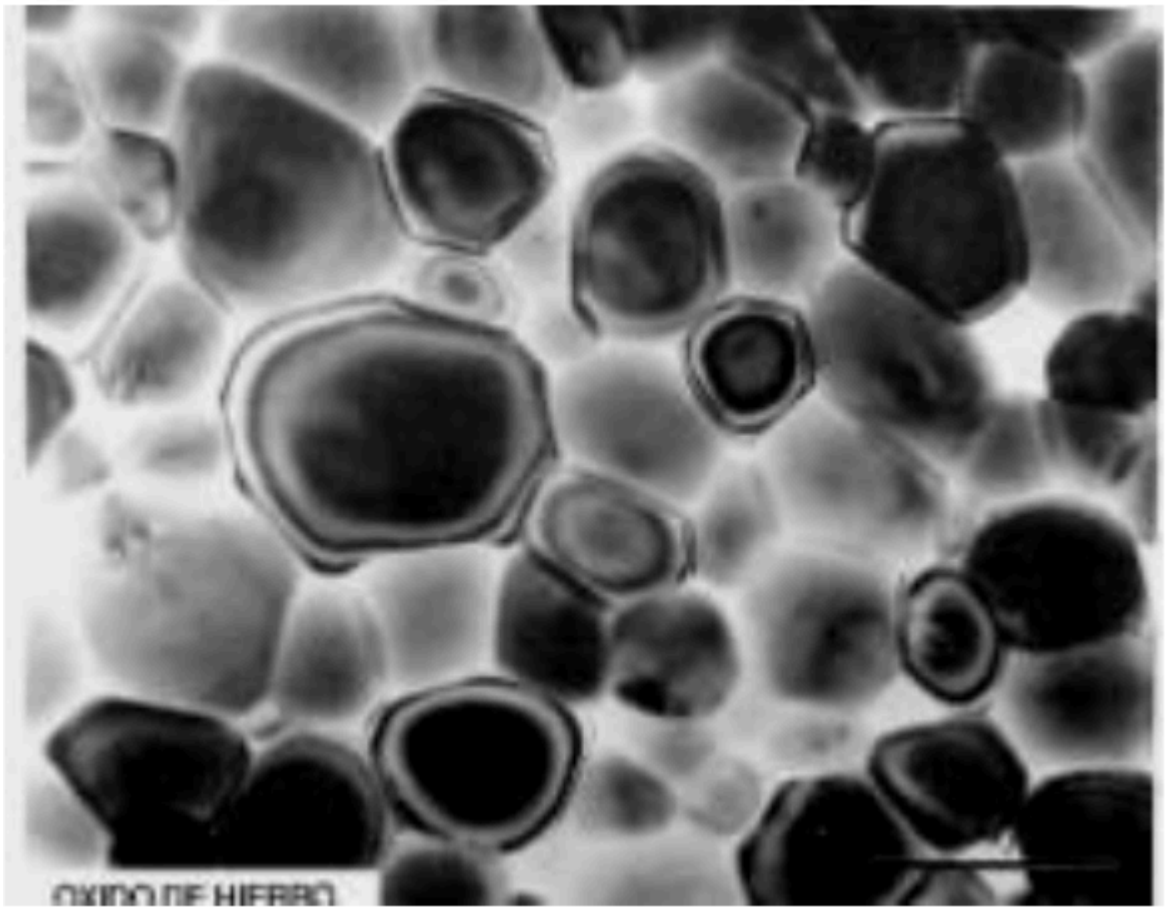


Figure 19. Micrograph of a cored (microsegregated) structure: the rings in each grain are layers of different composition. From Session 7 Slide 24.

Removing the coring. Microsegregation can be eliminated with a **homogenisation heat treatment** at a temperature below the solidus (for example $T_{TT} < T_7$, below the lowest temperature at which liquid could exist). During the treatment, solid-state diffusion (Topic 3) evens out the composition inside the grains [Session 7 Slide 24].

WORKED EXAMPLE 5 — HOW MUCH LIQUID IS LEFT AT D′ WITH FAST COOLING [SESSION 7 SLIDE 23]

Data from the slide. At d′ the liquid is C_L = 24 wt% Ni, and the average composition of the cored solid is 38 wt% Ni (the dashed line). The alloy is C_0 = 35 wt% Ni.

Lever rule on the average composition. The tie line now ends on the non-equilibrium solidus, at 38 wt% Ni:

$$W_L = \frac{38 - 35}{38 - 24} = \frac{3}{14} = 0.214 \qquad W_\alpha = \frac{35 - 24}{38 - 24} = \frac{11}{14} = 0.786$$

Comparison. In equilibrium (Worked example 4) the alloy is completely solid at d′. With fast cooling about 21 % of the alloy is still liquid there; solidification ends only at e′, where the average composition of the solid reaches 35 wt% Ni and $W_L = (35 - 35)/(35 - 21) = 0$. The slide gives the average compositions; the lever rule is used here with them exactly as in §6.

§ 8 Binary eutectic systems [Session 7 Slides 26-32]

8.1 Partial solid solubility and the eutectic reaction [Session 7 Slides 26-27]

When the two components have only **partial solid solubility**, two solid solutions appear: α, rich in A, and β, rich in B. Their regions are bounded by three kinds of line:

- the **liquidus**, above which everything is liquid;
- the **solidus**, below which there is no liquid;
- the **solvus**, which gives the maximum solubility of B in α (and of A in β) at each temperature in the solid state.

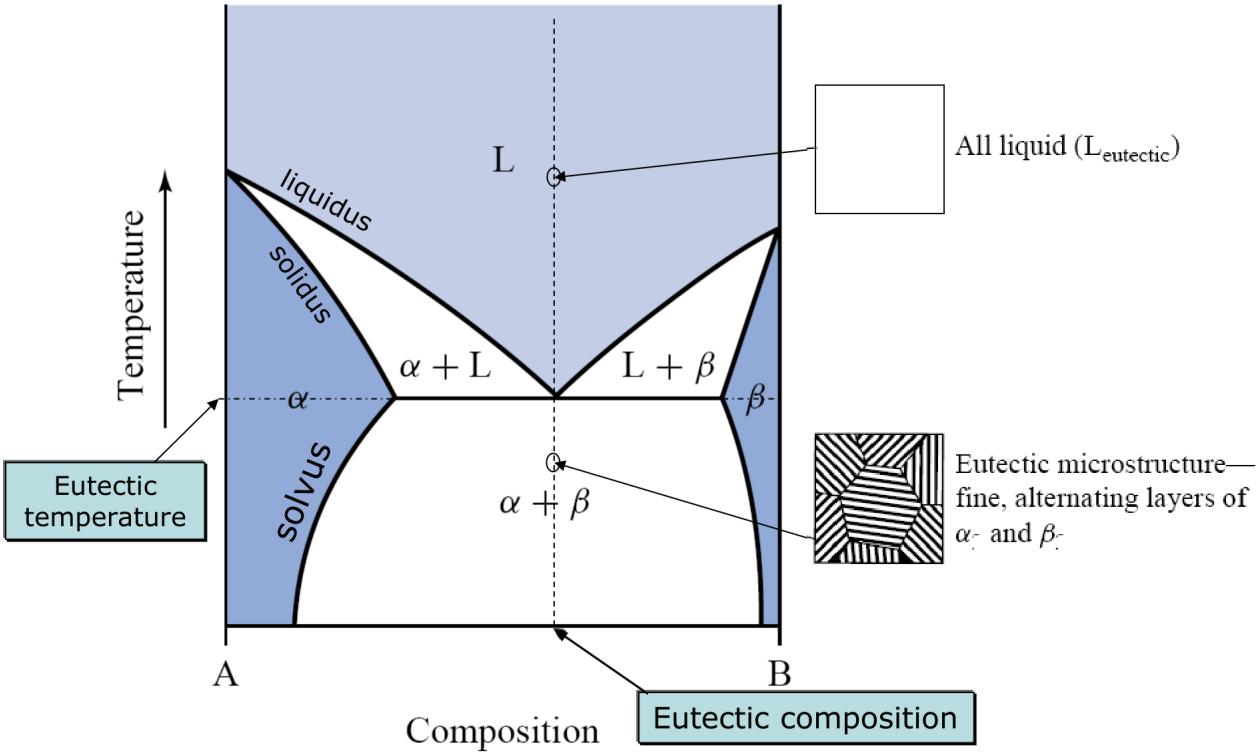


Figure 20. Binary eutectic diagram with liquidus, solidus and solvus, the regions L, α, β, α + L, L + β and α + β, the eutectic temperature and composition, and the eutectic microstructure of fine alternating layers of α and β. Copied from Session 7 Slide 26.

The two liquidus lines meet at the **eutectic point**, defined by the **eutectic temperature** and the **eutectic composition**. A liquid of eutectic composition does not pass through a two-phase L + solid region: at the eutectic temperature it transforms at once into **two** solids, in a fine microstructure of alternating layers of α and β.

Invariant reactions are produced at a fixed composition and a fixed temperature for a given system, and the result also remains fixed. In a system of two partially soluble components, a solidification reaction occurs at a fixed composition and temperature, with a cooling curve similar to that of a pure solid [Session 7 Slide 27]. This is the **eutectic reaction**:

$$L \rightarrow \alpha + \beta \quad \text{(on cooling, at the eutectic temperature)}$$

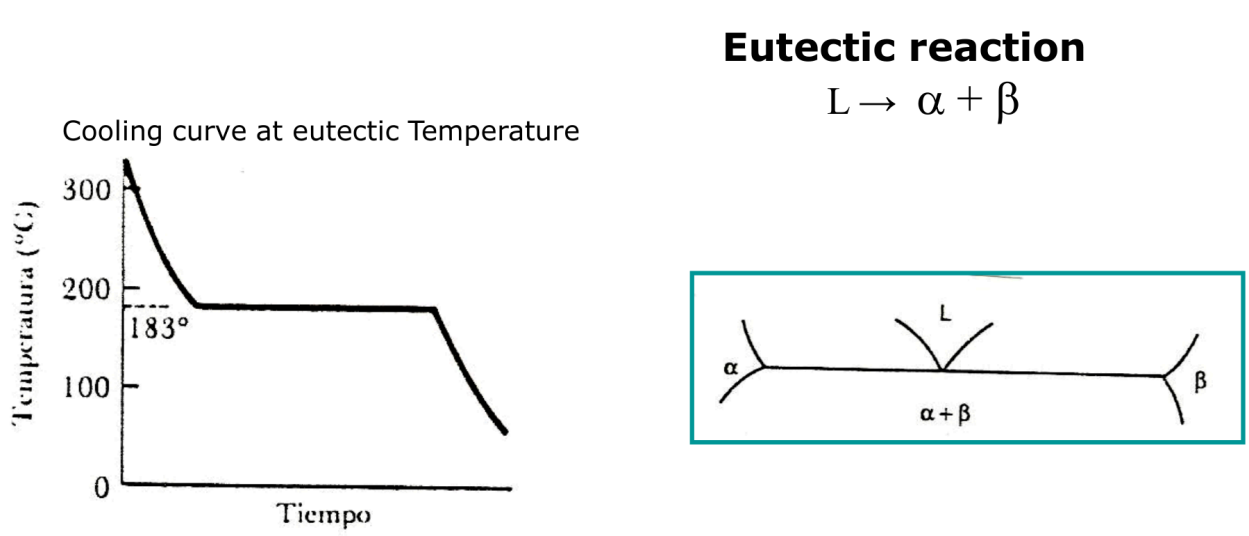


Figure 21. Cooling curve at the eutectic temperature (183 °C plateau; axes in Spanish, Temperatura and Tiempo) and the shape of the eutectic reaction L → α + β on the diagram. Copied from Session 7 Slide 27.

With the phase rule of §5 at 1 atm: at the eutectic point three phases (L, α, β) coexist in a two-component system, so $F = 2 - 3 + 1 = 0$. Temperature and the compositions of the three phases are all fixed, which is why the reaction is invariant and the cooling curve shows a plateau.

8.2 The Pb-Sn system and the eutectic microstructure [Session 7 Slides 28-29]

The Pb-Sn diagram is the example of the slides. Its eutectic point is at **61.9 wt% Sn and 183 °C**. At 183 °C, α can dissolve at most **18.3 wt% Sn** and β contains **97.8 wt% Sn** [Session 7 Slides 27, 32]. The melting points of the pure metals are 327 °C (Pb) and 232 °C (Sn) (Problems T4_PhaseDiagrams I.pdf, p. 1).

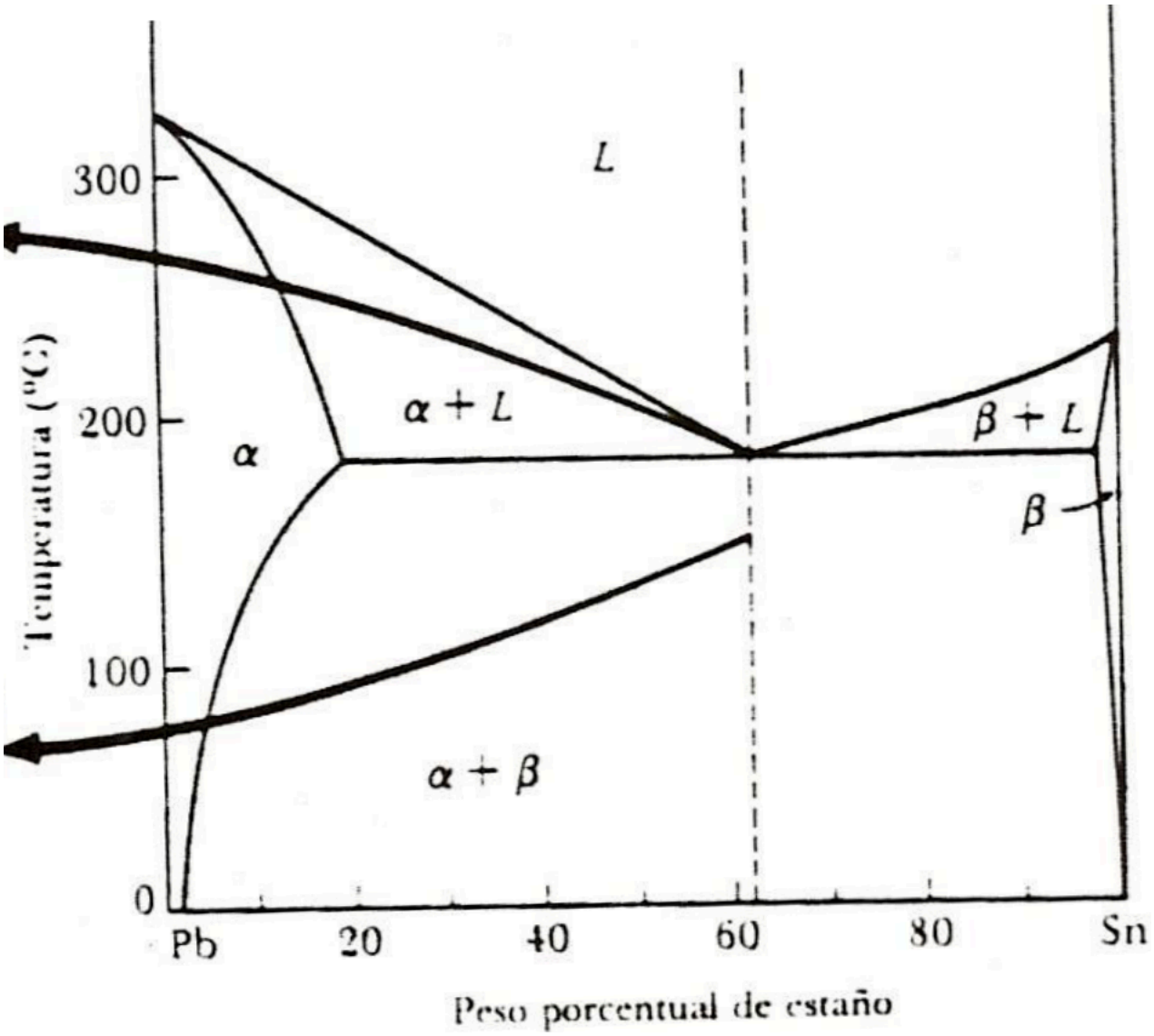


Figure 22. Pb-Sn phase diagram (axes in Spanish: Temperatura, Peso porcentual de estaño) with the eutectic at about 61.9 wt% Sn and 183 °C; the thick arrows point to the sketches of the eutectic reaction shown below. Copied from Session 7 Slide 28.

A liquid of 61.9 wt% Sn, cooled to 183 °C, forms colonies of α and β layers that grow together into the liquid until the whole structure is eutectic (the **eutectic product**). The layers alternate with a spacing λ:

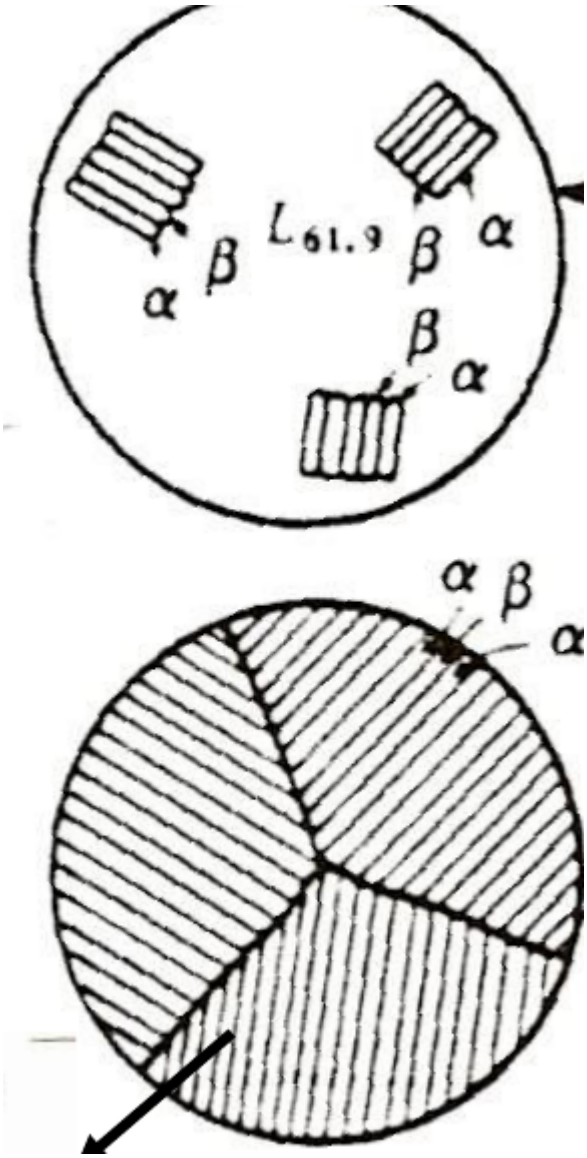


Figure 23. Pb-Sn eutectic: layers of α and β grow together from the liquid of 61.9 % Sn (top) until the whole structure is eutectic colonies (bottom). From Session 7 Slide 28.

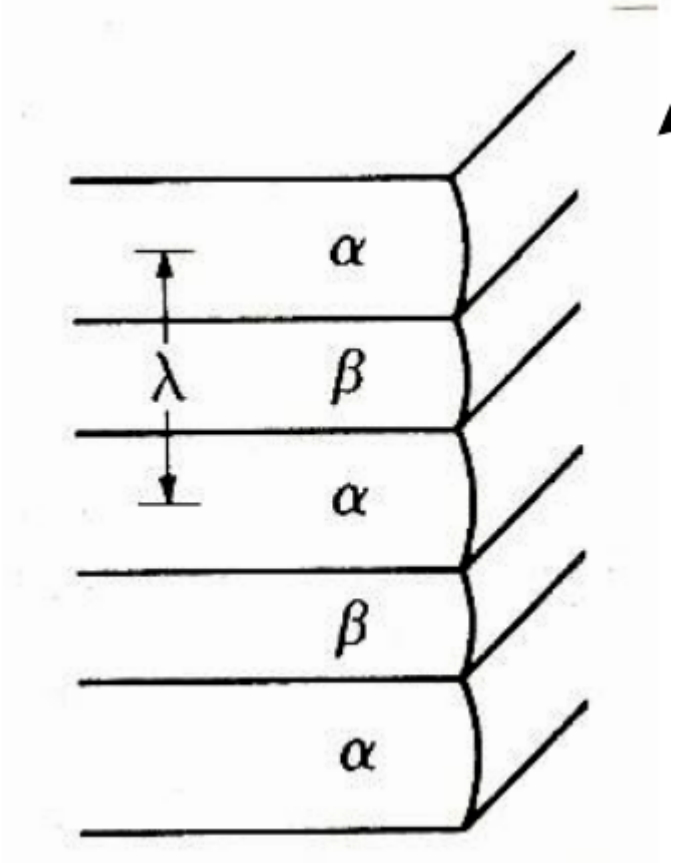


Figure 24. The eutectic product: alternating lamellae of α and β; λ is the spacing of the layers. From Session 7 Slide 28.

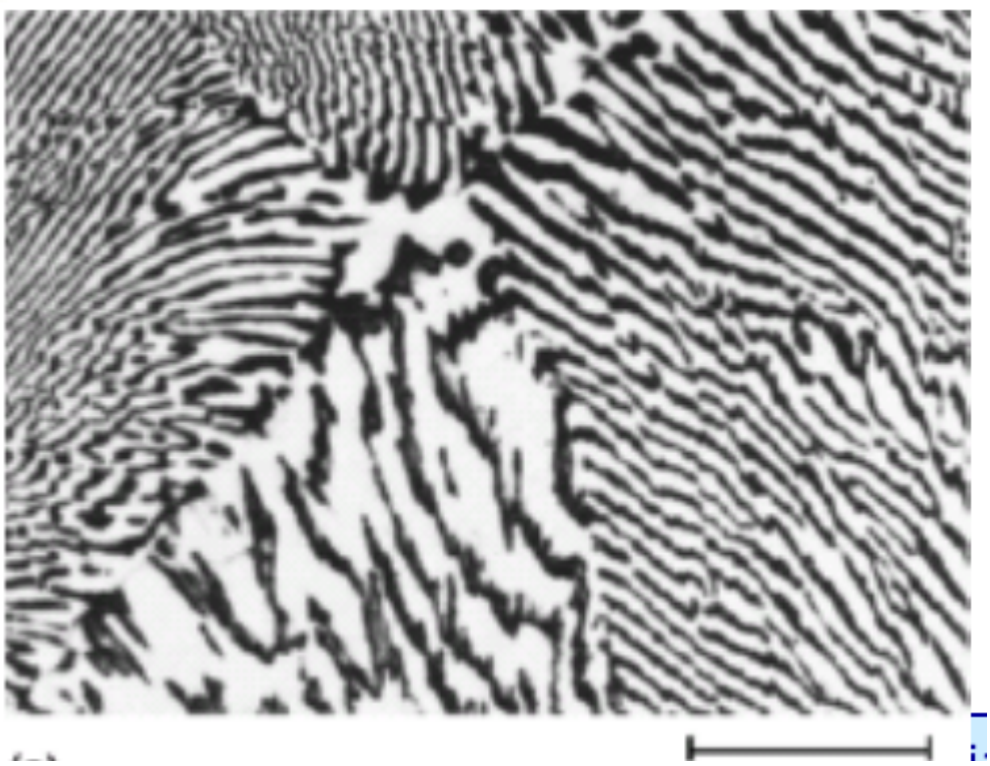


Figure 25. Micrograph of the Pb-Sn eutectic (scale bar 40 μm): dark and light layers are the two phases. From Session 7 Slide 28.

The shape of the eutectic changes from one system to another, as the four micrographs of Slide 29 show (Sn-50In, Al-13Si, Al-33Cu and Mg-37Sn): in them one phase appears as rods, plates, globules or needles inside the other (a description of those micrographs).

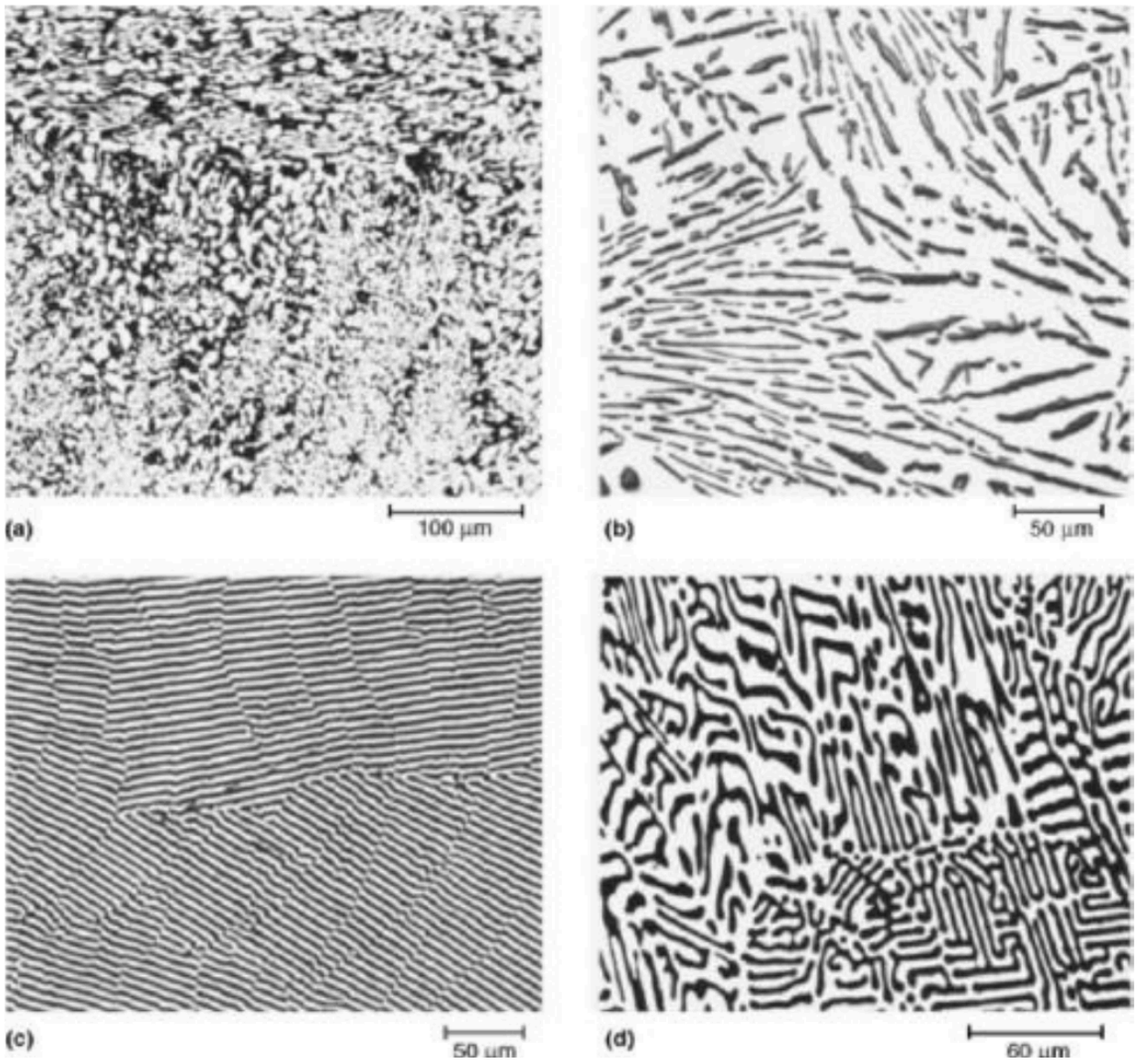


Figure 26. Eutectic microstructures: (a) Sn-50 % In, (b) Al-13 % Si, (c) Al-33 % Cu, (d) Mg-37 % Sn. From Session 7 Slide 29.

PB-SN VALUES IN THE PROBLEM SHEET

The diagram printed in the problem sheet (Problems T4_PhaseDiagrams I.pdf, p. 1) and the official solutions use 19.2 wt% Sn for α and 97.5 wt% Sn for β at 183 °C, instead of the 18.3 and 97.8 of Slide 32: they come from a different drawing of the same system. The worked examples of this page use the slide values; the solved problems use the values of the sheet.

WORKED EXAMPLE 6 — PHASES IN THE PB-SN EUTECTIC PRODUCT [SESSION 7 SLIDES 28 , 32]

Question. A Pb-61.9 wt% Sn alloy (eutectic composition) is cooled just below 183 °C. What are the amounts of α and β in the eutectic?

Tie line. Just below 183 °C the tie line crosses the α + β region from C_α = 18.3 to C_β = 97.8 wt% Sn; its length is $97.8 - 18.3 = 79.5$ wt% Sn.

Lever rule.

$$W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha} = \frac{97.8 - 61.9}{97.8 - 18.3} = \frac{35.9}{79.5} = 0.452 \qquad W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{61.9 - 18.3}{79.5} = \frac{43.6}{79.5} = 0.548$$

Check. $0.452 + 0.548 = 1$ and $0.452 \times 18.3 + 0.548 \times 97.8 = 8.27 + 53.59 = 61.9$ wt% Sn $= C_0$.

Answer. The alloy is 100 % eutectic microconstituent, and inside it 45.2 % of the mass is α and 54.8 % is β. Every eutectic of this system, whatever the alloy it comes from, has these proportions (used in Worked example 7).

8.3 Hypoeutectic and hypereutectic alloys [Session 7 Slides 30-31]

An alloy between the α limit and the eutectic composition is **hypoeutectic**; one between the eutectic composition and the β limit is **hypereutectic**.

Hypoeutectic alloy (the slide uses Pb-Sn with about 30 wt% Sn, a reading of the figure):

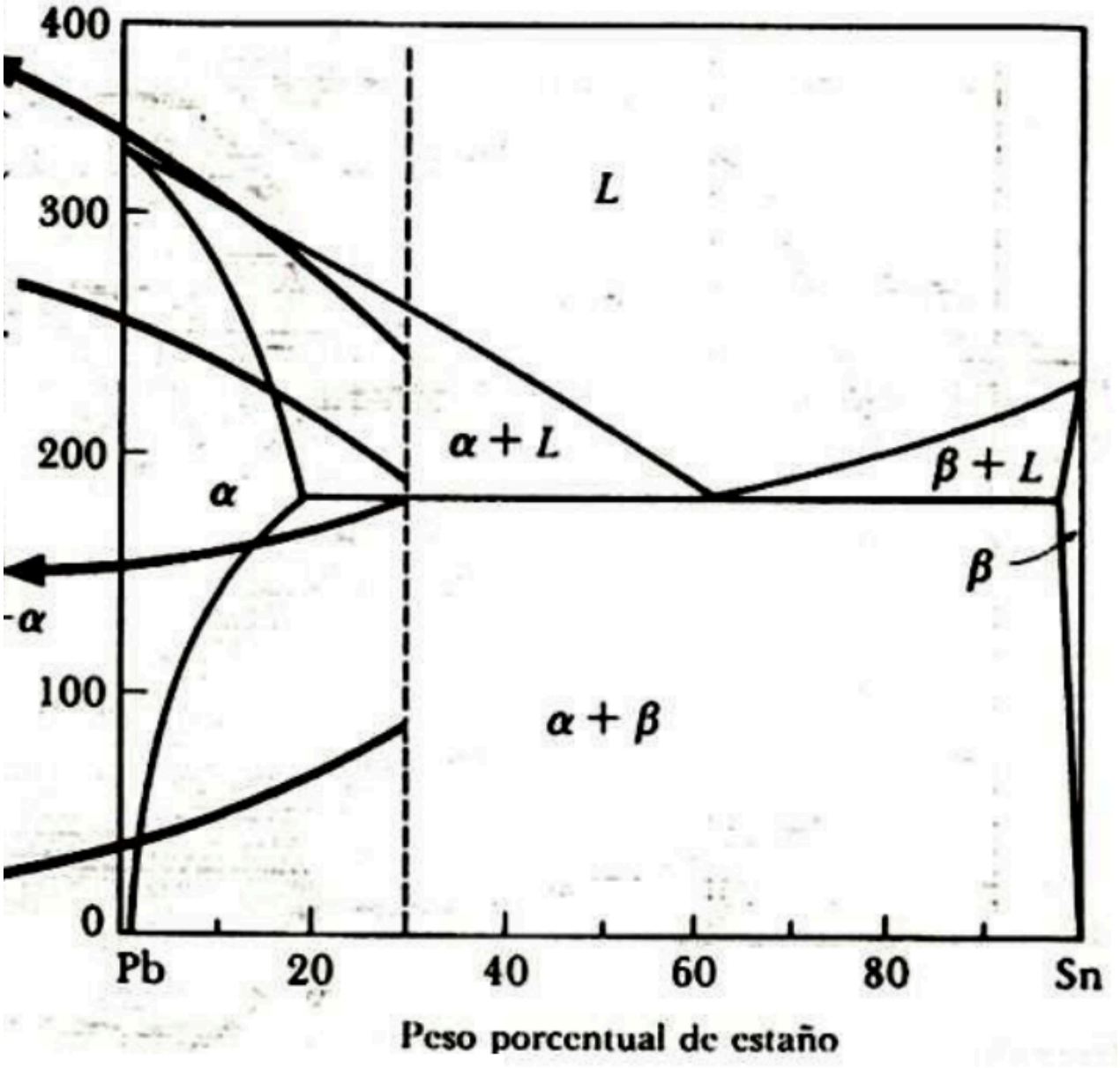


Figure 27. Pb-Sn diagram with the dashed line of the hypoeutectic alloy (about 30 wt% Sn); the arrows point to the microstructure sketches shown below. Copied from Session 7 Slide 30.

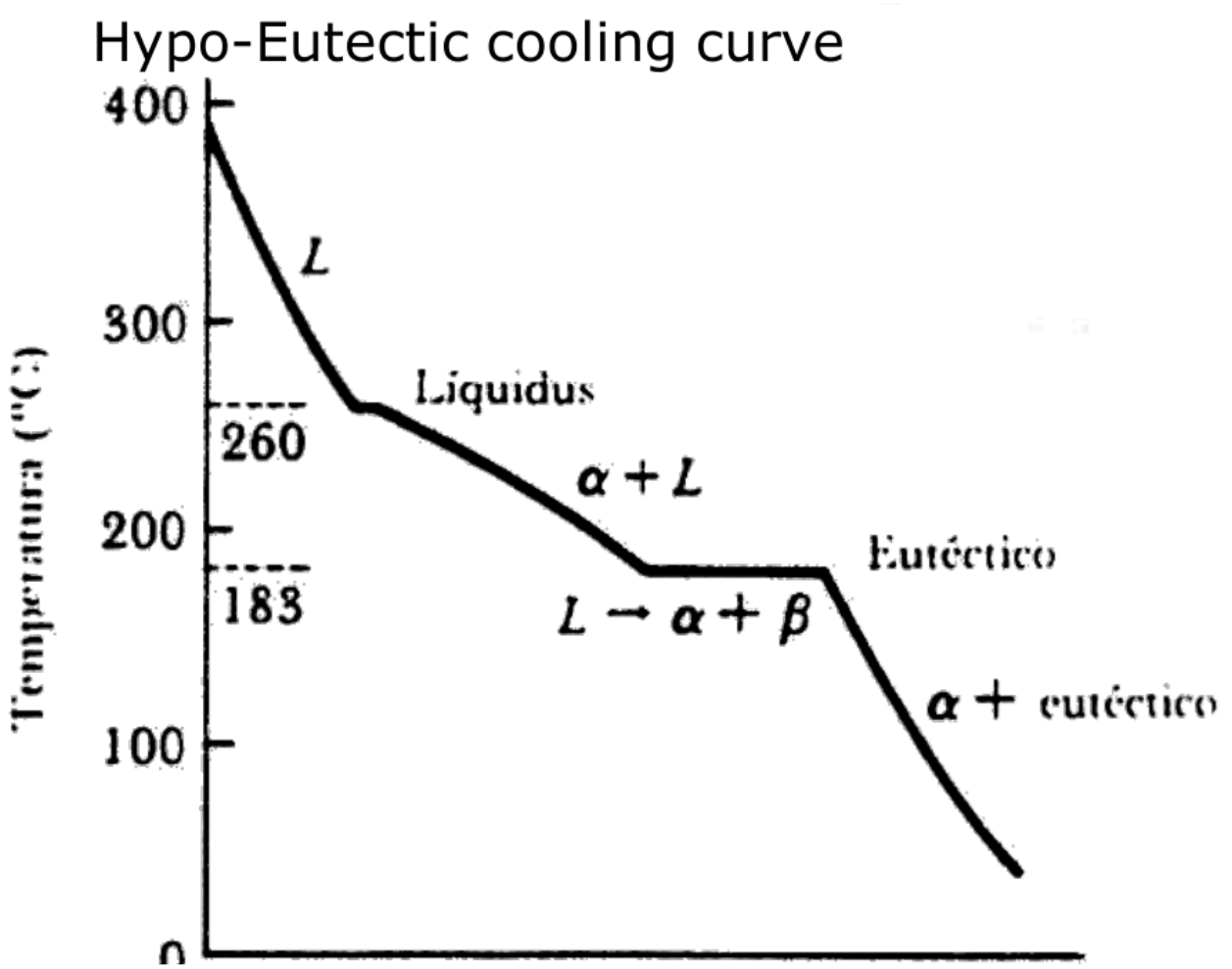


Figure 28. Cooling curve of the hypoeutectic alloy: liquidus at 260 °C, α + L, the eutectic plateau at 183 °C (L → α + β) and α + eutectic below it. Copied from Session 7 Slide 30.

1. Above the liquidus the alloy is liquid. The cooling curve shows the liquidus at 260 °C, where crystals of α start to form.
2. Between 260 and 183 °C, α and liquid coexist; α grows, and the liquid becomes richer in Sn along the liquidus.
3. Just above 183 °C the solid is α of 18.3 wt% Sn and the liquid has reached the eutectic composition, 61.9 wt% Sn.
4. At 183 °C that liquid transforms at constant temperature into the eutectic ($L \rightarrow \alpha + \beta$): a plateau in the cooling curve.
5. Below 183 °C the structure is **primary (proeutectic) α** surrounded by **eutectic**.

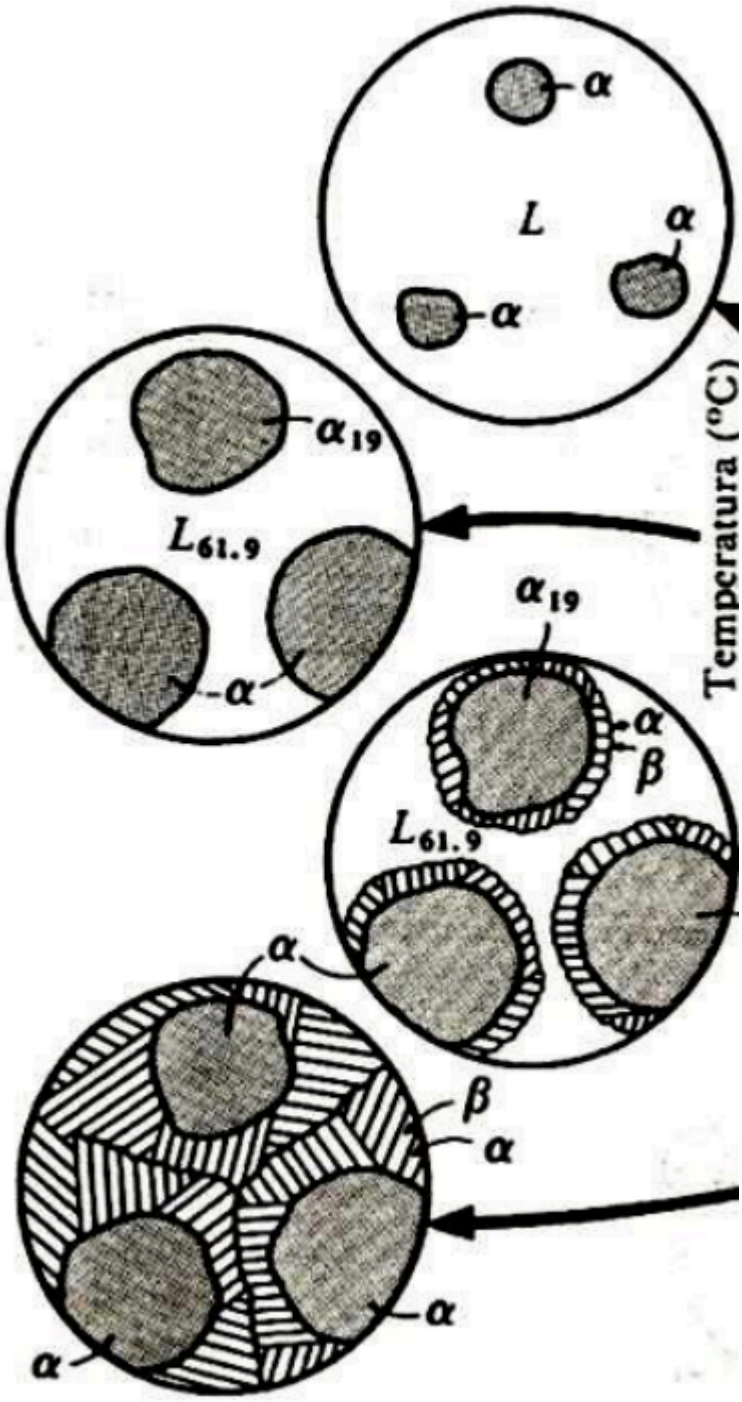


Figure 29. Hypoeutectic Pb-Sn alloy, from top to bottom: α crystals in the liquid; α of 19 % Sn in liquid of 61.9 % Sn just above the eutectic temperature; eutectic forming around the α; primary α in a eutectic matrix. From Session 7 Slide 30.

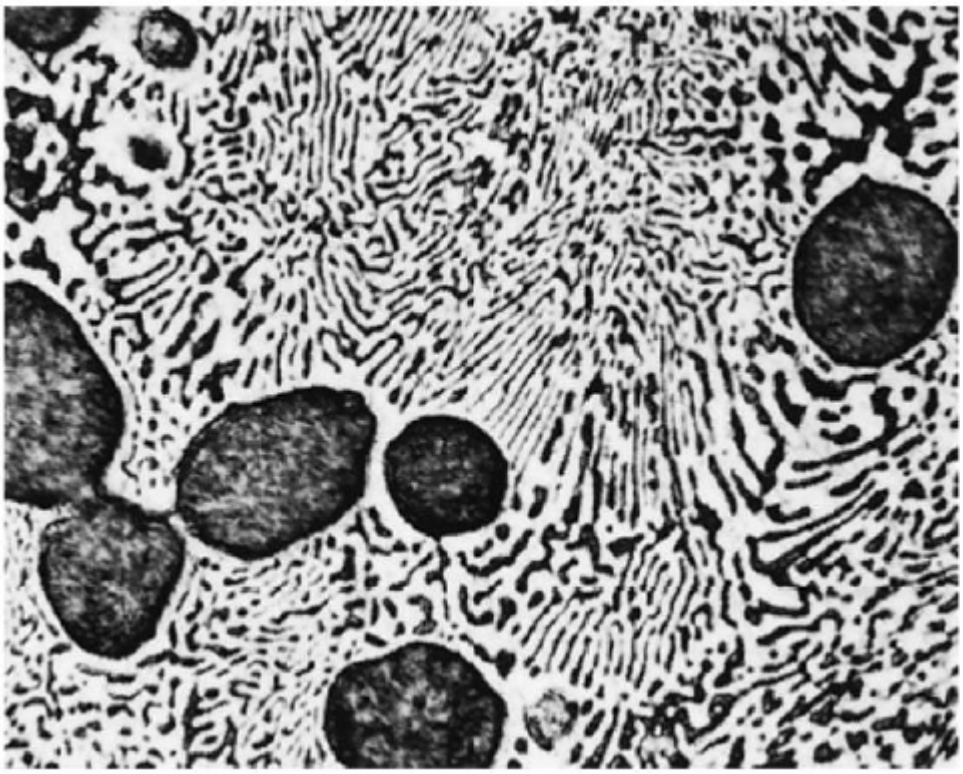


Figure 30. Micrograph of a hypoeutectic Pb-Sn alloy: dark rounded primary α in the lamellar eutectic. From Session 7 Slide 30.

In a casting, the primary α grows from the mould wall, and the liquid trapped between the crystals is the last to solidify, with the eutectic composition:

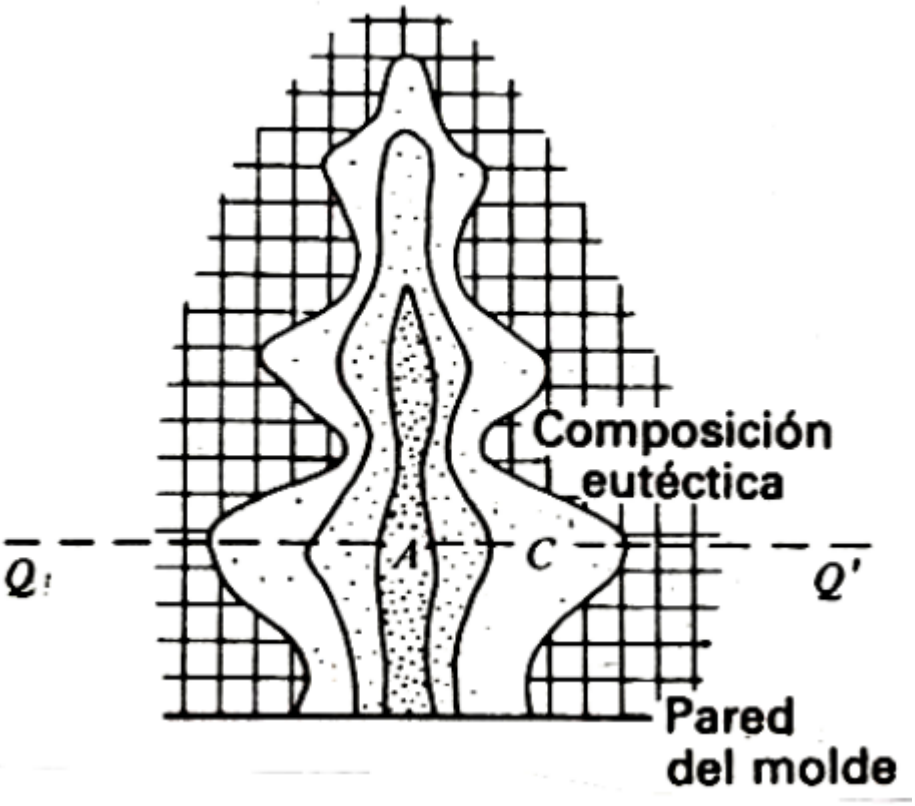


Figure 31. Hypoeutectic solidification in a mould: primary α grows from the mould wall and the last liquid, between the crystals, reaches the eutectic composition. From Session 7 Slide 30.

Hypereutectic alloy: the same sequence with the roles of the solids exchanged. Primary β forms between the liquidus and the eutectic temperature, the remaining liquid gives the eutectic at the eutectic temperature, and the final structure is **primary β plus eutectic**.

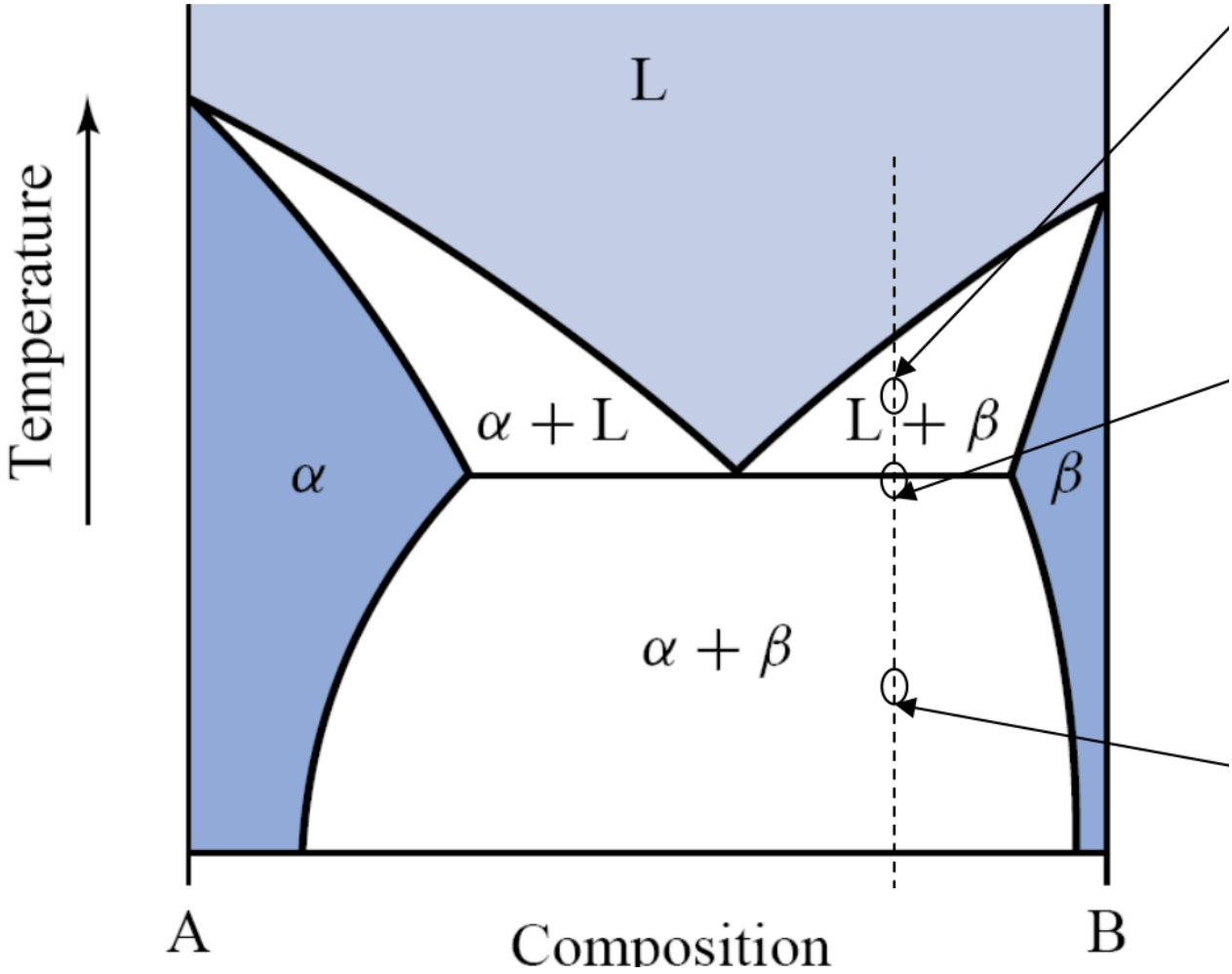


Figure 32. Eutectic diagram with the dashed line of a hypereutectic alloy and its points in L, L + β and α + β; the arrows point to the microstructure sketches shown below. Copied from Session 7 Slide 31.

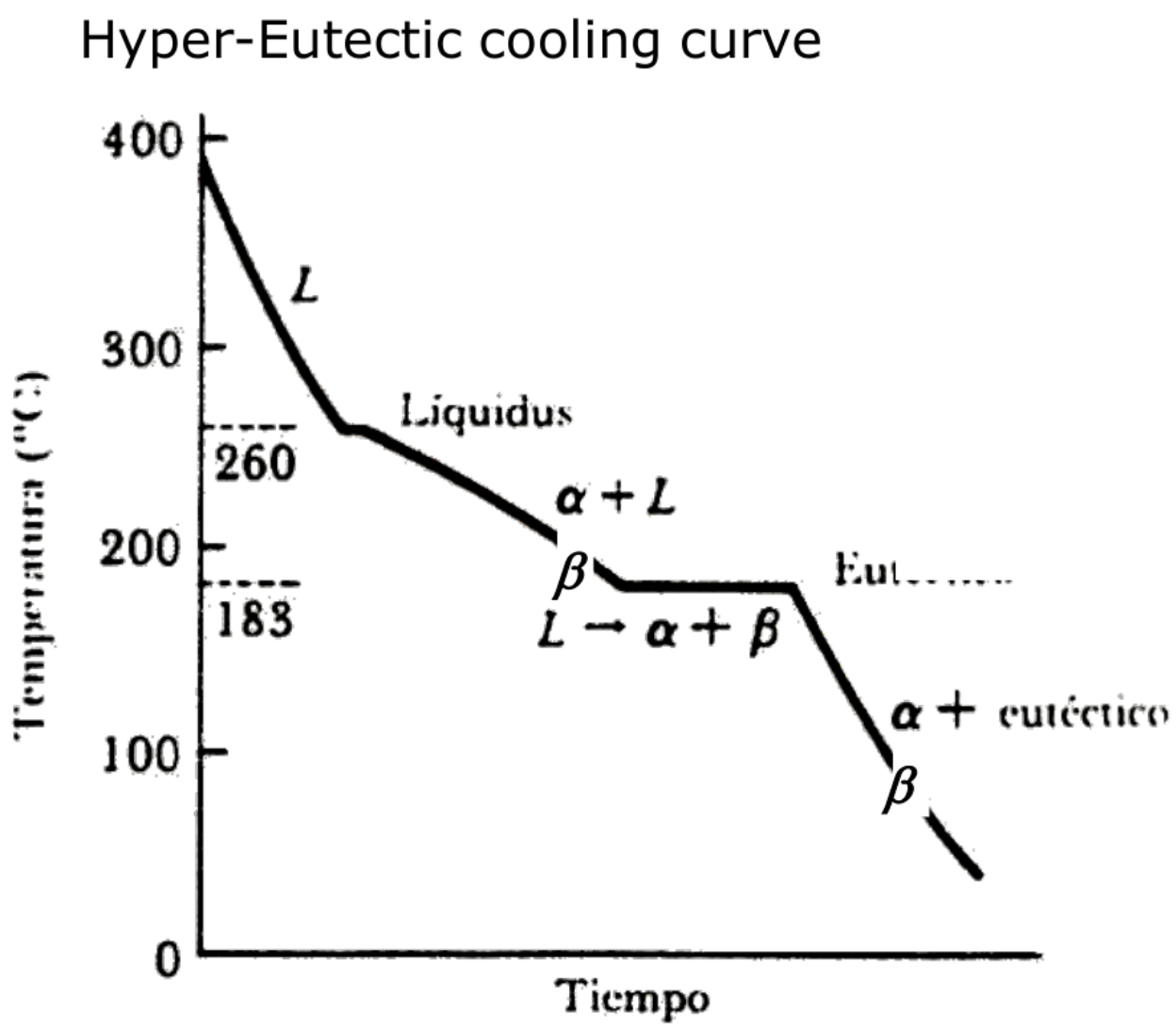


Figure 33. Cooling curve of the hypereutectic alloy: primary β forms below the liquidus, then the eutectic plateau (L → α + β) and β + eutectic below it; the slide repeats the labels 260 and 183 °C of the hypoeutectic curve. Copied from Session 7 Slide 31.

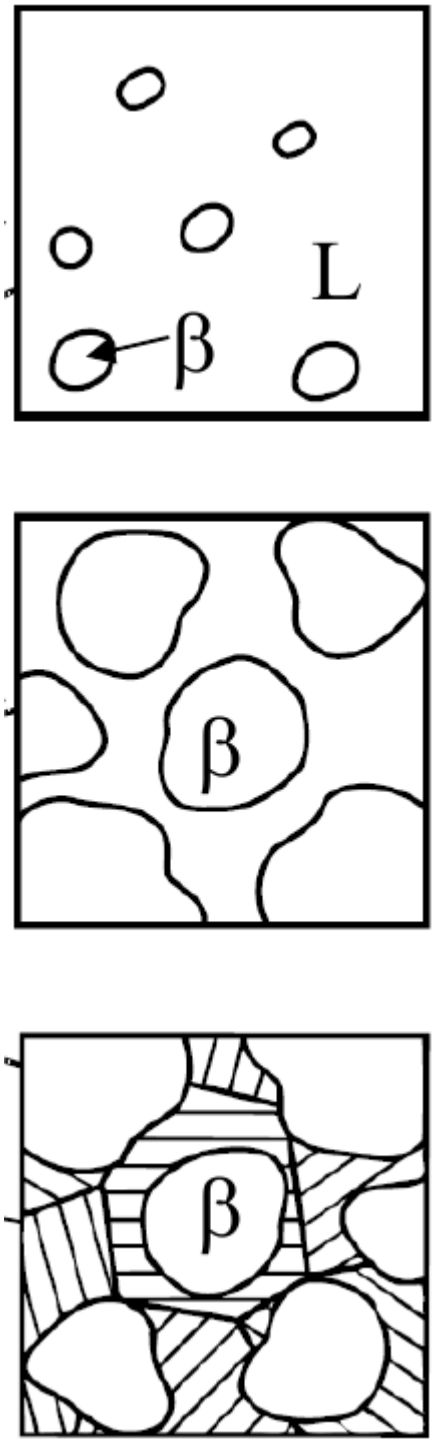


Figure 34. Hypereutectic alloy, from top to bottom: β crystals in the liquid; primary β just above the eutectic temperature; primary β in a eutectic matrix. From Session 7 Slide 31.

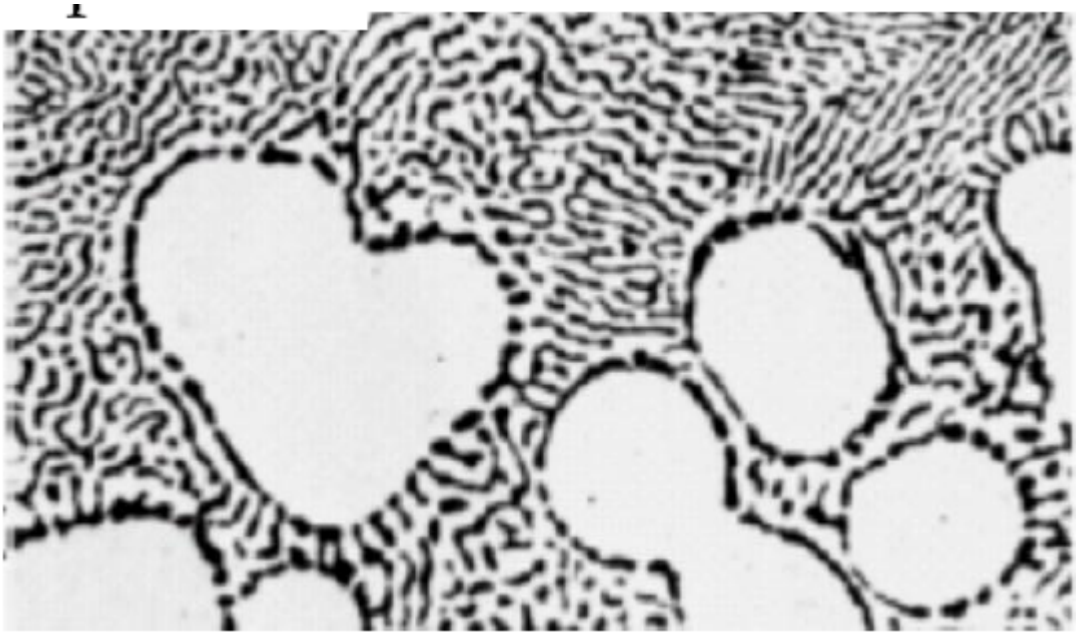


Figure 35. Micrograph of a hypereutectic alloy: light rounded primary β in the eutectic. From Session 7 Slide 31.

(The cooling curve of Slide 31 repeats the temperature labels of the hypoeutectic curve, 260 and 183 °C; for a hypereutectic Pb-Sn alloy the liquidus is below 232 °C, so read them only as the liquidus temperature T_L and the eutectic temperature T_E of the alloy.)

8.4 Lever rule in binary eutectic diagrams [Session 7 Slide 32]

Just below the eutectic temperature, the tie line of a hypoeutectic alloy C_4' is cut by the alloy and by the eutectic composition into three segments, P, Q and R:

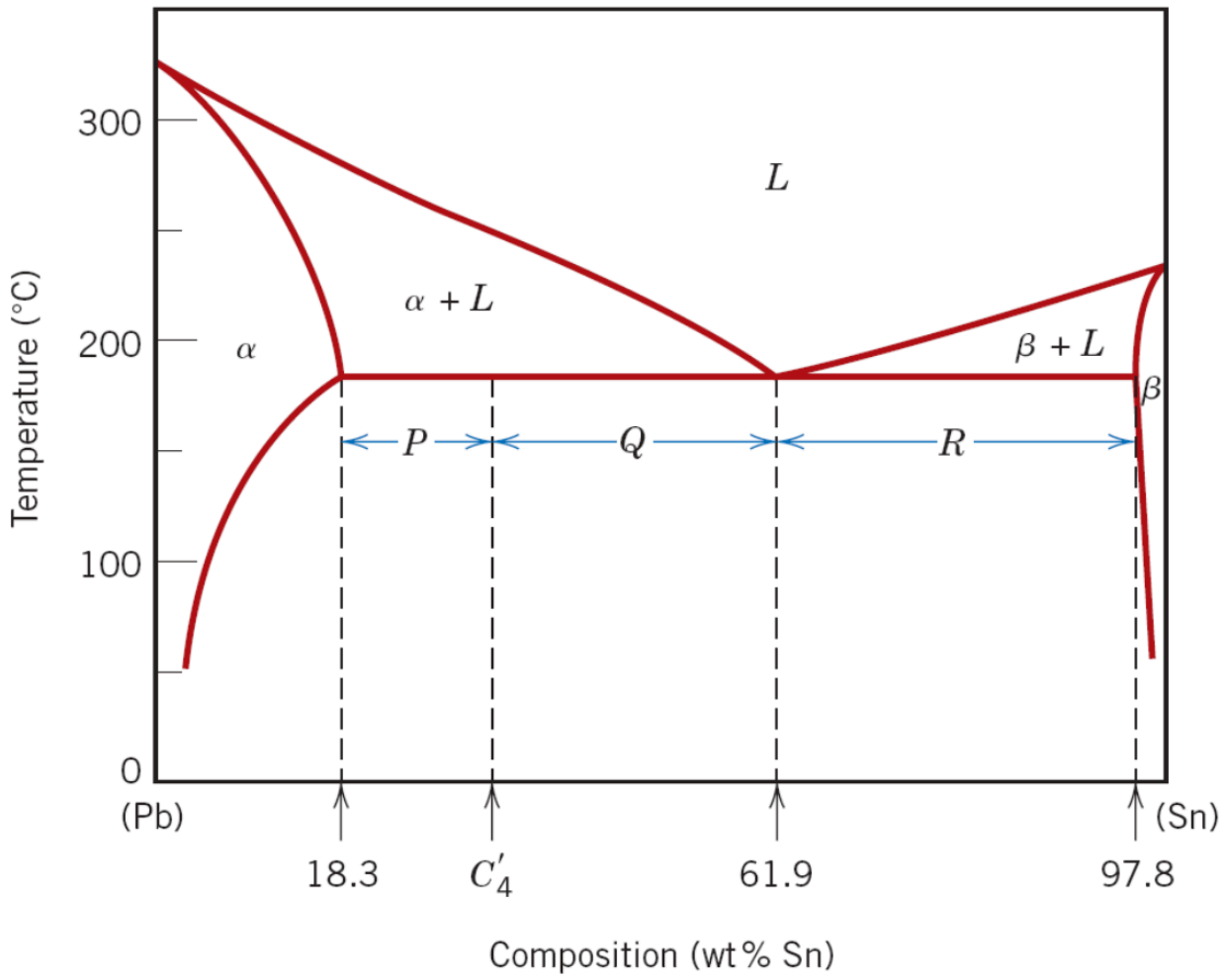


Figure 36. Pb-Sn diagram with the segments P, Q and R of the tie line just below 183 °C, from 18.3 wt% Sn (α) through the alloy C'4 and the eutectic 61.9 wt% Sn to 97.8 wt% Sn (β). Copied from Session 7 Slide 32.

$$W_{\alpha'} = \frac{Q}{P+Q} \qquad W_e = W_L = \frac{P}{P+Q} \qquad W_\alpha = \frac{Q+R}{P+Q+R} \qquad W_\beta = \frac{P}{P+Q+R}$$

- W_e: fraction of eutectic microconstituent, equal to W_L, the fraction of liquid just above the eutectic temperature (all that liquid becomes eutectic);
- $W_{\alpha'}$: fraction of α′ phase, the **primary** α;
- W_α: **total** fraction of α (primary α plus the α inside the eutectic);
- W_β: fraction of β phase (all of it inside the eutectic).

The first two use the short tie line between α and the eutectic liquid (microconstituents); the last two use the whole tie line between α and β (phases).

WORKED EXAMPLE 7 — MICROCONSTITUENTS AND PHASES OF PB-30 WT% SN [SESSION 7 SLIDES 30 , 32]

Data. Alloy C_0 = 30 wt% Sn (the hypoeutectic alloy of Slide 30), just below 183 °C, with the values of Slide 32: C_α = 18.3, C_e = 61.9, C_β = 97.8 wt% Sn.

Step 1, the segments. $P = C_0 - C_\alpha = 30 - 18.3 = 11.7$; $Q = C_e - C_0 = 61.9 - 30 = 31.9$; $R = C_\beta - C_e = 97.8 - 61.9 = 35.9$; so $P + Q = 43.6$ and $P + Q + R = 79.5$ (all in wt% Sn).

Step 2, microconstituents (short tie line, from α to the eutectic liquid):

$$W_{\alpha'} = \frac{Q}{P+Q} = \frac{31.9}{43.6} = 0.732 \qquad W_e = \frac{P}{P+Q} = \frac{11.7}{43.6} = 0.268$$

Step 3, phases (whole tie line, from α to β):

$$W_\alpha = \frac{Q+R}{P+Q+R} = \frac{67.8}{79.5} = 0.853 \qquad W_\beta = \frac{P}{P+Q+R} = \frac{11.7}{79.5} = 0.147$$

Step 4, the α inside the eutectic. It is the total α minus the primary α: $W_{\alpha,e} = W_\alpha - W_{\alpha'} = 0.853 - 0.732 = 0.121$.

Check, phases against microconstituents. By Worked example 6 every eutectic is 45.2 % α and 54.8 % β. Then $W_e \times 0.452 = 0.268 \times 0.452 = 0.121 = W_{\alpha,e}$ and $W_e \times 0.548 = 0.268 \times 0.548 = 0.147 = W_\beta$: both descriptions of the alloy agree.

Answer. Just below 183 °C the alloy is 73.2 % primary α and 26.8 % eutectic. Counting phases, it is 85.3 % α (73.2 % primary plus 12.1 % in the eutectic) and 14.7 % β. Just above 183 °C the same lever gives 73.2 % α of 18.3 wt% Sn and 26.8 % liquid of 61.9 wt% Sn: that liquid is what becomes the eutectic.

Part II — Phase diagrams II (Session 8)

§ 9 Solid-state precipitation [Session 8 Slides 2-4]

9.1 Below the solid solubility limit [Session 8 Slide 2]

An alloy whose composition stays inside the α region all the way down to room temperature never crosses the solvus: once solid, it is single-phase α. In the Pb-Sn example, a Pb-1 wt% Sn alloy (C_1) is:

- (a) liquid at 350 °C;
- (b) at about 330 °C, on the liquidus: the solid phase α begins to form (α + L);
- (c) only α at 270 °C, and it stays α on further cooling.

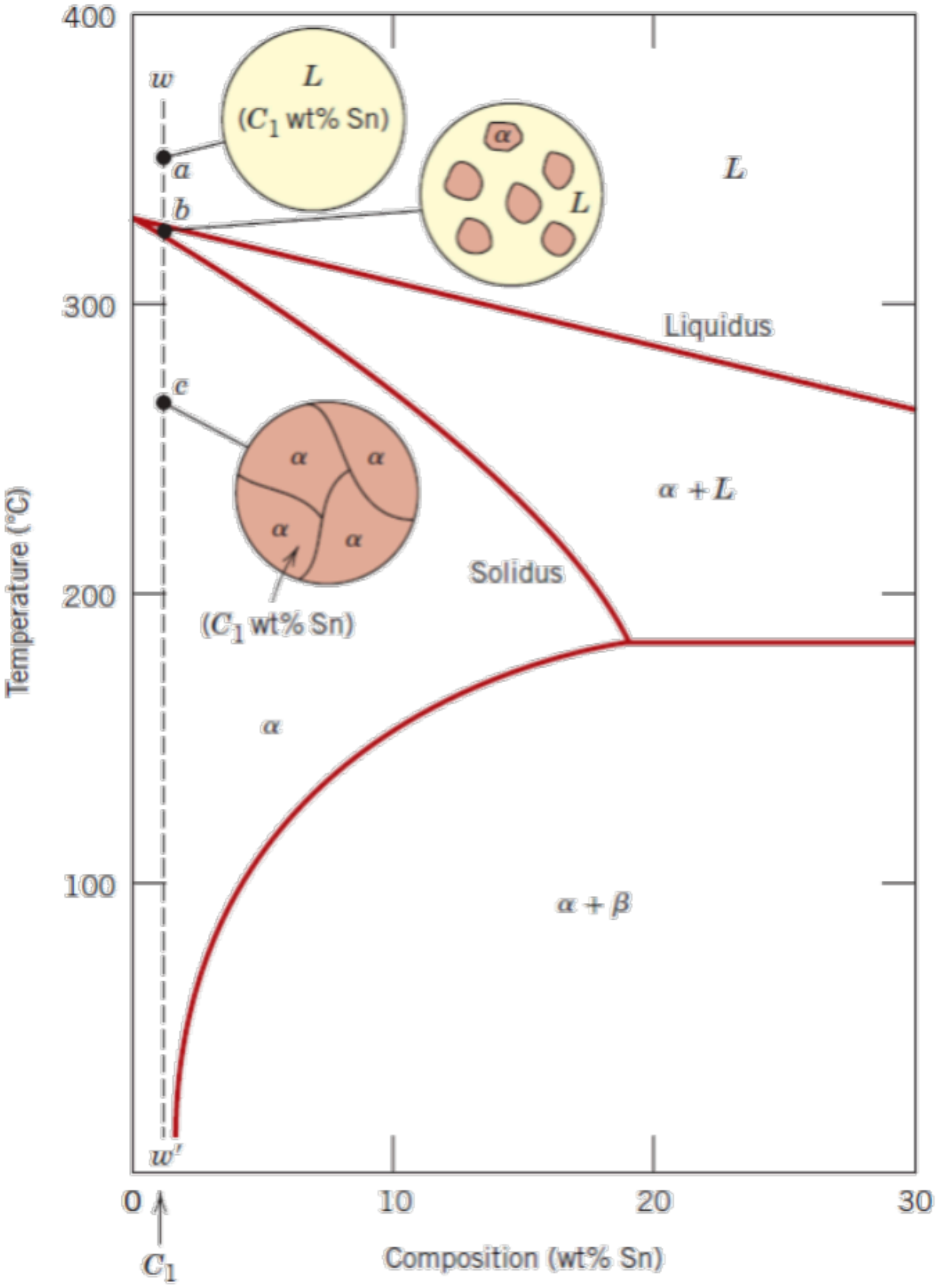


Figure 37. Pb-Sn diagram with the alloy C1 (1 wt% Sn) and its microstructures at a (liquid), b (α + L) and c (only α). Copied from Session 8 Slide 2.

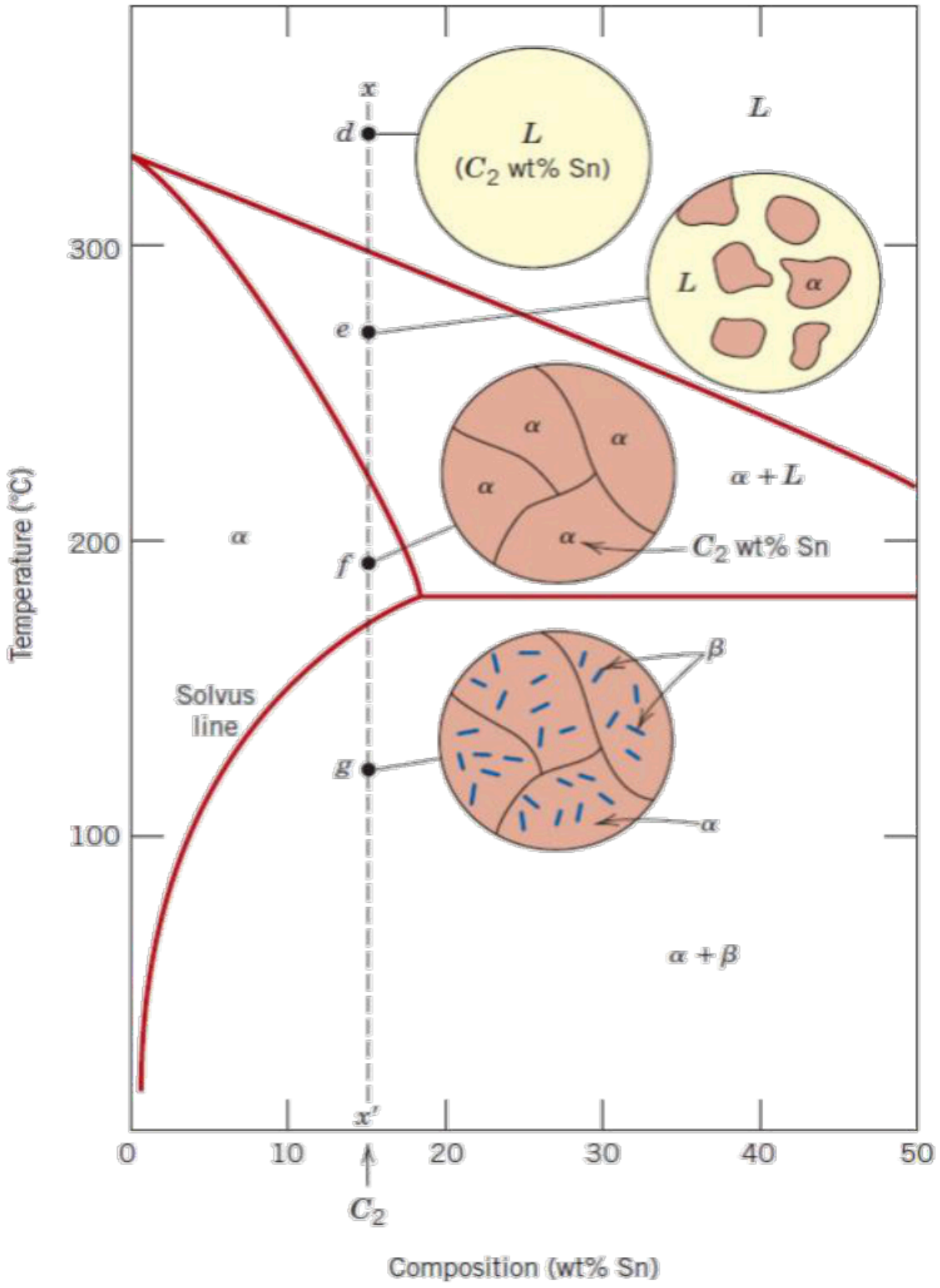


Figure 38. Pb-Sn diagram with the alloy C2 (16 wt% Sn) and its microstructures at d (liquid), e (α + L), f (only α) and g (α + β precipitates after crossing the solvus line). Copied from Session 8 Slide 4.

9.2 Above the solid solubility limit: precipitation [Session 8 Slides 3-4]

Precipitation in the solid state: when an alloy crosses the **solvus** line on cooling, the solid solubility of α is exceeded, and small particles of the β phase form; they **grow as the temperature decreases** [Session 8 Slide 3].

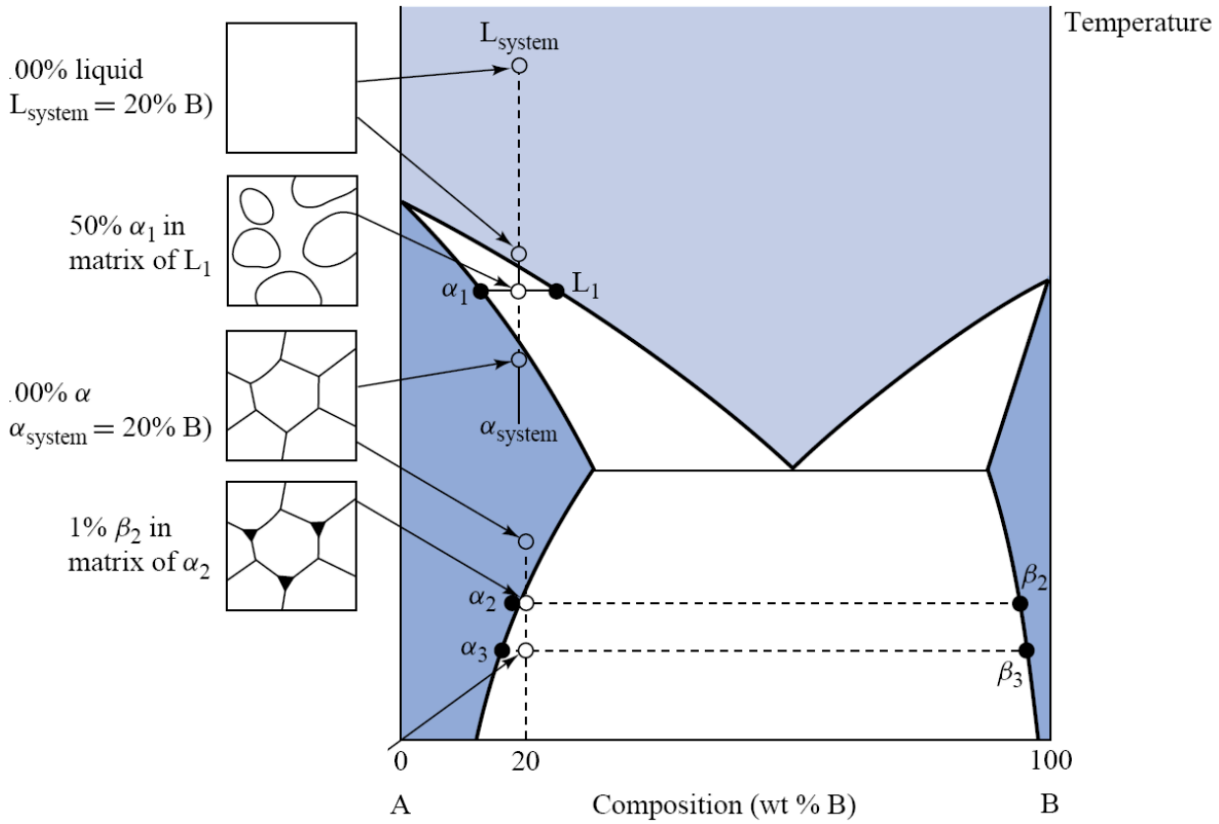


Figure 39. Cooling of an alloy of 20 % B: all liquid, α1 in a matrix of L1, all α, and β2 particles in a matrix of α2 after crossing the solvus (the slide prints '00%' for 100 %). Copied from Session 8 Slide 3.

For the alloy of 20 % B of the slide: all liquid; α_1 crystals in a matrix of liquid L_1; 100 % α with the composition of the alloy; and, below the solvus, small β_2 particles in a matrix of α_2 (mostly at the grain boundaries), which become more and grow at lower temperatures (α_3 + β_3).

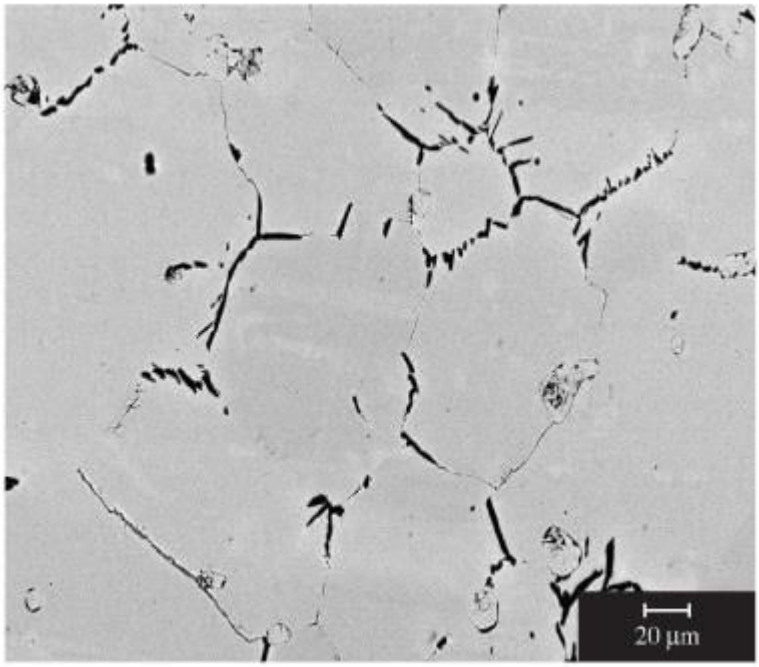


Figure 40. Micrograph of solid-state precipitates at the grain boundaries (scale bar 20 µm). From Session 8 Slide 3.

The Pb-Sn example of Slide 4 uses a Pb-16 wt% Sn alloy (C_2, also in the figure of §9.1):

- (d) liquid at 350 °C;
- (e) at 270 °C, solid α has begun to form: α + L;
- (f) only α at 192 °C;
- (g) α + β at 121 °C: the alloy has crossed the solvus and β has precipitated inside the α.

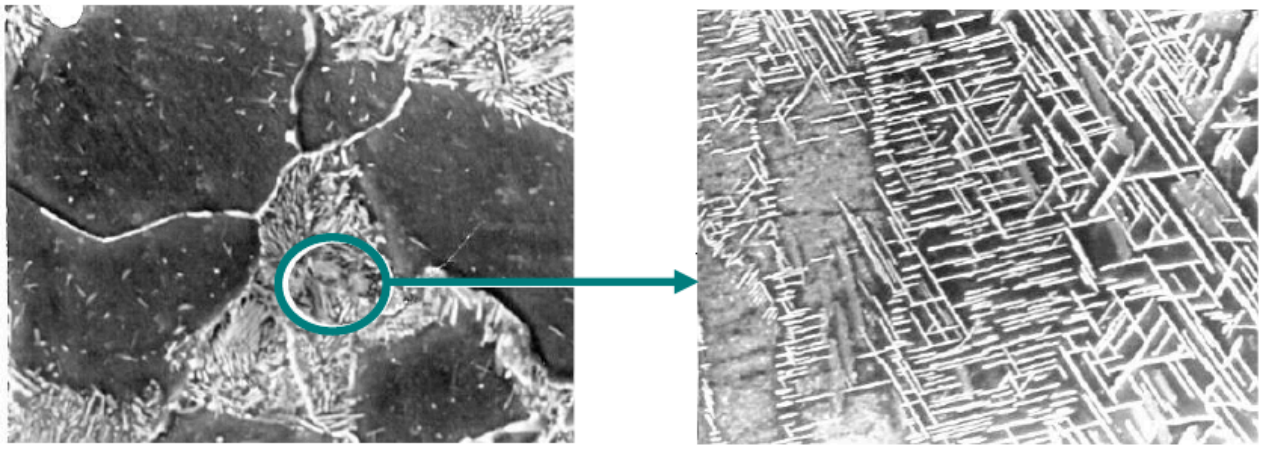


Figure 41. Precipitates inside the grains; on the right, a detail of the circled area. From Session 8 Slide 4.

WORKED EXAMPLE 8 — HOW MUCH B PRECIPITATES IN PB-16 WT% SN [SESSION 8 SLIDE 4]

Readings of the figure. On Slide 4 the vertical line of C_2 = 16 wt% Sn meets the solvus at about 172 °C; at point g (121 °C) the solvus is at about 6 wt% Sn. The β boundary at that temperature is close to pure Sn; Slide 32 of Session 7 places it at about 99 wt% Sn.

Above about 172 °C (point f, 192 °C) the alloy is single-phase α of 16 wt% Sn: $W_\alpha = 1$.

At 121 °C (point g), lever rule on the tie line from 6 to 99 wt% Sn:

$$W_\beta = \frac{C_0 - C_\alpha}{C_\beta - C_\alpha} = \frac{16 - 6}{99 - 6} = \frac{10}{93} = 0.108 \qquad W_\alpha = \frac{99 - 16}{99 - 6} = \frac{83}{93} = 0.892$$

Interpretation. About 11 % of the alloy has turned into β particles between 172 and 121 °C. As the temperature falls further the solvus moves to lower Sn contents, so the α holds less Sn and more β precipitates: the particles grow, as the slide says. These numbers depend on readings of the figure and are approximate.

§ 10 Invariant reactions [Session 8 Slides 6-12]

10.1 The five reactions [Session 8 Slide 6]

An invariant reaction happens at a fixed temperature, with fixed compositions of the three phases involved ($F = 2 - 3 + 1 = 0$ at constant pressure, §5). The slides list five types; the eutectoid and the peritectoid happen entirely in the **solid state**:

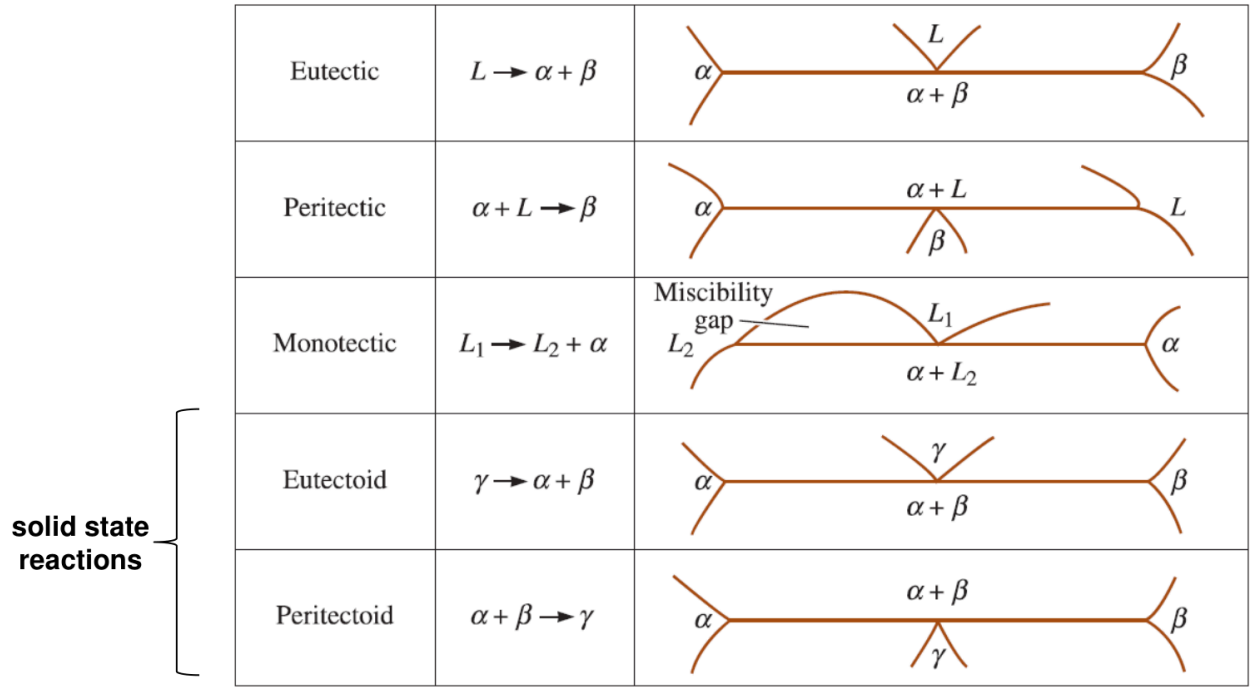


Figure 42. Eutectic, peritectic, monotectic, eutectoid and peritectoid reactions with their equations on cooling and their shape on the diagram; the last two are solid-state reactions. Copied from Session 8 Slide 6.

Reaction	On cooling	Shape on the diagram
Eutectic	$L \rightarrow \alpha + \beta$	a liquid region ends in a V on the horizontal line
Peritectic	$\alpha + L \rightarrow \beta$	a solid region ends in an inverted V below the line
Monotectic	$L_1 \rightarrow L_2 + \alpha$	a liquid miscibility gap above the line
Eutectoid	$\gamma \rightarrow \alpha + \beta$	like the eutectic, with a solid γ instead of L
Peritectoid	$\alpha + \beta \rightarrow \gamma$	like the peritectic, with two solids

10.2 Eutectoid reaction [Session 8 Slide 7]

Eutectoid reaction: an invariant reaction where **a solid phase transforms into two solid phases**: $\gamma \rightarrow \alpha + \beta$.

Example, Cu-Zn at 560 °C (74 % Zn; here and in the next examples the slides print the compositions as % of the second element, with no basis):

$$\delta\,(74\%\,\mathrm{Zn}) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} \gamma\,(70\%\,\mathrm{Zn}) + \varepsilon\,(78\%\,\mathrm{Zn})$$

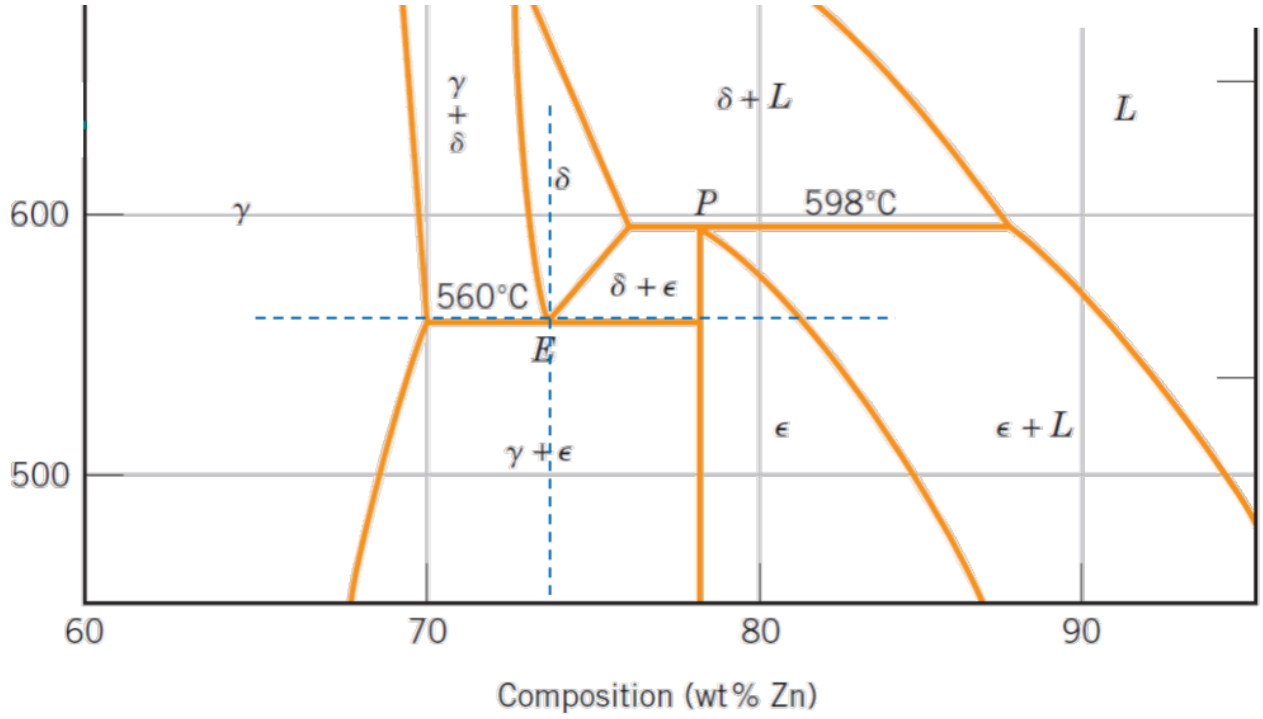


Figure 43. Detail of the Cu-Zn diagram around the eutectoid point E (560 °C, about 74 % Zn), where δ gives γ + ε, and the peritectic point P (598 °C). Copied from Session 8 Slide 7.

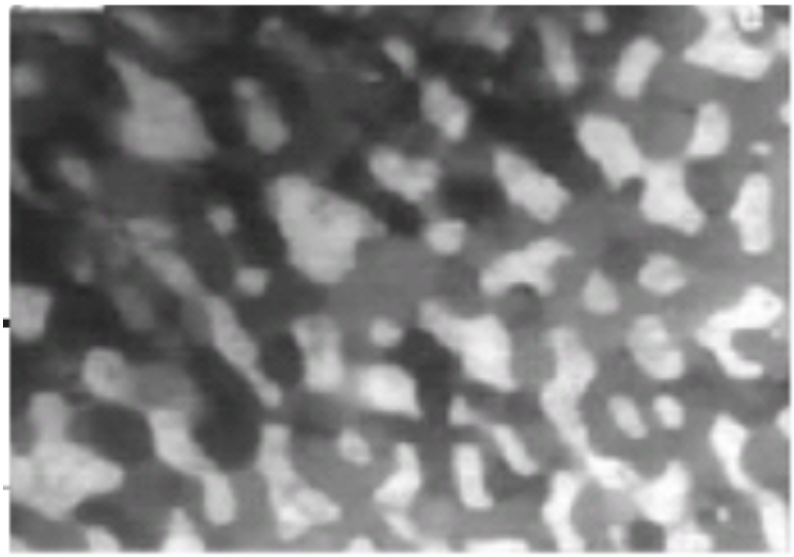


Figure 44. Micrograph of the eutectoid product of the Cu-Zn alloy, a fine mixture of γ and ε. From Session 8 Slide 7.

WORKED EXAMPLE 9 — THE CU-ZN EUTECTOID PRODUCT [SESSION 8 SLIDE 7]

Question. A Cu-74 wt% Zn alloy (all δ) is cooled just below 560 °C. How much γ and how much ε does it contain?

Lever rule on the tie line from γ (70 % Zn) to ε (78 % Zn), length $78 - 70 = 8$:

$$W_\gamma = \frac{C_\varepsilon - C_0}{C_\varepsilon - C_\gamma} = \frac{78 - 74}{78 - 70} = \frac{4}{8} = 0.50 \qquad W_\varepsilon = \frac{C_0 - C_\gamma}{C_\varepsilon - C_\gamma} = \frac{74 - 70}{8} = 0.50$$

Check: $0.50 \times 70 + 0.50 \times 78 = 74$ % Zn. **Answer:** the eutectoid product is half γ and half ε, a fine mixture of the two solids, as in the micrograph. On heating the reaction runs backwards and the mixture turns into single-phase δ.

10.3 Peritectic reaction [Session 8 Slides 8-9]

Peritectic reaction: an invariant reaction where **a solid and a liquid phase transform into a different solid phase**, $\alpha + L \rightarrow \beta$. The **peritectic point** is the invariant point where this happens [Session 8 Slide 8].

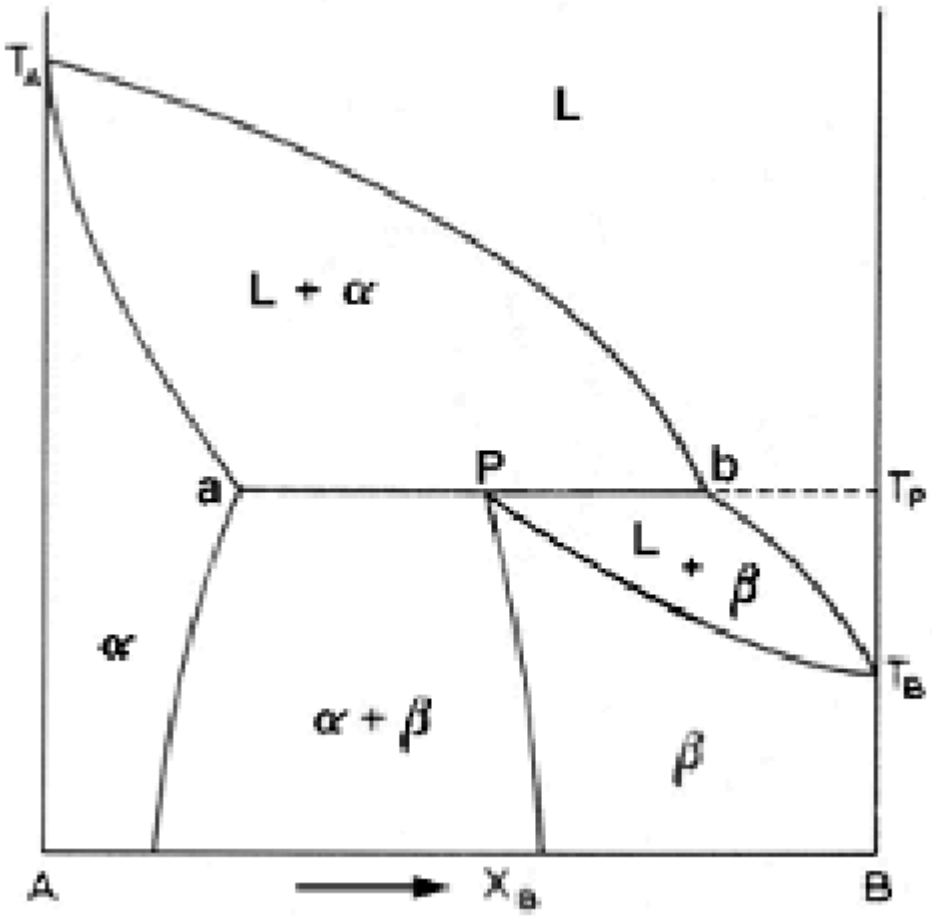


Figure 45. Generic peritectic diagram: the reaction α + L → β at the peritectic point P, temperature T_P, between the points a and b. Copied from Session 8 Slide 8.

Example, Fe-Ni at 1517 °C and 4.3 % Ni:

$$\delta\,(4.0\%\,\mathrm{Ni}) + L\,(5.4\%\,\mathrm{Ni}) \xrightarrow[\text{cooling}]{1517^\circ\mathrm{C}} \gamma\,(4.3\%\,\mathrm{Ni})$$

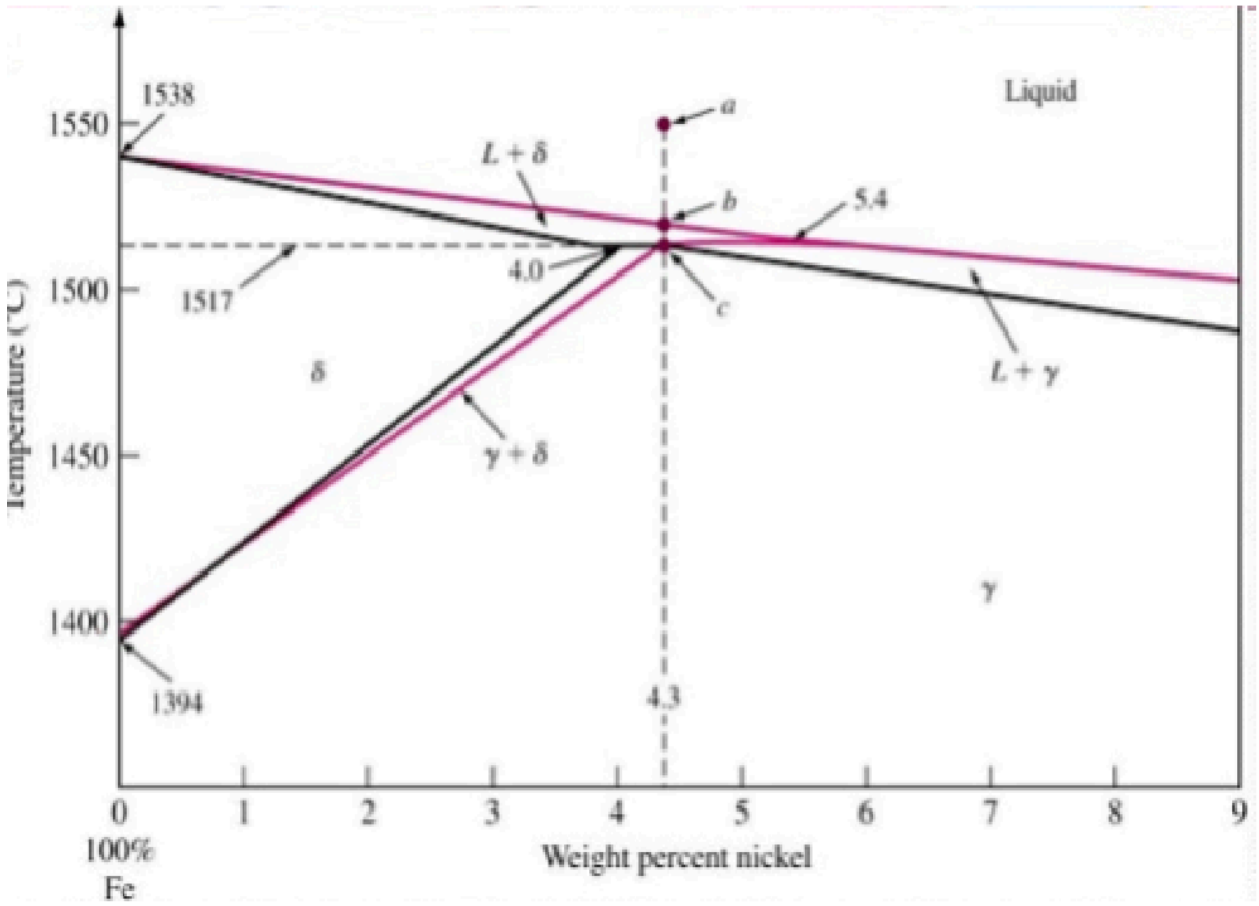


Figure 46. Fe-Ni diagram near the peritectic at 1517 °C: δ (4.0 % Ni) + L (5.4 % Ni) → γ (4.3 % Ni), with the points a, b and c of the 4.3 % Ni alloy. Copied from Session 8 Slide 8.

WORKED EXAMPLE 10 — FE-4.3 WT% NI AROUND 1517 °C [SESSION 8 SLIDE 8]

Just above 1517 °C the alloy is δ (4.0 % Ni) + L (5.4 % Ni); tie line length $5.4 - 4.0 = 1.4$:

$$W_\delta = \frac{5.4 - 4.3}{5.4 - 4.0} = \frac{1.1}{1.4} = 0.786 \qquad W_L = \frac{4.3 - 4.0}{1.4} = \frac{0.3}{1.4} = 0.214$$

Just below 1517 °C both react and give γ of 4.3 % Ni, the composition of the alloy, so the alloy is 100 % γ.

Check: $0.786 \times 4.0 + 0.214 \times 5.4 = 3.14 + 1.16 = 4.30$ % Ni. The liquid and the δ are used up together because the alloy has exactly the peritectic composition.

Second example, Pt-Ag at 1186 °C (42.4 % Ag) [Session 8 Slide 9]:

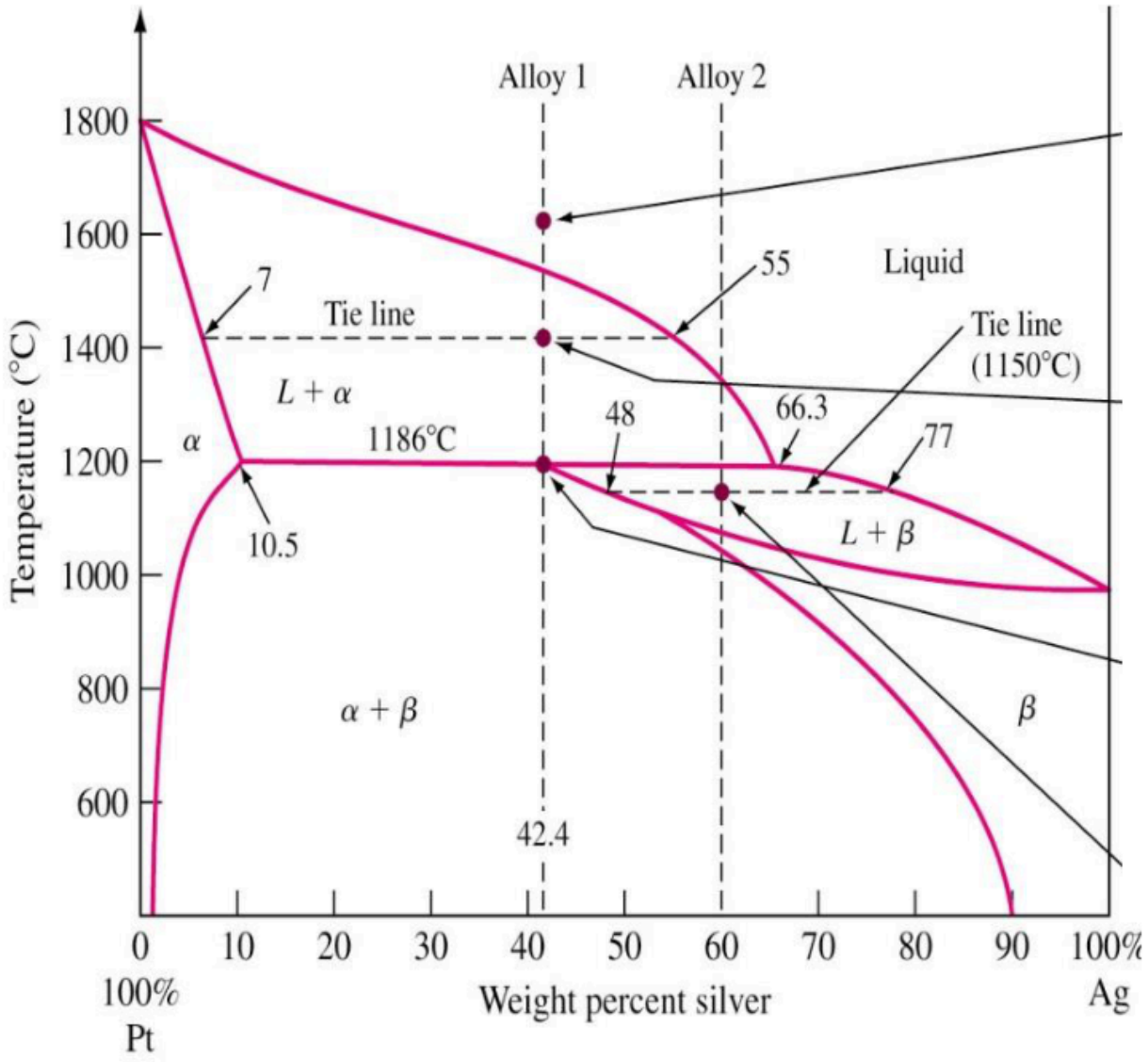


Figure 47. Pt-Ag diagram with alloy 1 (42.4 % Ag) and alloy 2 (about 60 % Ag), the tie lines at 1400 °C and 1150 °C and the peritectic line at 1186 °C; the arrows lead to the microstructure sketches of the slide. Copied from Session 8 Slide 9.

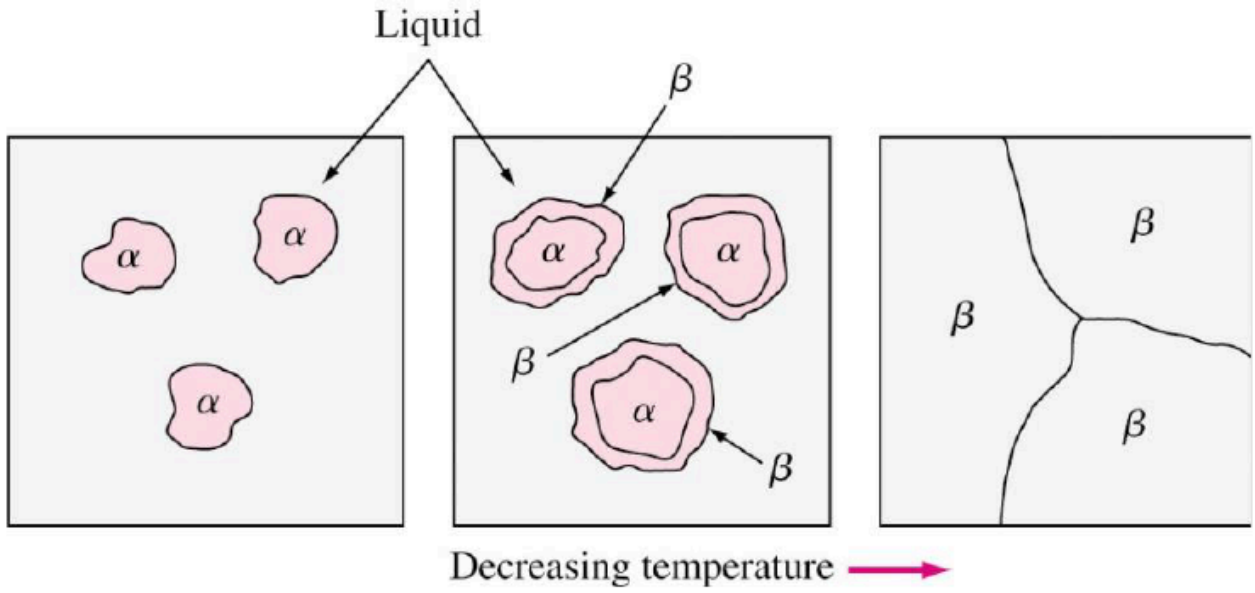


Figure 48. Peritectic reaction with decreasing temperature: α crystals in liquid; β forms around them; all β. From Session 8 Slide 9.

WORKED EXAMPLE 11 — THE PT-AG PERITECTIC, ALLOYS 1 AND 2 [SESSION 8 SLIDE 9]

Alloy 1, 42.4 % Ag, at 1400 °C. Liquid (55 % Ag) + α (7 % Ag); tie line length $55 - 7 = 48$:

$$W_L = \frac{42.4 - 7}{55 - 7} = \frac{35.4}{48} = 0.7375 \qquad W_\alpha = \frac{55 - 42.4}{48} = \frac{12.6}{48} = 0.2625$$

Alloy 1 at 1186 °C + ΔT. Liquid (66.3 % Ag) + α (10.5 % Ag); length $66.3 - 10.5 = 55.8$:

$$W_L = \frac{42.4 - 10.5}{66.3 - 10.5} = \frac{31.9}{55.8} = 0.572 \qquad W_\alpha = \frac{66.3 - 42.4}{55.8} = \frac{23.9}{55.8} = 0.428$$

Alloy 1 at 1186 °C − ΔT. Only β, of 42.4 % Ag: 100 %. The liquid and the α have reacted completely ($\alpha + L \to \beta$).

Alloy 2 (about 60 % Ag, a reading of the figure) **at 1150 °C**: L + β, with the tie line of the slide from β (48 % Ag) to L (77 % Ag), length 29:

$$W_\beta = \frac{77 - 60}{77 - 48} = \frac{17}{29} = 0.586 \qquad W_L = \frac{60 - 48}{29} = \frac{12}{29} = 0.414$$

Alloy 2 has more Ag than the peritectic composition, so after the peritectic reaction some liquid is left (L + β); it solidifies completely on further cooling.

FORMULA ON SLIDE 9

For 1186 °C + ΔT the slide writes "Alpha: (66.3 − 10.5)/(66.3 − 10.5) = 43 %". The numerator must be $66.3 - 42.4 = 23.9$ (the arm on the side of the liquid); $(66.3 - 10.5)/(66.3 - 10.5)$ would be 1. The value 43 % is right.

Rapid (non-equilibrium) solidification in peritectic systems [Session 8 Slide 10]:

- **Surrounding or encasement:** during the peritectic reaction $L + \alpha \rightarrow \beta$, the β formed surrounds the primary α.
- β then acts as a **diffusion barrier** (B atoms from the liquid and A atoms from α must cross it), which results in **coring**: the reaction does not finish and the structure is not the equilibrium one.

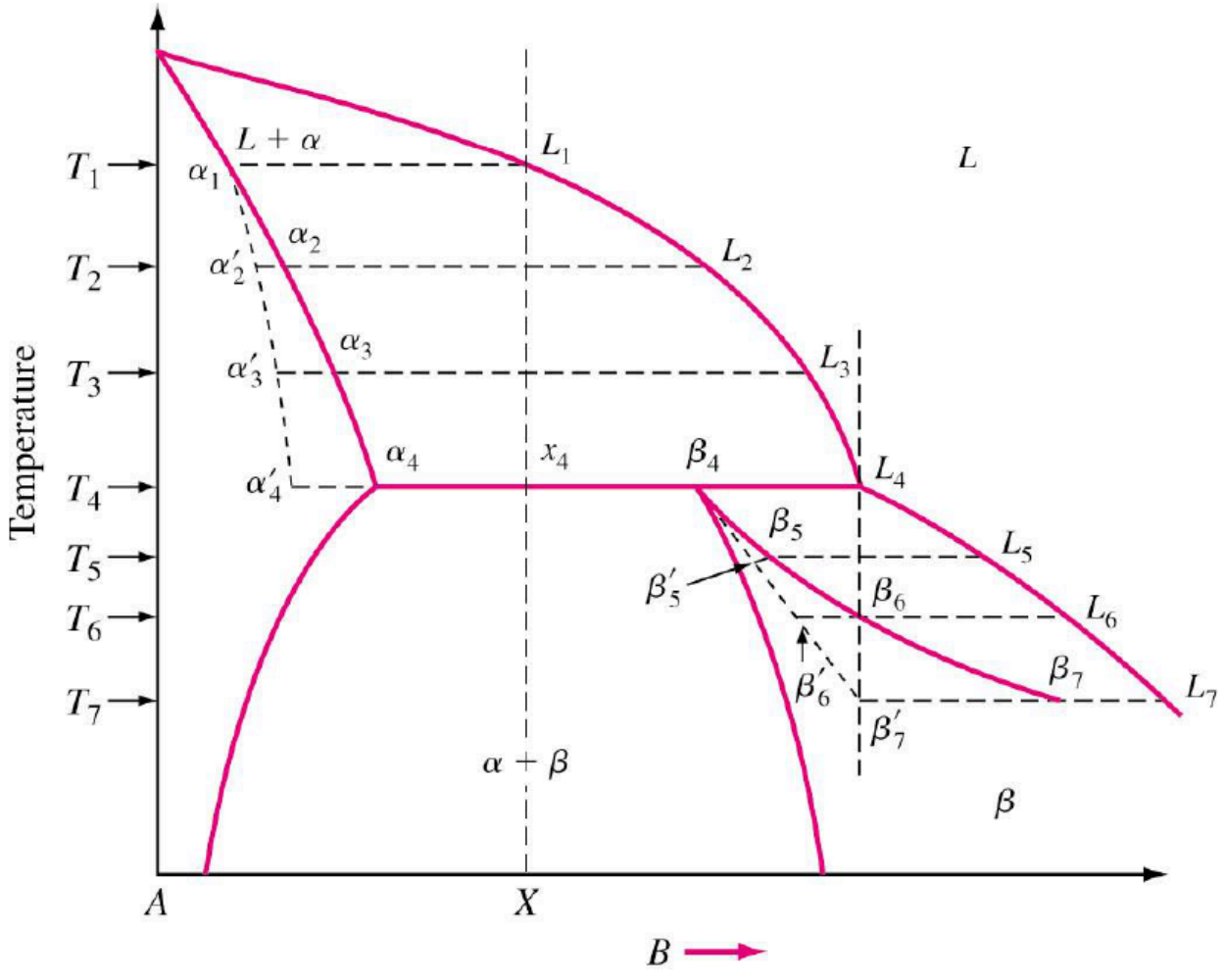


Figure 49. Rapid (non-equilibrium) solidification of alloy X: the solid follows non-equilibrium compositions (α′, β′) and the peritectic reaction is incomplete. From Session 8 Slide 10.

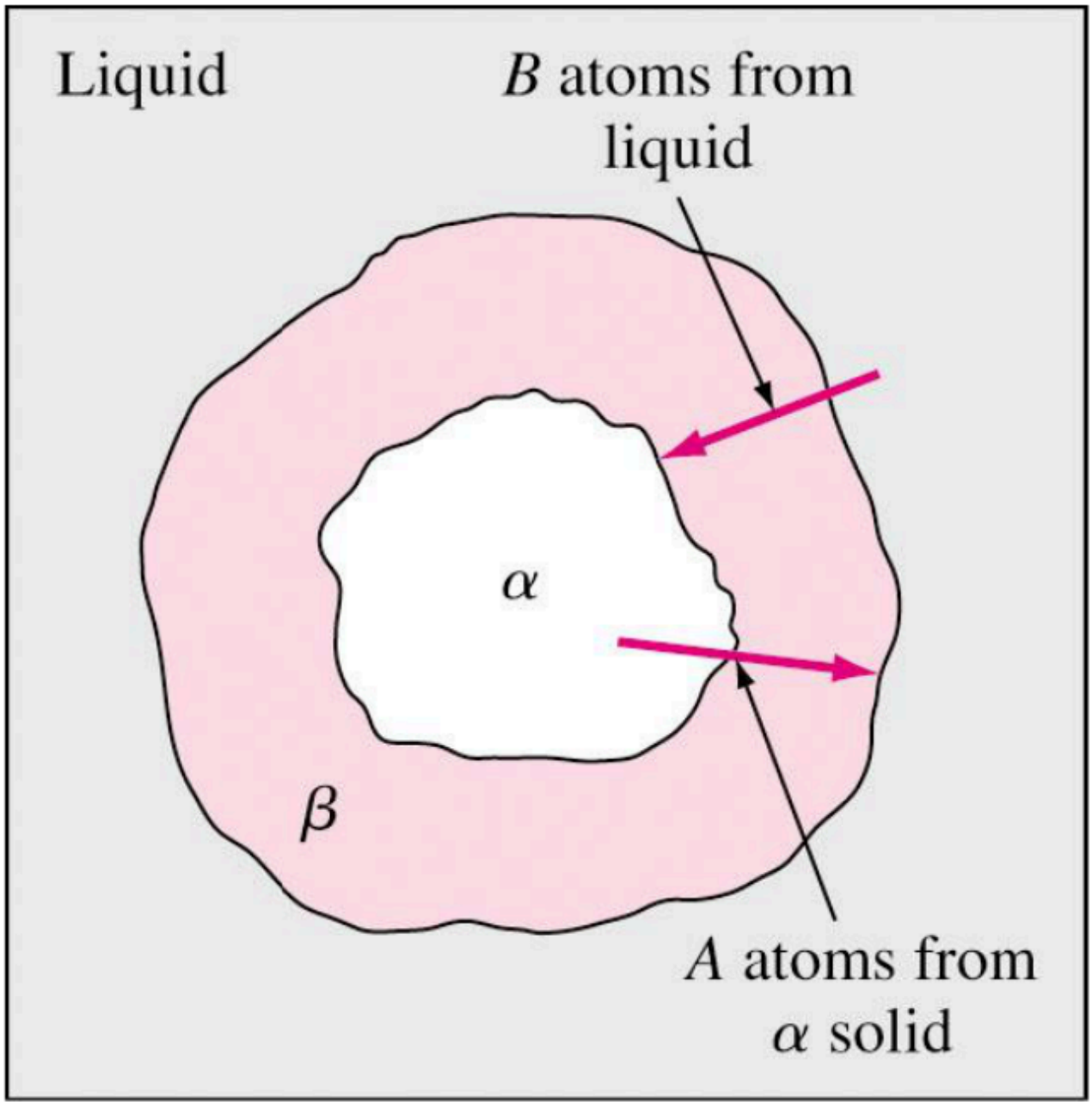


Figure 50. Encasement: the β formed by the peritectic reaction surrounds the primary α, so B atoms from the liquid and A atoms from α must diffuse through it. From Session 8 Slide 10.

10.4 Peritectoid reaction [Session 8 Slide 11]

Peritectoid reaction: an invariant reaction where **two solid phases transform into a new solid phase**, $\beta + \alpha \rightarrow \gamma$. The **peritectoid point** is the invariant point where it happens.

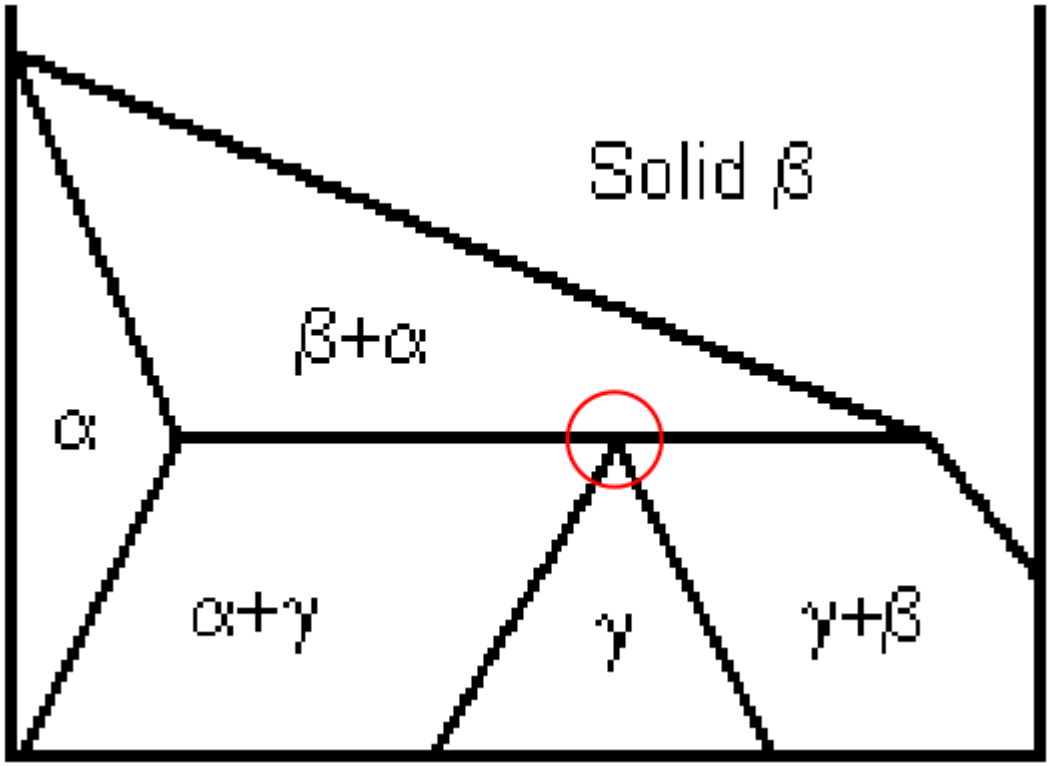


Figure 51. Generic peritectoid diagram: the solids α and β react to give γ at the circled peritectoid point. Copied from Session 8 Slide 11.

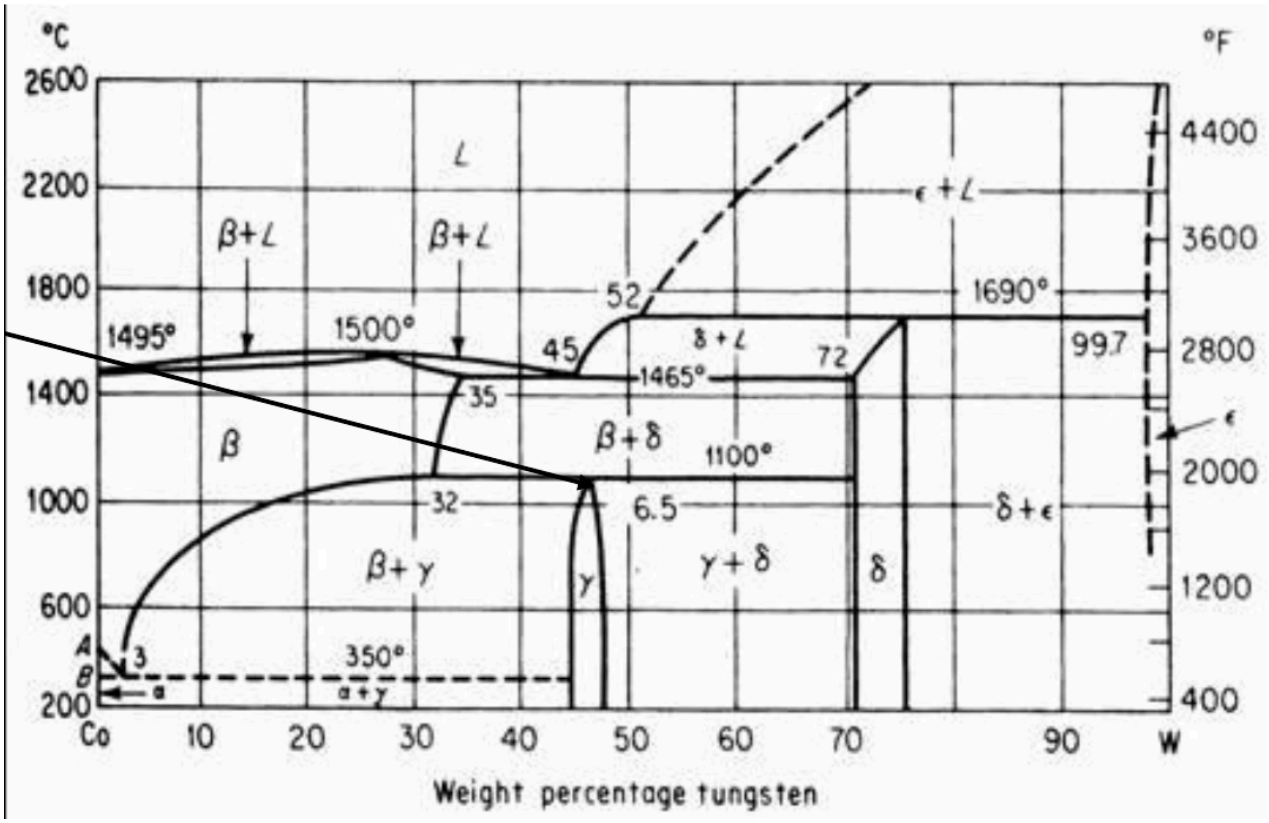


Figure 52. Co-W diagram: the arrow marks a peritectoid point, where two solid phases give the new solid γ. From Session 8 Slide 11.

10.5 Monotectic reaction [Session 8 Slide 12]

Monotectic reaction: an invariant reaction where **a liquid phase transforms into a solid phase plus another liquid** (the liquid has a **miscibility gap**): $L_1 \rightarrow L_2 + \alpha$. The **monotectic point** is the invariant point where this happens.

Example, Cu-Pb at 955 °C (36 % Pb):

$$L\,(36\%\,\mathrm{Pb}) \underset{\text{heating}}{\overset{\text{cooling}}{\rightleftharpoons}} L_2\,(87\%\,\mathrm{Pb}) + \alpha\,(0\%\,\mathrm{Pb})$$

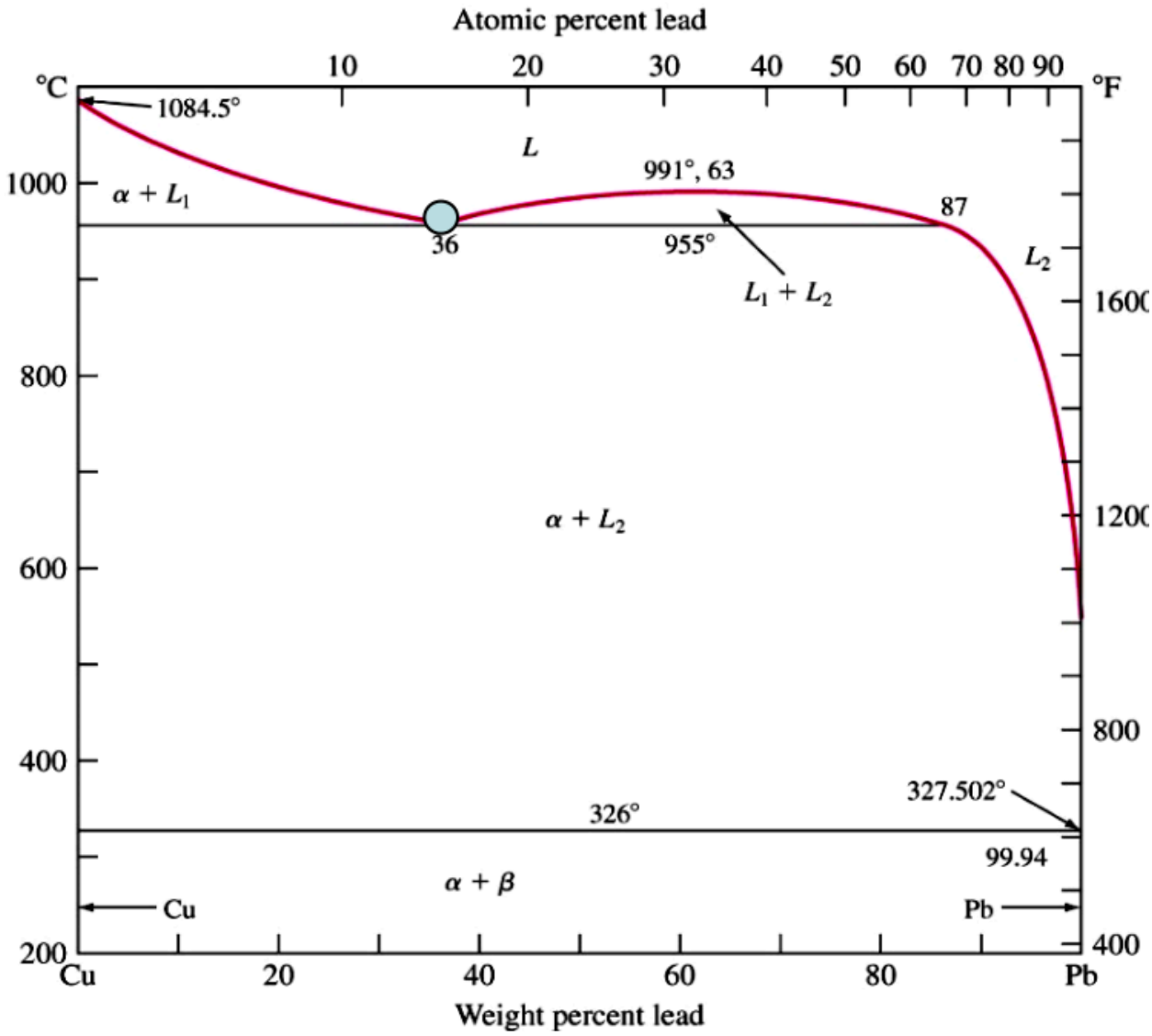


Figure 53. Cu-Pb diagram with the monotectic at 955 °C (36 % Pb, the blue dot), the miscibility gap of the liquid up to 991 °C and the line at 326 °C. Copied from Session 8 Slide 12.

WORKED EXAMPLE 12 — CU-36 WT% PB JUST BELOW 955 °C [SESSION 8 SLIDE 12]

Tie line from α (0 % Pb) to L_2 (87 % Pb), length 87:

$$W_\alpha = \frac{87-36}{87-0} = \frac{51}{87} = 0.586 \qquad W_{L_2} = \frac{36-0}{87} = \frac{36}{87} = 0.414$$

Interpretation. Just below 955 °C the alloy is 59 % solid copper and 41 % of a second liquid, rich in lead (87 %). That liquid stays liquid down to 326 °C, where L_2 undergoes a eutectic reaction, $L_2 \rightarrow \alpha + \beta$ (a reading of the diagram, which shows the 326 °C line with α + β below it; the slide text does not name it); the slide's diagram shows the region α + L2 between the two lines.

§ 11 Intermetallic compounds [Session 8 Slides 14-15]

An **intermetallic compound** is made up of two or more elements (components) and produces a **new phase with its own composition, crystal structure and properties** [Session 8 Slides 14-15].

- **Terminal phases** occur at the ends of the phase diagram (α and ζ in the figure; the slide text writes ξ).
- **Intermediate phases** occur in a composition range inside the diagram (β, γ, δ, ε).
- The percentage of ionic or covalent bonding in an intermetallic compound depends on the **electronegativity difference** between the elements (Topic 1).

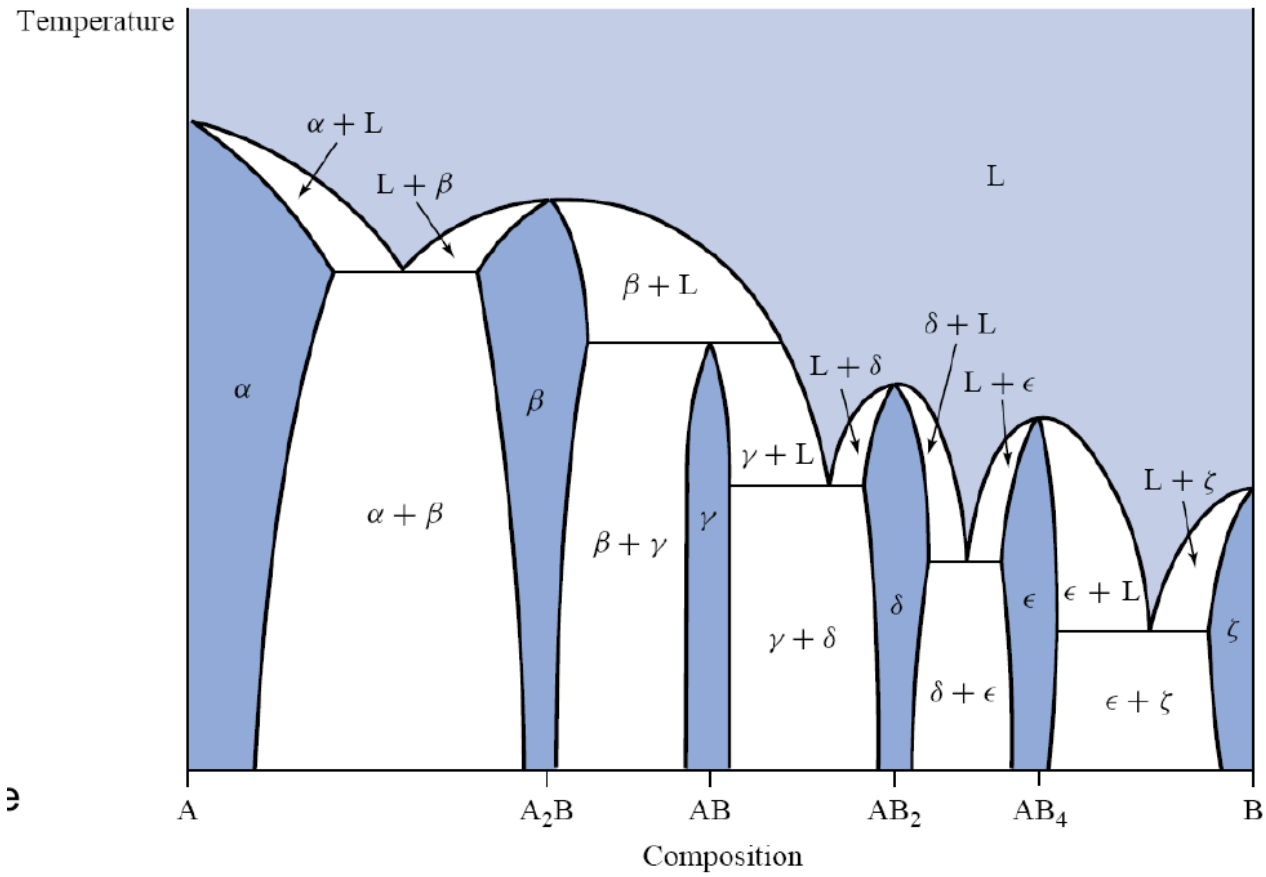


Figure 54. A diagram with terminal phases (α, ζ) at the ends and intermediate phases (β, γ, δ, ε) inside, around the compounds A_2B, AB, AB_2 and AB_4. From Session 8 Slide 14.

Two kinds of compound [Session 8 Slide 15]:

- **Stoichiometric:** fixed composition, a fixed ratio of the atoms involved; it is drawn as a **vertical line** in the phase diagram. Example in Ti-B: Ti_3B_4.
- **Non-stoichiometric:** a range of compositions, an intermediate solid solution. Examples in Ti-B: TiB and TiB_2.

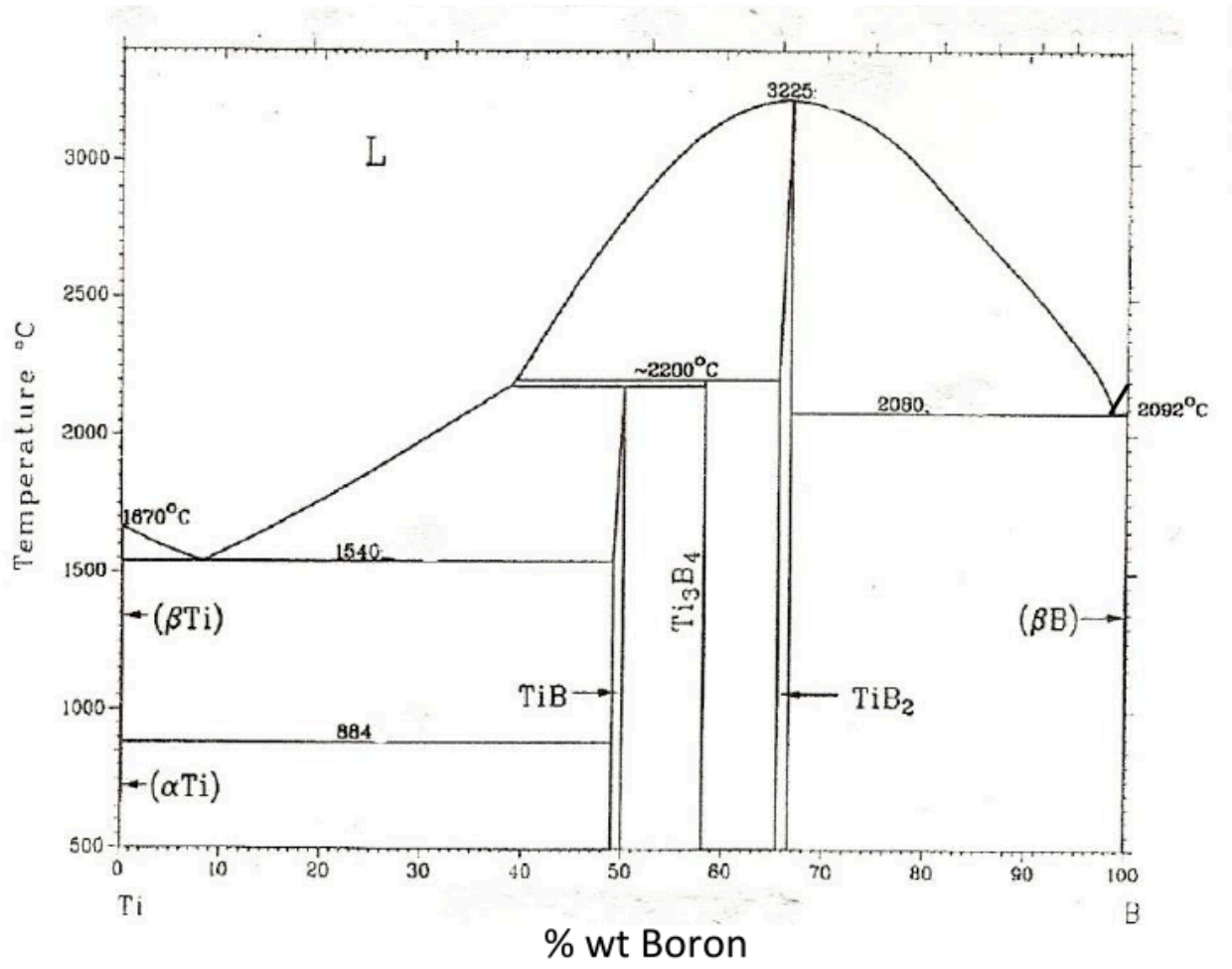


Figure 55. Ti-B: Ti_3B_4 is a stoichiometric compound (a vertical line); TiB and TiB_2 have a range of compositions. From Session 8 Slide 15.

COMPOSITION AXIS OF SLIDE 15

The label under the Ti-B diagram of Slide 15 reads "% wt Boron", but the compounds sit at about 50 % (TiB) and 67 % (TiB_2), which are their **atomic** percentages (one B per Ti, two B per Ti). The same diagram in the problem sheet (Problems T4_PhaseDiagrams I.pdf, p. 7) has the atomic scale on top and the weight scale below, where TiB is at about 18 wt% and TiB_2 at about 31 wt% B.

§ 12 Congruent and incongruent melting [Session 8 Slide 17]

- **Congruent melting:** an intermetallic compound melts congruently when the liquid formed on melting (solid → liquid) **keeps its composition**. Examples: $MgNi_2$ and the pure metals.
- **Incongruent melting:** the composition is **not kept** on melting, and a **different solid** appears: the compound gives a liquid plus another solid. Example: Mg_2Ni, which is formed on cooling by a peritectic reaction of the type $\alpha + L \rightarrow \beta$.

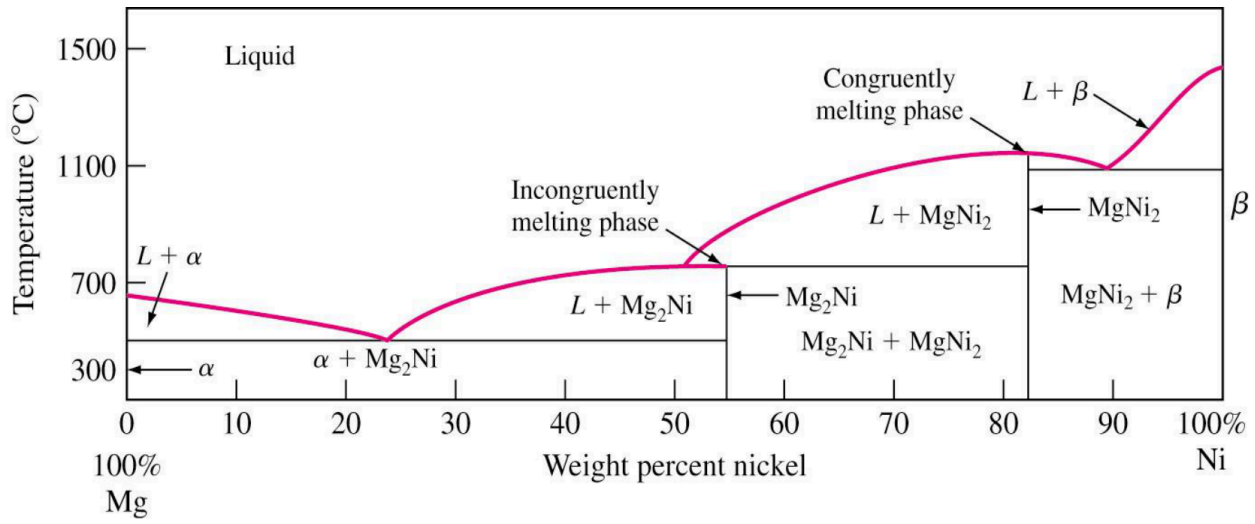


Figure 56. Mg-Ni diagram: MgNi2 melts congruently (at a maximum of the liquidus) and Mg2Ni incongruently (it ends at a horizontal peritectic line). Copied from Session 8 Slide 17.

On the diagram, a congruently melting compound sits at a **maximum of the liquidus** (it melts at a single temperature, like a pure metal), while an incongruently melting one ends at a **horizontal peritectic line**.

WORKED EXAMPLE 13 — CLASSIFYING THE TI-B COMPOUNDS [SESSION 8 SLIDES 15 , 17]

Reading the Ti-B diagram of Slide 15:

- TiB_2 reaches the top of the liquidus dome, at 3225 °C: the liquid formed there has the same composition as the solid, so TiB_2 **melts congruently**.
- Ti_3B_4 and TiB end at horizontal lines near 2200 °C, below the liquidus: on heating, each gives liquid plus a different solid (for Ti_3B_4, liquid + TiB_2). Both **melt incongruently**; on cooling they form by peritectic reactions.

This is the classification used in Problem 15 of the sheet.

Part III — Phase diagrams III: the Fe-C system (Session 9)

§ 13 The Fe-C system [Session 9 Slides 2-11]

13.1 Why it matters [Session 9 Slide 2]

- The **Fe-C system** is a binary alloy system of iron and carbon.
- It is the most important alloy system of our civilisation: it gives **steels** and **cast irons**.
- It is a **binary eutectic system**.
- Depending on the carbon content, temperature, cooling rate, pressure and so on, **different microstructures** can be developed, with different properties and applications.

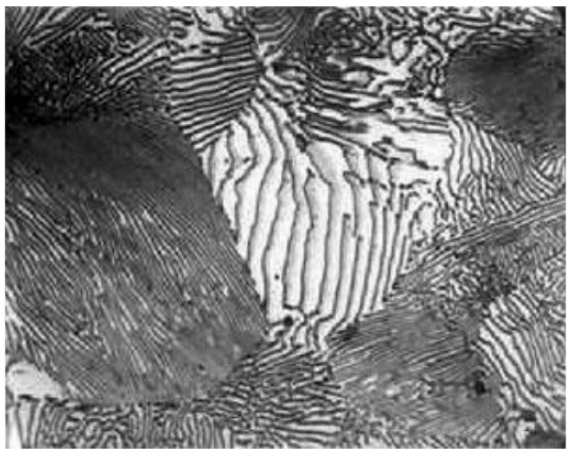


Figure 57. Different microstructures in the Fe-C system. From Session 9 Slide 2.

13.2 Allotropic phases of pure iron [Session 9 Slide 3]

Allotropy is the existence of a chemical element in two or more forms, which may differ in the arrangement of the atoms in the crystalline solid (Topic 2). Iron undergoes **two changes of crystal structure before it melts**:

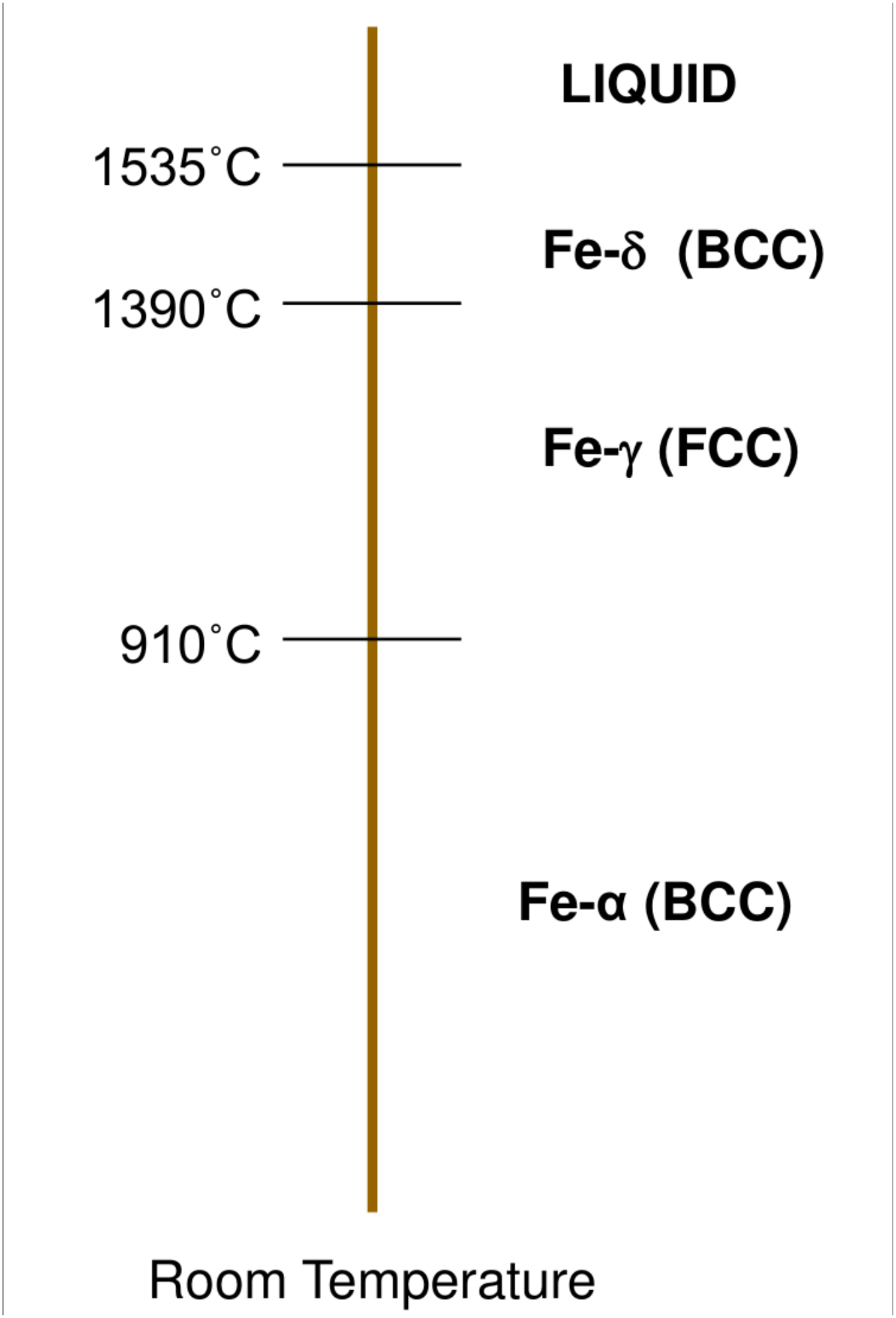

LIQUID
1535˚C
Fe-δ (BCC)
1390˚C
Fe-γ (FCC)
910˚C
Fe-α (BCC)
Room Temperature

Figure 58. Allotropic phases of pure iron from room temperature to the liquid: Fe-α (BCC) up to 910 °C, Fe-γ (FCC) up to 1390 °C and Fe-δ (BCC) up to 1535 °C. Copied from Session 9 Slide 3.

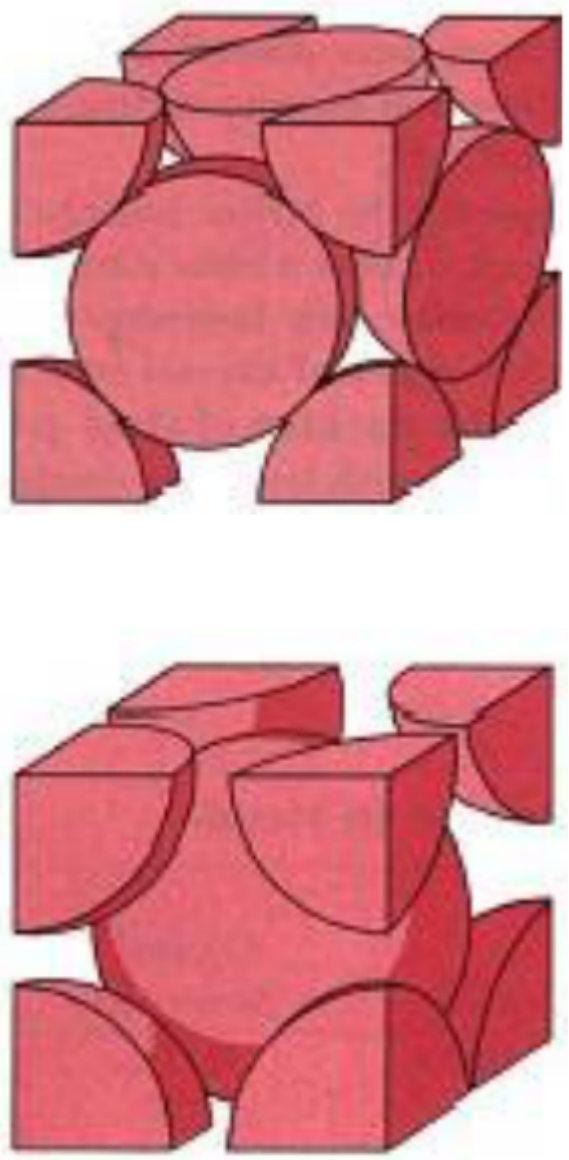


Figure 59. Unit cells of Fe-γ (FCC, top) and Fe-α (BCC, bottom). From Session 9 Slide 3.

The temperatures on the scale of Slide 3 (910, 1390 and 1535 °C) differ by a few degrees from those printed on the Fe-C diagram of the same session (912, 1394 and 1538 °C, Slide 4); the diagram values are used in the rest of this page.

13.3 The Fe-Fe_3C diagram [Session 9 Slide 4]

The diagram used in practice is **metastable**: its right-hand end is **6.67 wt% C**, the stoichiometric composition of **Fe_3C**, iron carbide or **cementite**. So the diagram is in fact **Fe-Fe_3C**. It has three invariant points (a peritectic, a eutectic and a eutectoid), and its composition axis is split into four ranges:

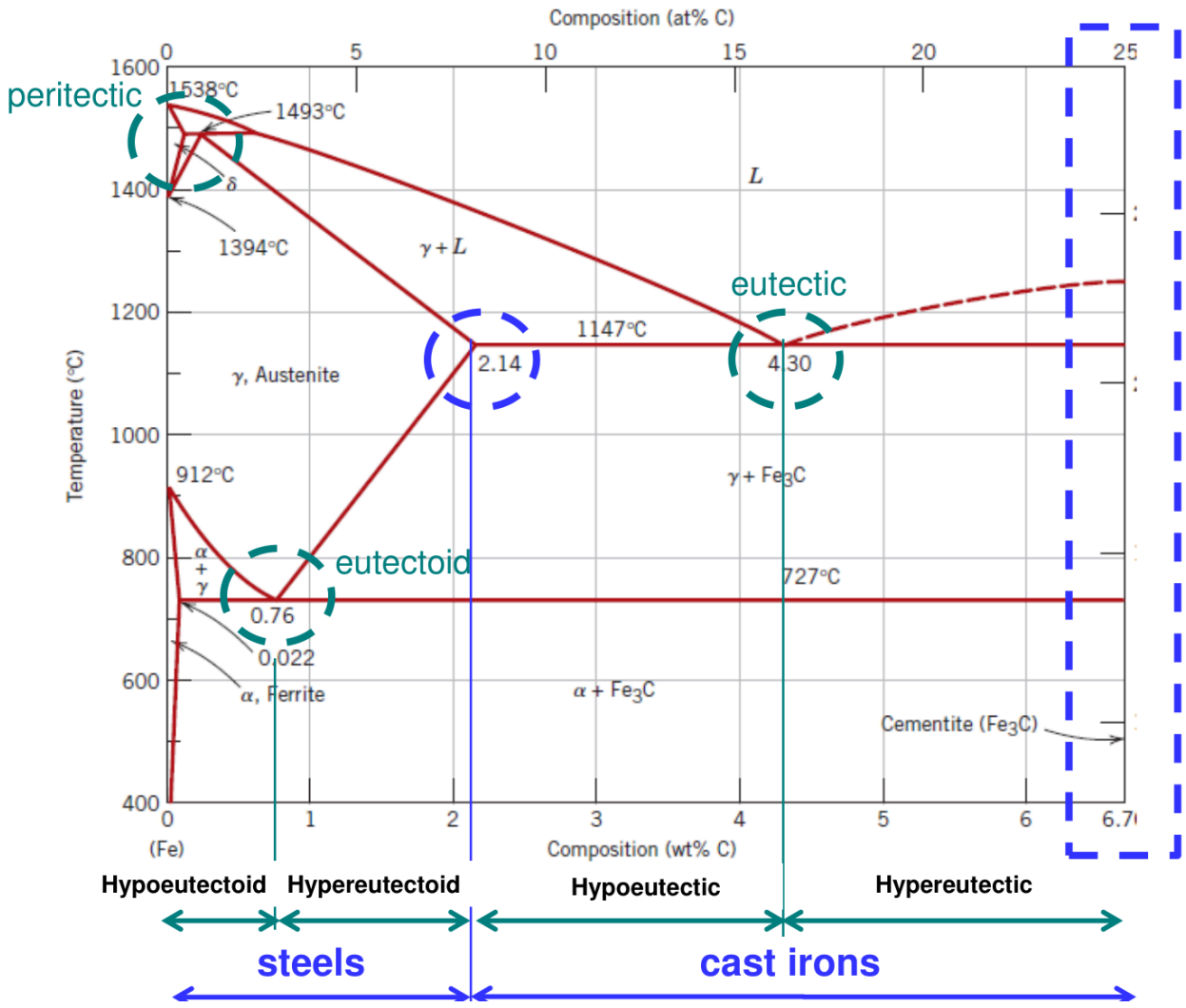


Figure 60. Fe-Fe3C diagram up to 6.67 wt% C with the peritectic, eutectic (4.30 % C, 1147 °C) and eutectoid (0.76 % C, 727 °C) points marked, the limit 2.14 % C, and the ranges of hypoeutectoid and hypereutectoid steels and of hypoeutectic and hypereutectic cast irons. Copied from Session 9 Slide 4.

Range	Name
up to 0.76 % C	hypoeutectoid steels
0.76 to 2.14 % C	hypereutectoid steels
2.14 to 4.30 % C	hypoeutectic cast irons
4.30 to 6.67 % C	hypereutectic cast irons

Steels have up to 2.14 % C; cast irons have more [Session 9 Slide 4].

13.4 Phases [Session 9 Slides 5-9]

- **α-ferrite (BCC):** interstitial solid solution of C in BCC iron. Low C content: maximum solubility **0.022 wt% at 727 °C**. A soft phase; density 7.88 g/cm³ [Slide 5].
- **Austenite, γ (FCC):** interstitial solid solution of C in FCC iron. Maximum solubility of C **2.14 wt% at 1147 °C**. Important for the phase transitions in the heat treatment of steels. Not magnetic [Slide 6].
- **δ-ferrite (BCC):** interstitial solid solution of C in BCC iron, with the same properties as α-ferrite but at higher temperatures [Slide 7].
- **Cementite, Fe_3C:** an intermetallic **stoichiometric compound**, 6.67 % C. It forms when the solubility limit of C in α-Fe is exceeded below 727 °C. **Hard and brittle**. A **metastable** compound [Slide 8].

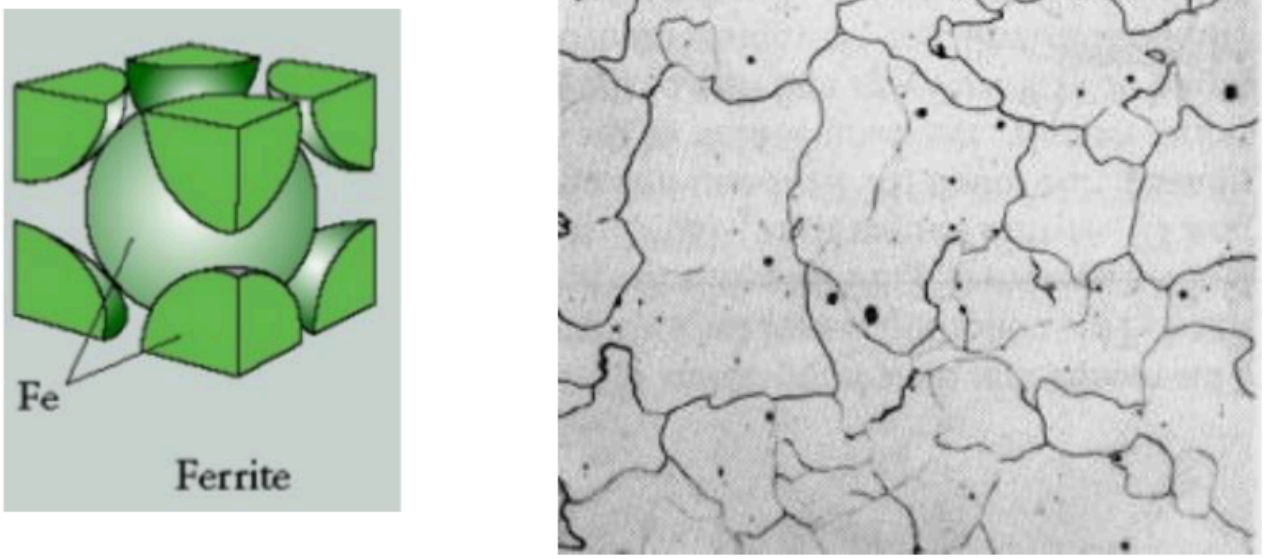


Figure 61. α-ferrite: BCC cell and micrograph of its grains. From Session 9 Slide 5.

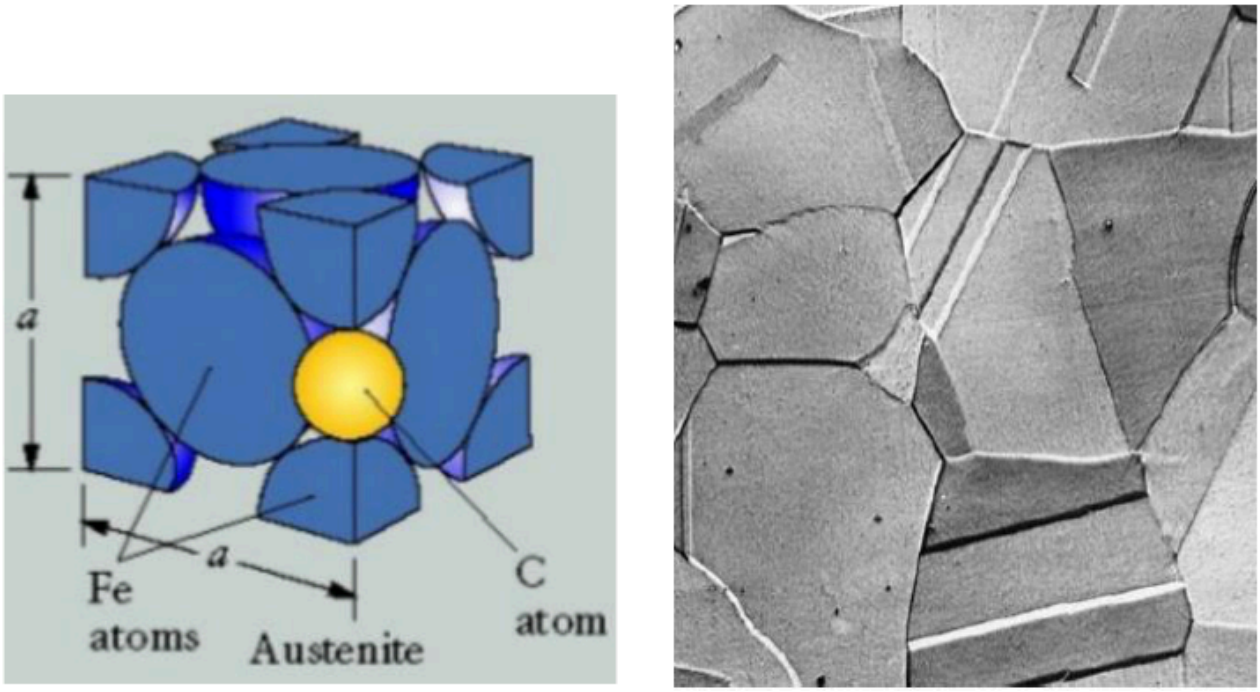


Figure 62. Austenite: FCC cell with a C atom in an interstitial site, and micrograph. From Session 9 Slide 6.

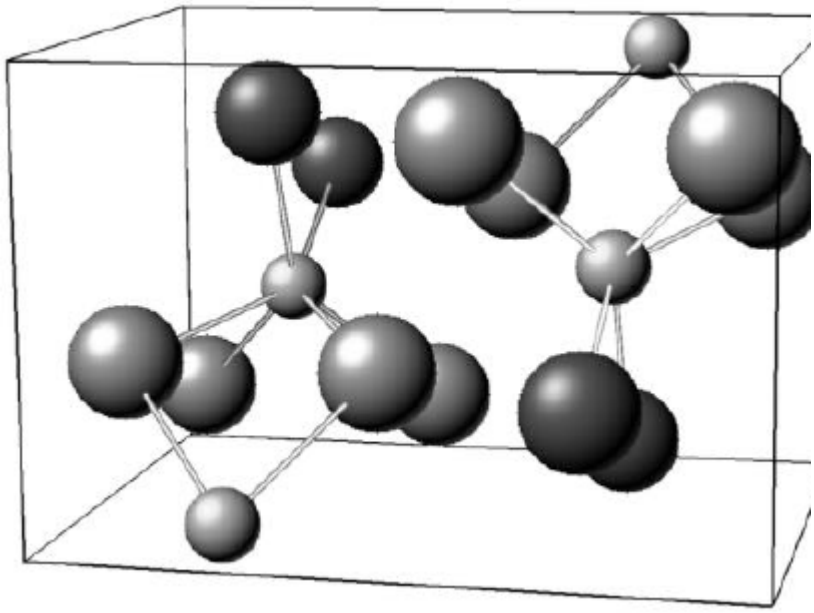


Figure 63. Unit cell of cementite, Fe_3C. From Session 9 Slide 8.

Austenite dissolves much more carbon than ferrite (2.14 against 0.022 wt%), because the interstitial sites of FCC iron are larger than those of BCC iron (extension: the Session 9 slides state the solubilities but not this reason).

Two-phase regions [Slide 9]: α-ferrite + Fe3C below 727 °C; γ-austenite + Fe3C between 727 and 1147 °C.

13.5 Invariant reactions [Session 9 Slide 10]

Reaction	Equation (wt% C)	Product
Eutectoid, 727 °C	$\gamma\,(0.76) \rightleftharpoons \alpha\,(0.022) + \mathrm{Fe_3C}\,(6.67)$	pearlite
Eutectic, 1147 °C	$L\,(4.3) \rightleftharpoons \gamma\,(2.14) + \mathrm{Fe_3C}\,(6.67)$	ledeburite
Peritectic, 1493 °C	$\delta\,(0.09) + L\,(0.53) \rightleftharpoons \gamma\,(0.16)$	austenite

WORKED EXAMPLE 14 — THE MAKE-UP OF LEDEBURITE [SESSION 9 SLIDE 10]

Question. How much austenite and how much cementite does the eutectic product of Fe-C (ledeburite) contain just below 1147 °C?

Tie line from γ (2.14 % C) to Fe3C (6.67 % C), length $6.67 - 2.14 = 4.53$; the liquid had $C_0 = 4.30$ % C:

$$W_{\mathrm{Fe_3C}} = \frac{C_0 - C_\gamma}{C_{\mathrm{Fe_3C}} - C_\gamma} = \frac{4.30 - 2.14}{6.67 - 2.14} = \frac{2.16}{4.53} = 0.477 \qquad W_\gamma = \frac{6.67 - 4.30}{4.53} = \frac{2.37}{4.53} = 0.523$$

Check: $0.523 \times 2.14 + 0.477 \times 6.67 = 1.12 + 3.18 = 4.30$ % C. **Answer:** ledeburite is about 52 % austenite and 48 % cementite. Below 727 °C its austenite becomes pearlite ("transformed ledeburite", Slide 11).

13.6 Microconstituents [Session 9 Slide 11]

The microconstituents of the system are **pearlite**, proeutectoid **ferrite** and proeutectoid **cementite** in steels, and **ledeburite** (transformed below 727 °C) with cementite and pearlite in cast irons. The lines of the diagram have names used in heat treatment: A_1 is the eutectoid line (727 °C), A_3 the boundary between γ and α + γ, and A_{cm} the boundary between γ and γ + Fe3C.

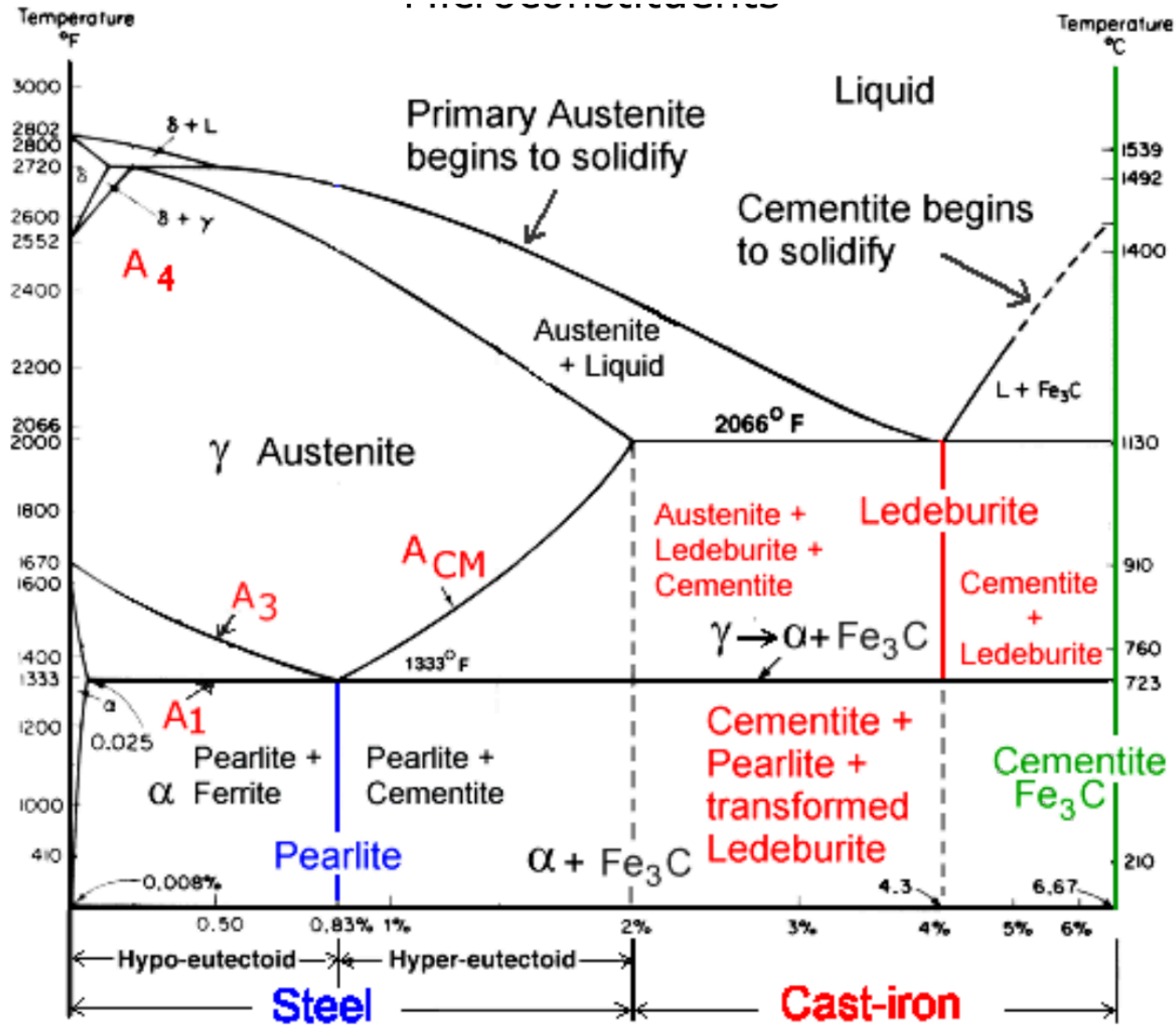


Figure 64. Microconstituents of the Fe-C system: pearlite, ferrite and cementite in steels; ledeburite in cast irons. The figure uses older values (0.83 % C, 723 and 1130 °C). From Session 9 Slide 11.

§ 14 Equilibrium transformations of steels [Session 9 Slides 12-20]

14.1 Eutectoid steels, C_0 = 0.76 % C [Session 9 Slides 12-13]

Pearlitic transformation: austenite of 0.76 % C gives a single microconstituent, **pearlite**, formed by layers (lamellae) of two phases, α + Fe3C [Slide 12].

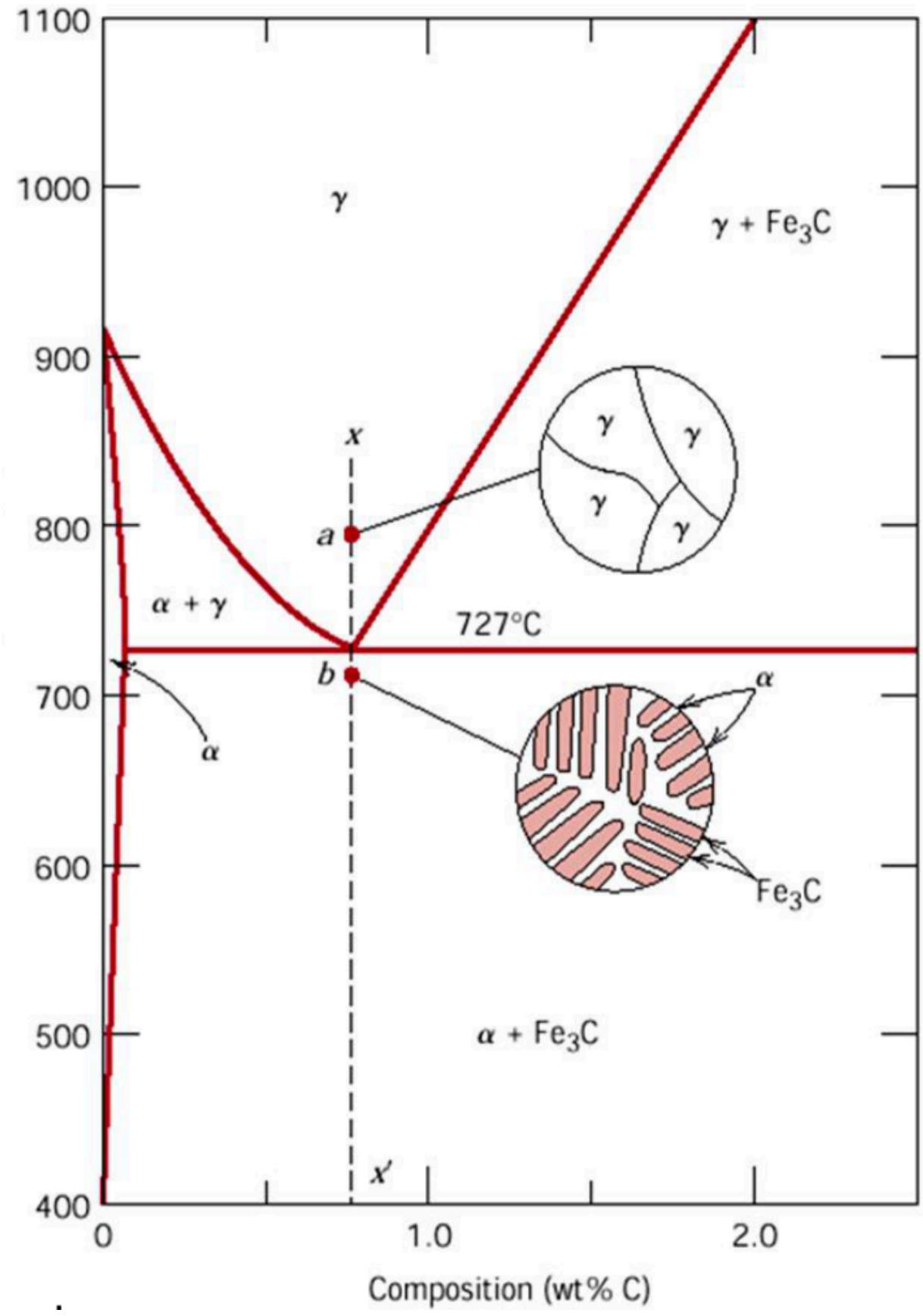


Figure 65. Eutectoid steel: at a (above 727 °C) only austenite; at b (just below 727 °C) the austenite has become pearlite, layers of α and Fe3C. Copied from Session 9 Slide 12.

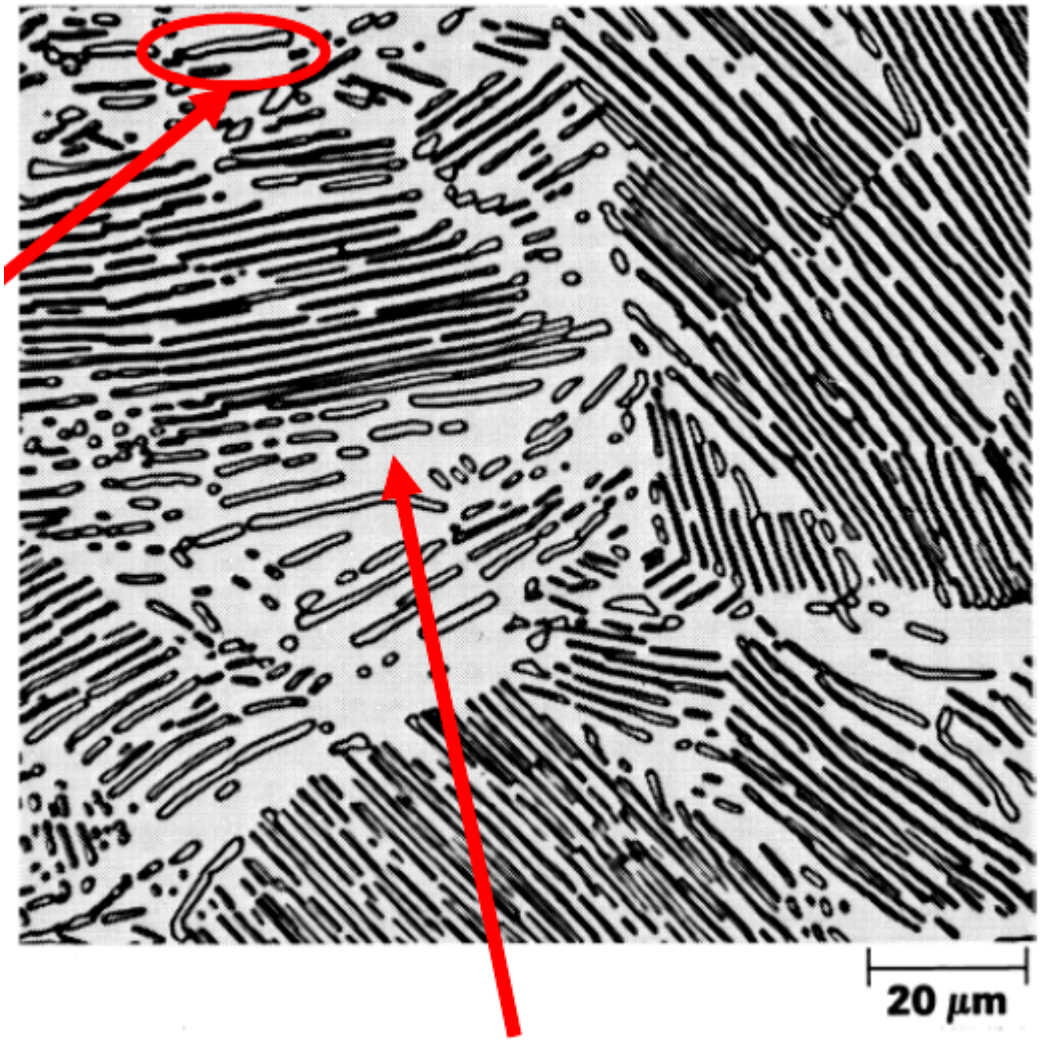


Figure 66. Pearlite: alternating layers of ferrite (light) and cementite (dark). From Session 9 Slide 12.

Ferrite nucleates on the grain boundaries of the austenite and displaces the carbon to the neighbouring zones by diffusion; there cementite forms. **Decreasing the spacing of the pearlite lamellae increases its hardness** [Slide 12].

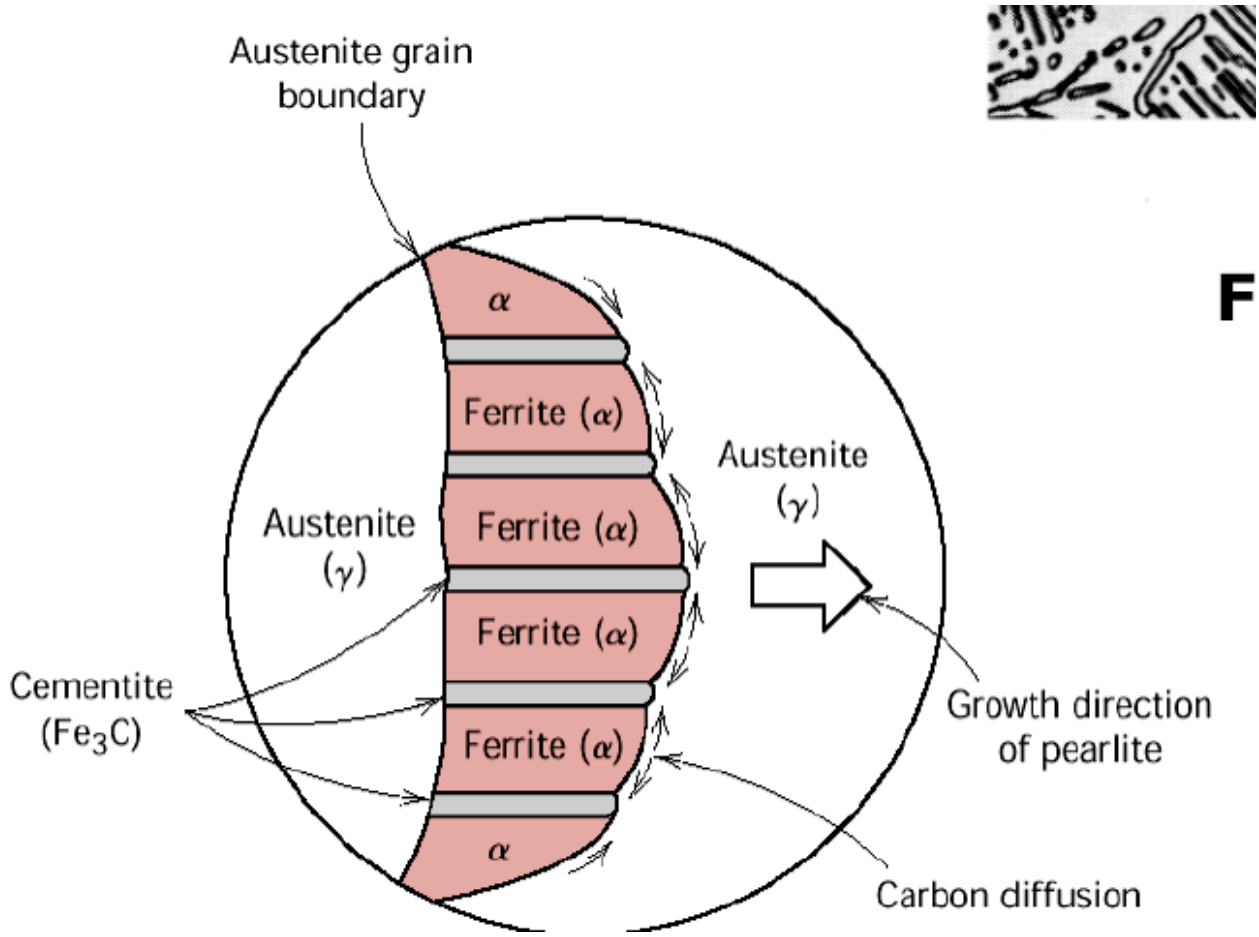


Figure 67. Growth of pearlite from an austenite grain boundary: ferrite rejects carbon, which diffuses to the neighbouring zones where cementite forms. From Session 9 Slide 12.

WORKED EXAMPLE 15 — PHASES OF A EUTECTOID STEEL AT POINT B [SESSION 9 SLIDE 13]

Point b (just below 727 °C): one microconstituent, pearlite; two phases, ferrite (C_α = 0.022 % C) and cementite (C_{Fe_3C} = 6.67 % C). Tie line length $6.67 - 0.022 = 6.648$:

$$W_\alpha = \frac{C_{Fe_3C} - C_0}{C_{Fe_3C} - C_\alpha} = \frac{6.67 - 0.76}{6.67 - 0.022} = \frac{5.91}{6.648} = 0.889 \qquad W_{Fe_3C} = 1 - 0.889 = 0.111$$

Check: $0.889 \times 0.022 + 0.111 \times 6.67 = 0.020 + 0.740 = 0.76$ % C. **Answer:** pearlite is 89 % ferrite and 11 % cementite, as on the slide. Every pearlite, in any steel, has this make-up.

14.2 Hypoeutectoid steels, C_0 < 0.76 % C [Session 9 Slides 14-18]

Two microconstituents: **pearlite** and **proeutectoid (primary) ferrite**, α_P [Slide 14].

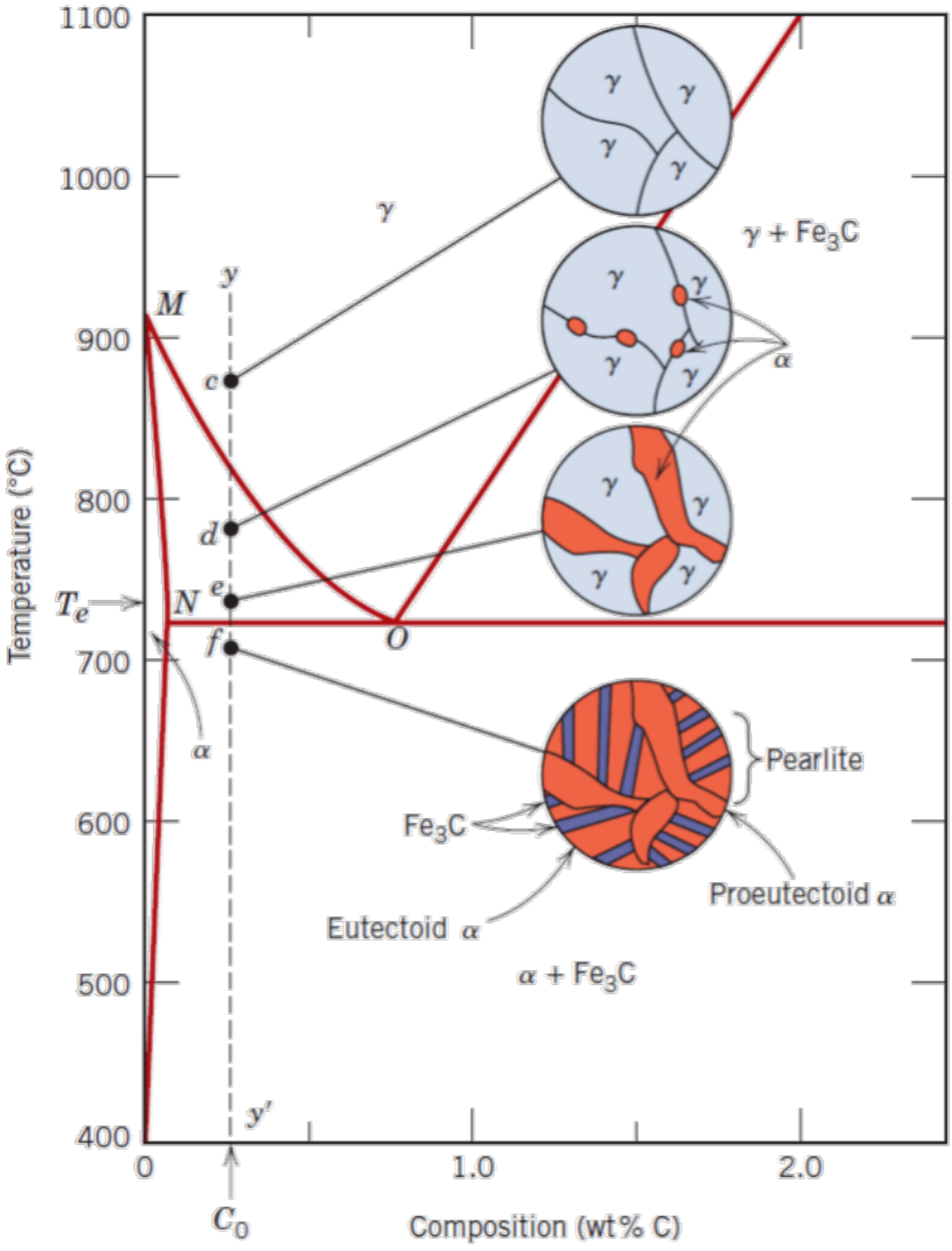


Figure 68. Hypoeutectoid steel C0: c only γ; d and e, proeutectoid α growing at the γ grain boundaries between the A3 line M-O and 727 °C; f, proeutectoid α and pearlite. Copied from Session 9 Slide 14.

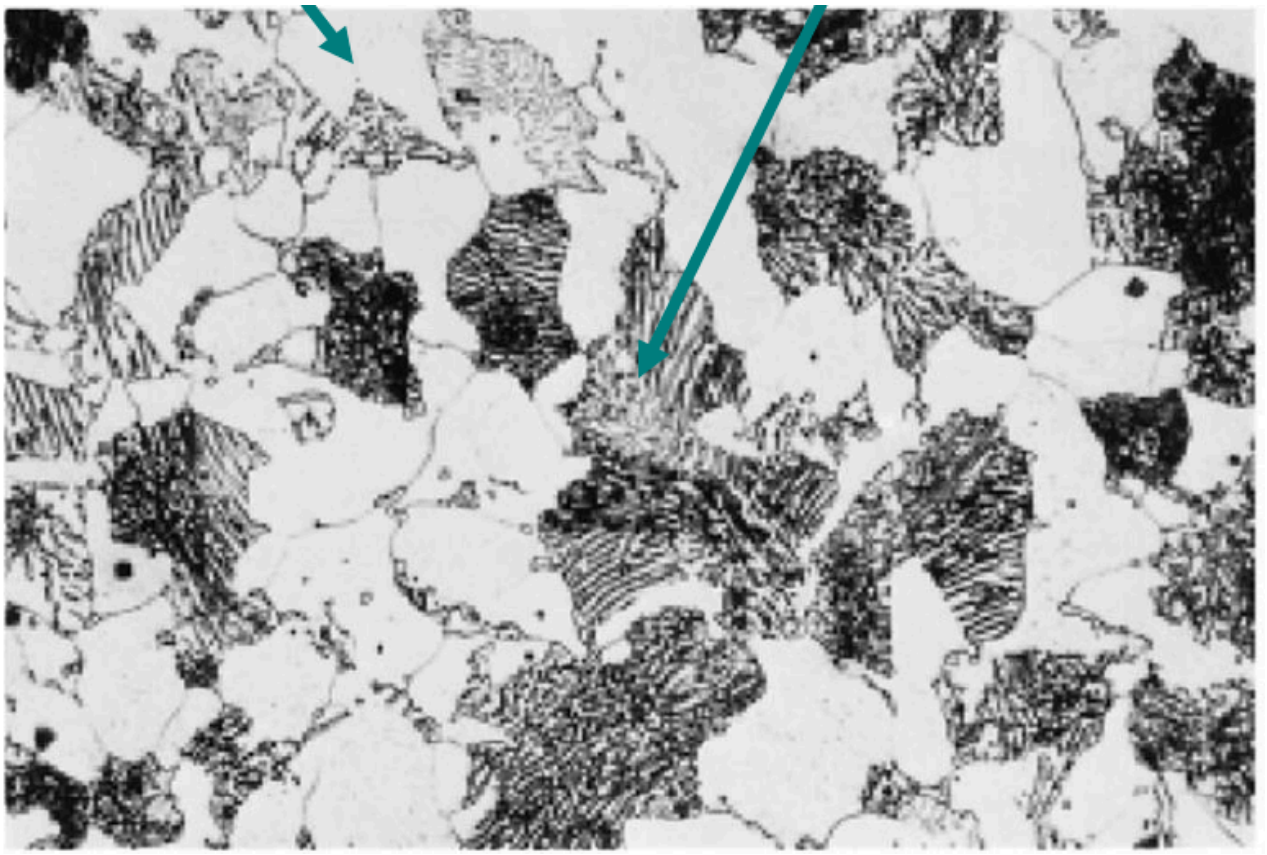


Figure 69. Hypoeutectoid steel: light proeutectoid (primary) ferrite and dark pearlite. From Session 9 Slide 14.

On cooling a steel of 0.3 % C: at c it is only γ; between the A_3 line (M-O) and 727 °C ferrite nucleates at the austenite grain boundaries and grows (d, e) while the austenite becomes richer in carbon along A_3; at 727 °C the remaining austenite, now of 0.76 % C, becomes pearlite (f).

WORKED EXAMPLE 16 — A 0.3 % C STEEL AT POINTS C, D, E AND F [SESSION 9 SLIDES 15-18]

Point c: one phase, austenite, $C_\gamma = C_0 = 0.3$ % C, $W_\gamma = 100$ %.

Point d: ferrite (C_α = 0.02 % C) and austenite (C_γ = 0.4 % C), length 0.38:

$$W_\alpha = \frac{C_\gamma - C_0}{C_\gamma - C_\alpha} = \frac{0.4 - 0.3}{0.4 - 0.02} = \frac{0.1}{0.38} = 0.263 \qquad W_\gamma = \frac{C_0 - C_\alpha}{C_\gamma - C_\alpha} = \frac{0.3 - 0.02}{0.38} = 0.737$$

Point e (just above 727 °C): C_α = 0.02, C_γ = 0.76 % C, length 0.74 (Slide 17 prints C_α = 0.02 at d and e; at f the diagram value 0.022 is used):

$$W_\alpha = \frac{0.76 - 0.3}{0.76 - 0.02} = \frac{0.46}{0.74} = 0.622 \qquad W_\gamma = \frac{0.3 - 0.02}{0.74} = 0.378$$

Point f (just below 727 °C): two microconstituents (primary α and pearlite) and two phases (α and Fe3C):

$$W_{\alpha,\text{total}} = \frac{6.67 - 0.3}{6.67 - 0.022} = \frac{6.37}{6.648} = 0.958 \qquad W_{\alpha_p} = \frac{C_E - C_0}{C_E - C_\alpha} = \frac{0.76 - 0.3}{0.76 - 0.022} = \frac{0.46}{0.738} = 0.623$$

$$W_{\alpha_E} = W_{\alpha,\text{total}} - W_{\alpha_p} = 0.958 - 0.623 = 0.335 \qquad W_{\text{Fe}_3\text{C}} = 1 - 0.958 = 0.042 \qquad W_{\text{pearlite}} = 1 - 0.623 = 0.377$$

Checks: the primary ferrite at f equals the ferrite at e (0.62): it does not change at 727 °C. The pearlite, 0.377, equals the austenite at e (0.378; the 0.001 gap, here and in the primary ferrite 0.622/0.623, comes only from the 0.02 printed at e against 0.022 at f), and it is $0.377 \times 0.889 = 0.335$ ferrite and $0.377 \times 0.111 = 0.042$ cementite, the numbers above. The slides round these to 26/74, 62/38 and 96, 62, 34 and 4 %.

14.3 Hypereutectoid steels, C_0 > 0.76 % C [Session 9 Slides 19-20]

Two microconstituents: **pearlite** and **proeutectoid cementite**, $(\text{Fe}_3\text{C})_P$, which forms a network along the boundaries of the former austenite grains when the steel crosses the A_{cm} line.

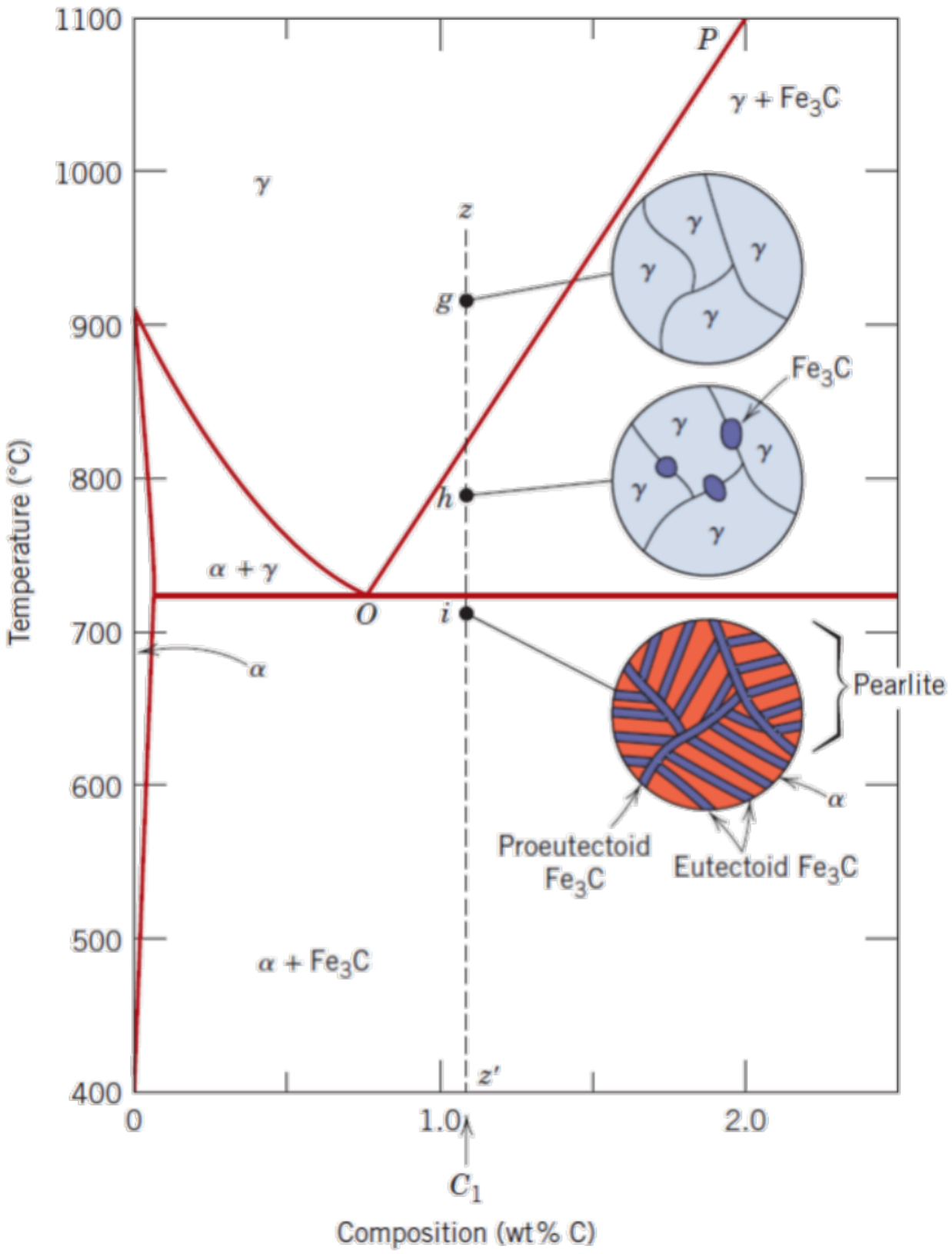


Figure 70. Hypereutectoid steel C1: g only γ; h, proeutectoid Fe3C at the γ grain boundaries below the Acm line; i, proeutectoid Fe3C and pearlite. Copied from Session 9 Slide 19.

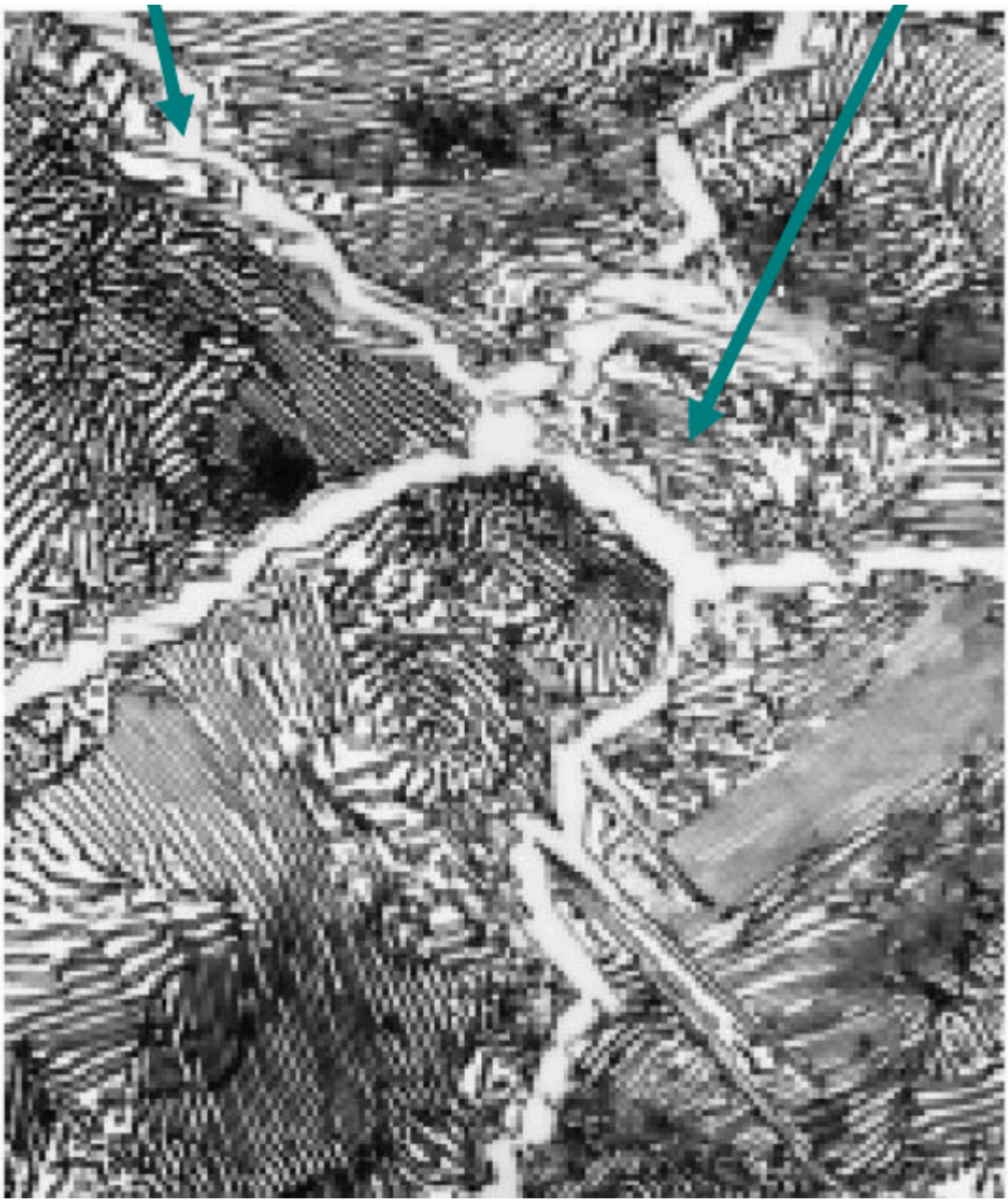


Figure 71. Hypereutectoid steel: a network of proeutectoid cementite (light) around pearlite colonies. From Session 9 Slide 19.

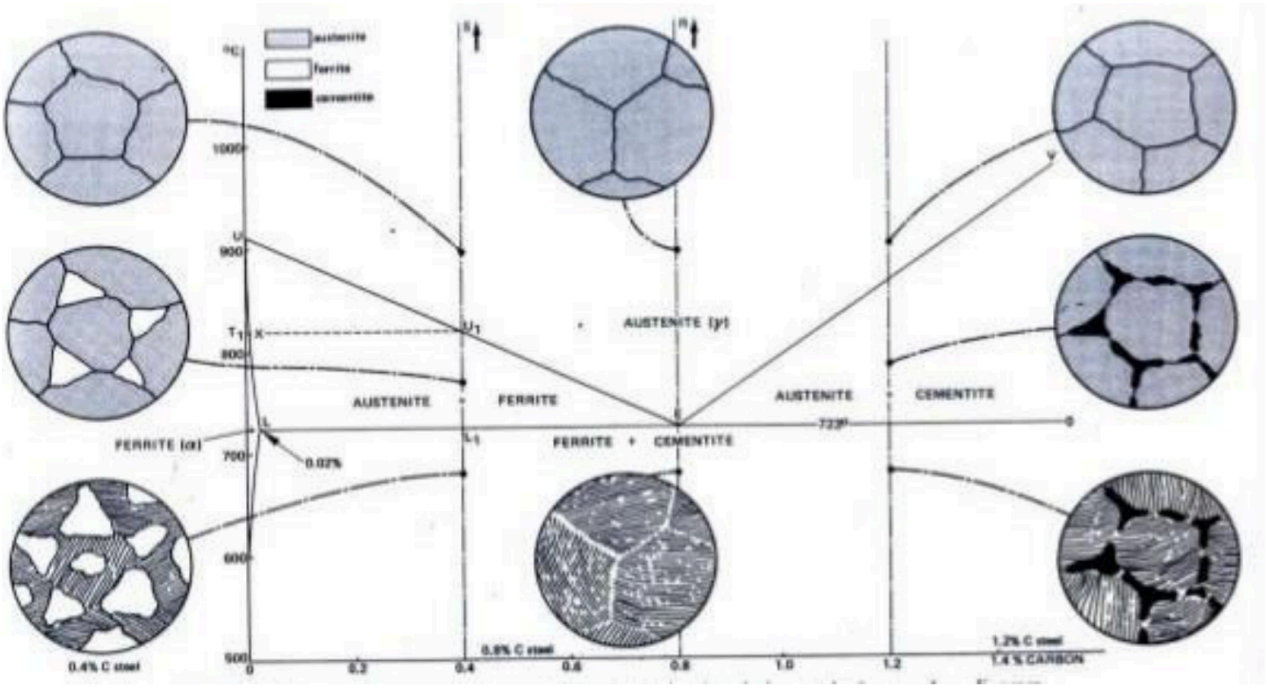


Figure 72. Microstructures on cooling of steels with 0.4, 0.8 and 1.2 % C. The figure uses older values (0.8 % C, 723 °C). From Session 9 Slide 20.

WORKED EXAMPLE 17 — THE 1.2 % C STEEL OF SLIDE 20 [SESSION 9 SLIDES 4 , 19-20]

Just below 727 °C, with the values of the diagram of Slide 4 (0.022, 0.76 and 6.67 % C).

Microconstituents (short tie line from the eutectoid austenite, 0.76, to Fe3C, 6.67; length 5.91):

$$W_{\mathrm{Fe_3C},p} = \frac{C_0 - C_E}{C_{\mathrm{Fe_3C}} - C_E} = \frac{1.2 - 0.76}{6.67 - 0.76} = \frac{0.44}{5.91} = 0.074 \qquad W_{\text{pearlite}} = \frac{6.67 - 1.2}{5.91} = \frac{5.47}{5.91} = 0.926$$

Phases (long tie line, 0.022 to 6.67):

$$W_{\mathrm{Fe_3C}} = \frac{1.2 - 0.022}{6.67 - 0.022} = \frac{1.178}{6.648} = 0.177 \qquad W_{\alpha} = 1 - 0.177 = 0.823$$

Check: the cementite inside the pearlite is $0.926 \times 0.111 = 0.103$, and $0.074 + 0.103 = 0.177$.
Answer: a thin network of proeutectoid cementite (7 %) around pearlite (93 %); even this hard steel is 82 % ferrite by mass. (The drawing of Slide 20 uses older values, 0.8 % C and 723 °C, for the eutectoid point.)

§ 15 Pearlitic transformations and cooling rate [Session 9 Slide 21]

If the cooling rate increases, the eutectoid point moves towards **lower temperatures and lower carbon contents**. Consequences: the percentage of **proeutectoid (primary) ferrite decreases** and the **spacing of the pearlite lamellae decreases**, which increases its hardness (fine pearlite instead of coarse pearlite).

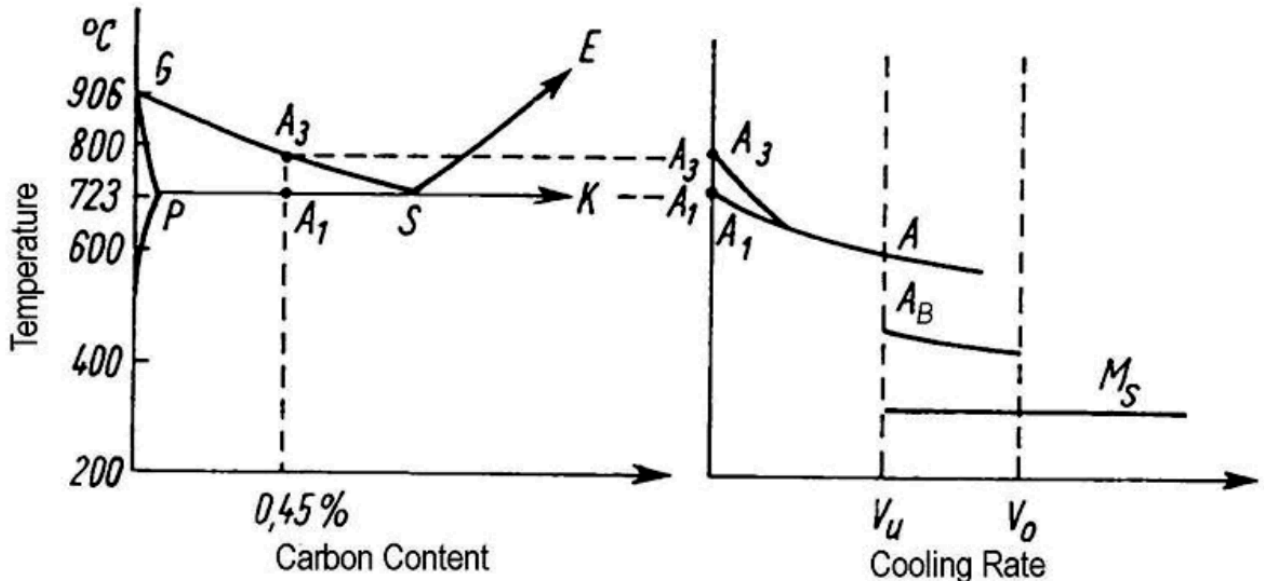


Figure 73. With a higher cooling rate the eutectoid point moves to lower temperatures and lower carbon contents. From Session 9 Slide 21.

Coarse Pearlite

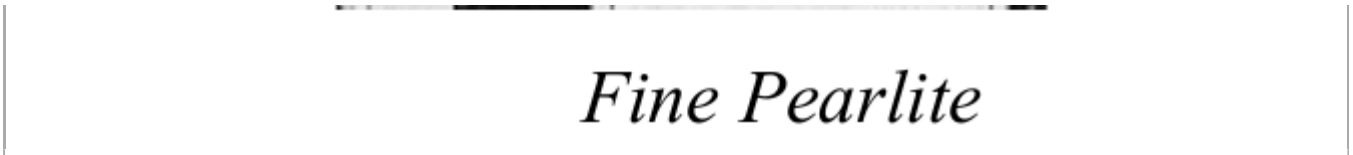


Figure 74. Coarse pearlite (slow cooling, top) and fine pearlite (fast cooling, bottom). From Session 9 Slide 21.

§ 16 Influence of the alloying elements [Session 9 Slides 22-23]

Other elements in the Fe-C system change the temperature and the composition of the eutectoid point:

- **Elements favouring α** (α-stabilisers) displace the eutectoid to **higher temperature** and **lower %C**: Si, P, Al, Be, Sn, Sb, As, Ti, Nb, V, Ta, Mo, W, Cr.
- **Elements favouring γ** (γ-stabilisers) displace the eutectoid to **lower temperature** and **lower %C**: Mn, Co, Ni, Cu, Zn, Au.

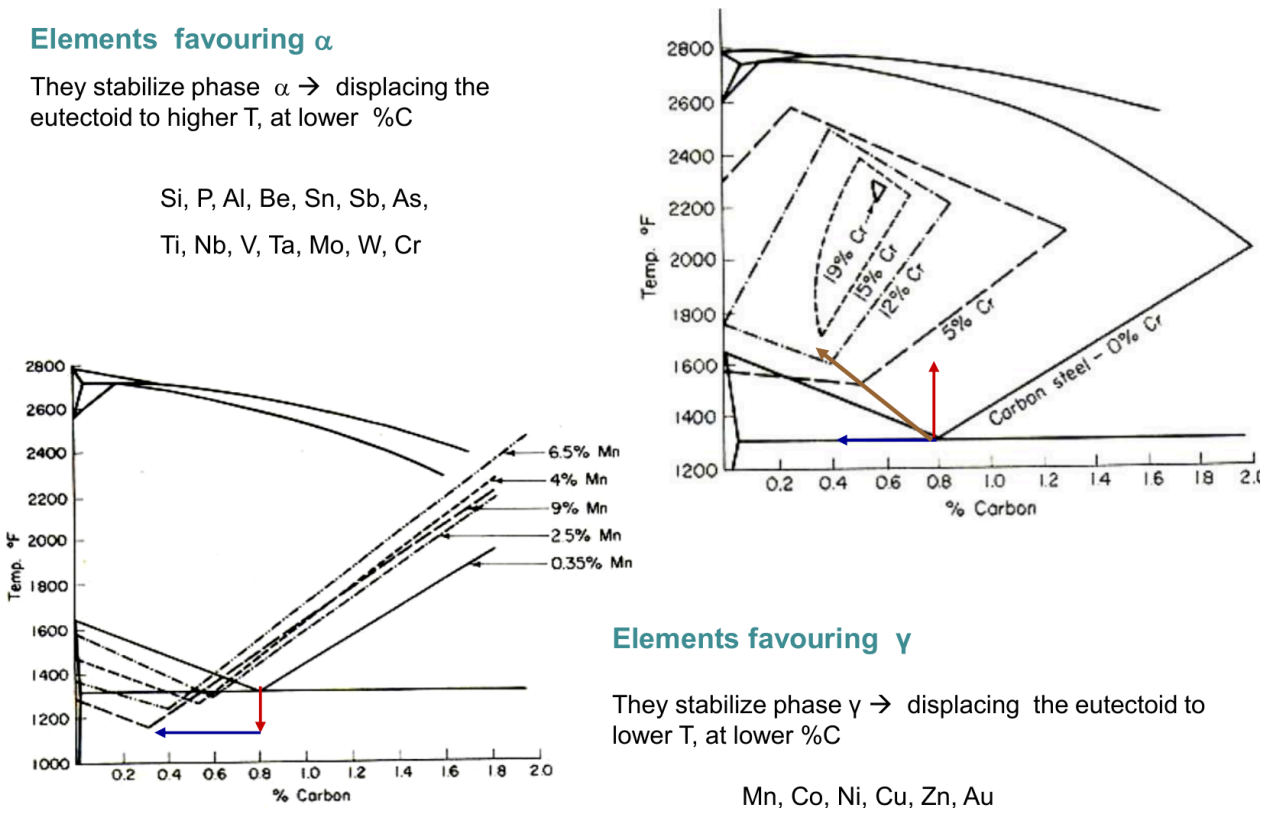
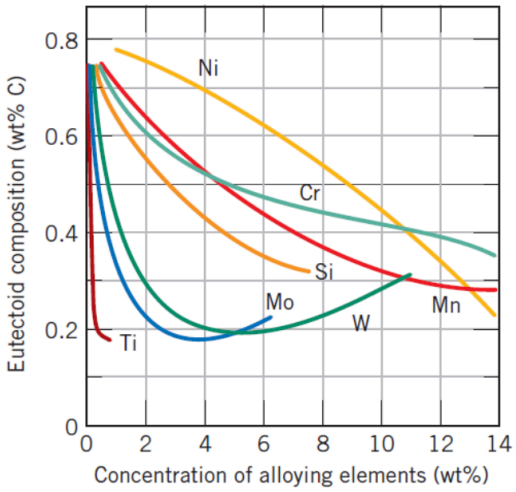
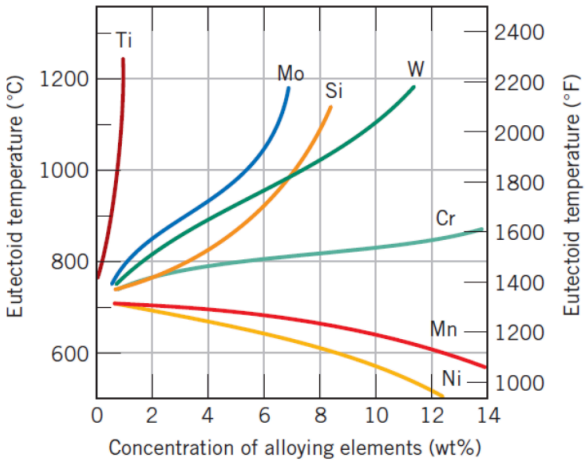


Figure 75. α-stabilising elements such as Cr shift the eutectoid to higher temperature and lower %C; γ-stabilising elements such as Mn shift it to lower temperature and lower %C. From Session 9 Slide 22.



Influence of alloying elements on the eutectoid composition (%C)



Influence of alloying elements on the eutectoid temperature

Figure 76. Influence of alloying elements on the eutectoid composition (left) and temperature (right). From Session 9 Slide 23.

In both cases the eutectoid composition falls, so a given steel lies closer to (or beyond) the eutectoid point: a hypoeutectoid steel with a γ-stabiliser such as Ni has less proeutectoid ferrite and more pearlite than the plain carbon steel of the same carbon content (used in Problem FC08).

Part IV — Phase diagrams IV: ceramic and ternary diagrams (Session 10)

§ 17 Ceramic phase diagrams [Session 10 Slides 2-4]

Ceramic systems are read with the same tools as metallic ones: the components are now compounds (oxides), and the phases can be solid solutions, intermediate compounds and the liquid.

17.1 Al_2O_3-Cr_2O_3: complete solid solubility [Session 10 Slide 2]

Al_2O_3 and Cr_2O_3 form a **complete interval of solid solution**, $Al_{2-x}Cr_xO_3$, below the solidus temperature (about 2100 °C) and down to 950 °C: the diagram has the lens shape of an isomorphous system (§4), between the melting points of Al_2O_3 (2045 ± 5 °C) and Cr_2O_3 (2275 ± 25 °C).

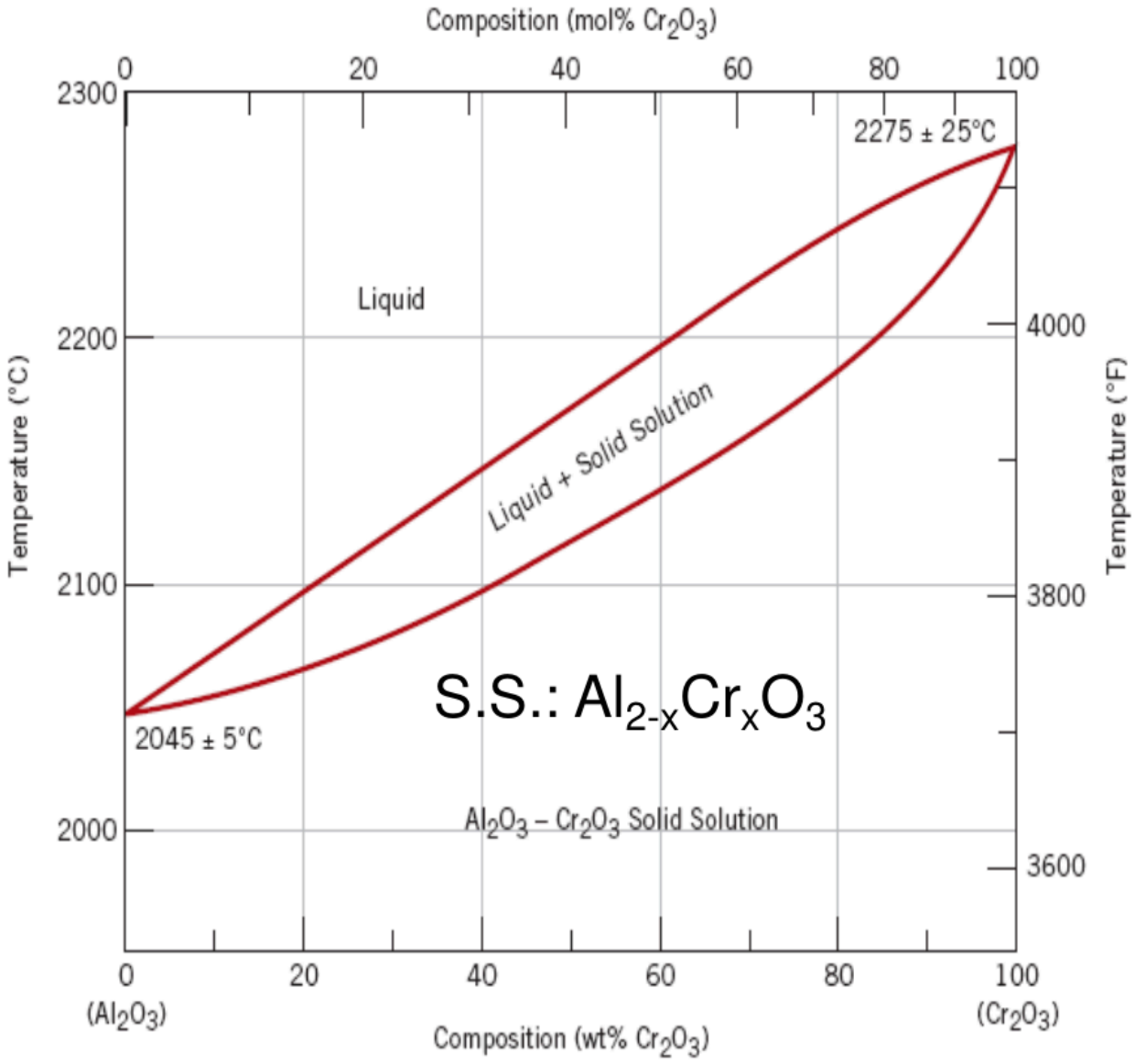


Figure 77. Al2O3-Cr2O3: a lens-shaped liquid + solid solution region between the melting points of Al2O3 (2045 ± 5 °C) and Cr2O3 (2275 ± 25 °C), and the solid solution Al2-xCrxO3 below it. Copied from Session 10 Slide 2.

Below 950 °C there is **immiscibility**: two crystalline phases, or solid solutions, exist (the dome at the bottom of the second diagram of the slide).

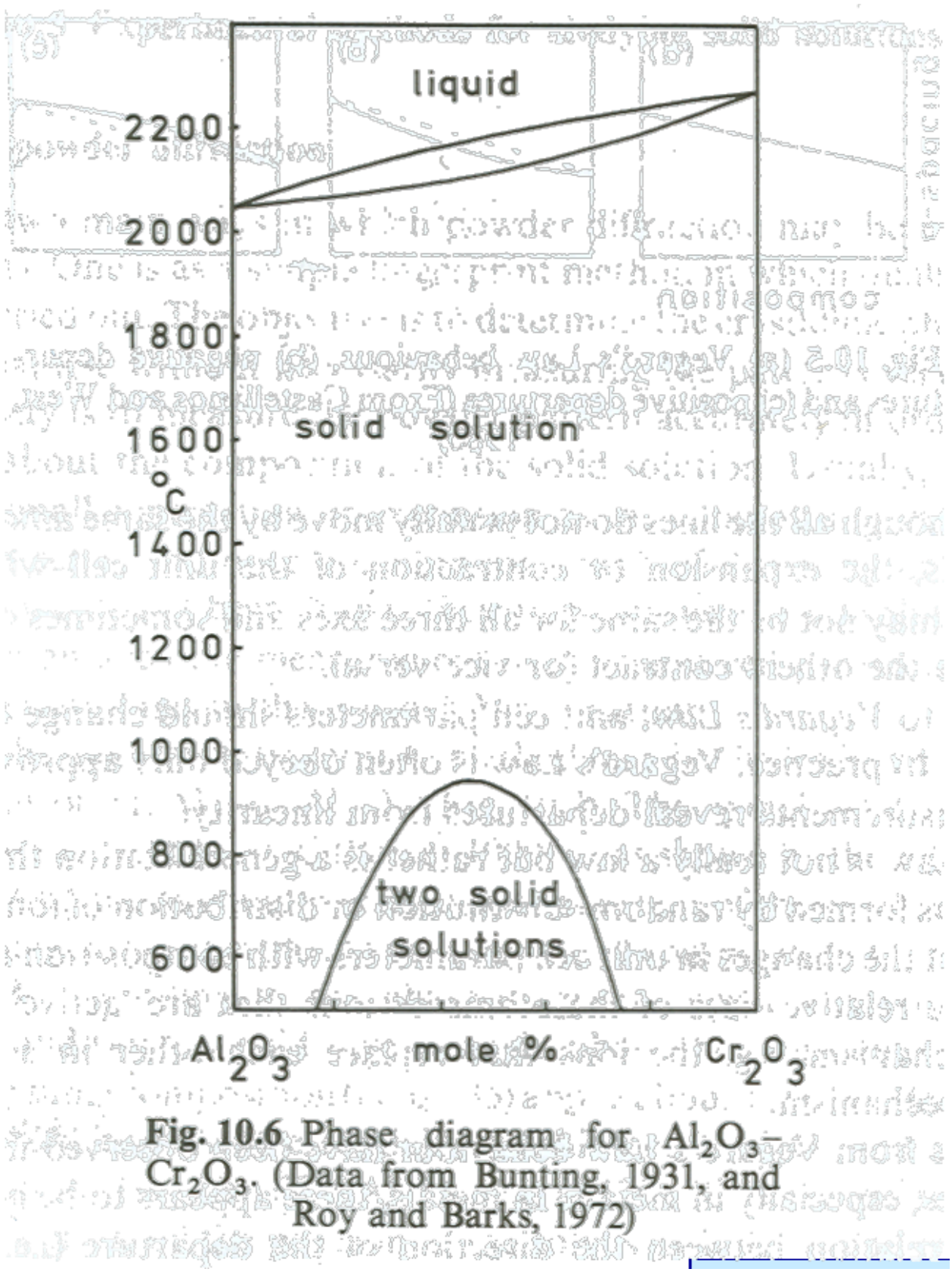


Fig. 10.6 Phase diagram for Al_2O_3–Cr_2O_3. (Data from Bunting, 1931, and Roy and Barks, 1972)

Figure 78. Al_2O_3-Cr_2O_3 over a wider temperature range: complete solid solution between about 2100 and 950 °C, and two solid solutions (immiscibility) below 950 °C. From Session 10 Slide 2.

17.2 Al_2O_3-SiO_2 and mullite [Session 10 Slides 3-4]

The Al2O3-SiO2 system has an **intermediate phase**, **mullite**, which includes the compound 3Al2O3·2SiO2 and exists as a solid solution over a small range of compositions. The other solid phases are **alumina** (Al2O3) and **cristobalite** (SiO2).

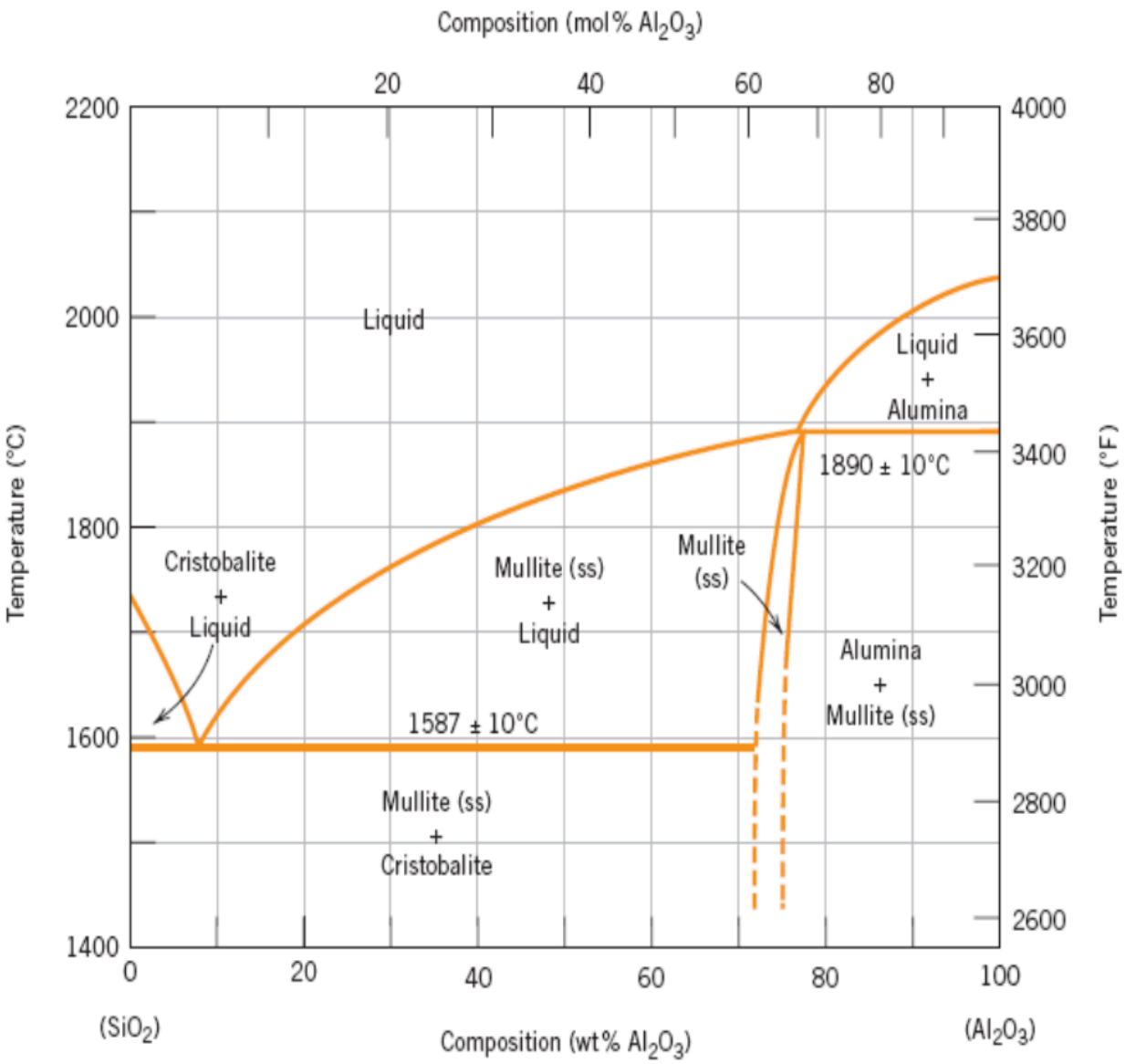


Figure 79. Al2O3-SiO2 with the eutectic at 1587 ± 10 °C, the mullite solid solution and the line at 1890 ± 10 °C where mullite melts incongruently. Copied from Session 10 Slide 3.

- **Invariant eutectic reaction at 1587 °C:**

$$L\,(7.7\ \text{wt\%}\ Al_2O_3) \rightarrow 3Al_2O_3 \cdot 2SiO_2 + SiO_2$$

- **Invariant peritectic reaction at 1828 °C:**

$$L\,(77\ \text{wt\%}\ \mathrm{Al_2O_3}) + \mathrm{Al_2O_3} \rightarrow 3\mathrm{Al_2O_3}\cdot 2\mathrm{SiO_2}$$

- **Mullite (ss) has an incongruent melting point at 1828 °C:** $3\mathrm{Al_2O_3}\cdot 2\mathrm{SiO_2} \rightarrow L\,(77\ \text{wt\%}\ \mathrm{Al_2O_3}) + \mathrm{Al_2O_3}$, the same reaction read on heating (§12).

> **TEMPERATURES OF THE AL_2O_3-SIO_2 DIAGRAMS**
> The text of Slide 3 gives 1828 °C for the peritectic, while the diagram on the same slide marks that line at 1890 ± 10 °C; the diagram of Slide 4, from another source, gives 1840 °C and about 1850 °C, and 1590 °C for the eutectic. They are different measurements of the same system; the figure above is the diagram of Slide 3. Slide 4 also writes "Al2O2 - SiO2" for Al2O3-SiO2.

The diagram of Slide 4 (composition in wt% SiO2, reversed with respect to Slide 3) shows the refractories made in this system and their microstructures: alumina refractory, mullite refractory, firebrick and silica brick.

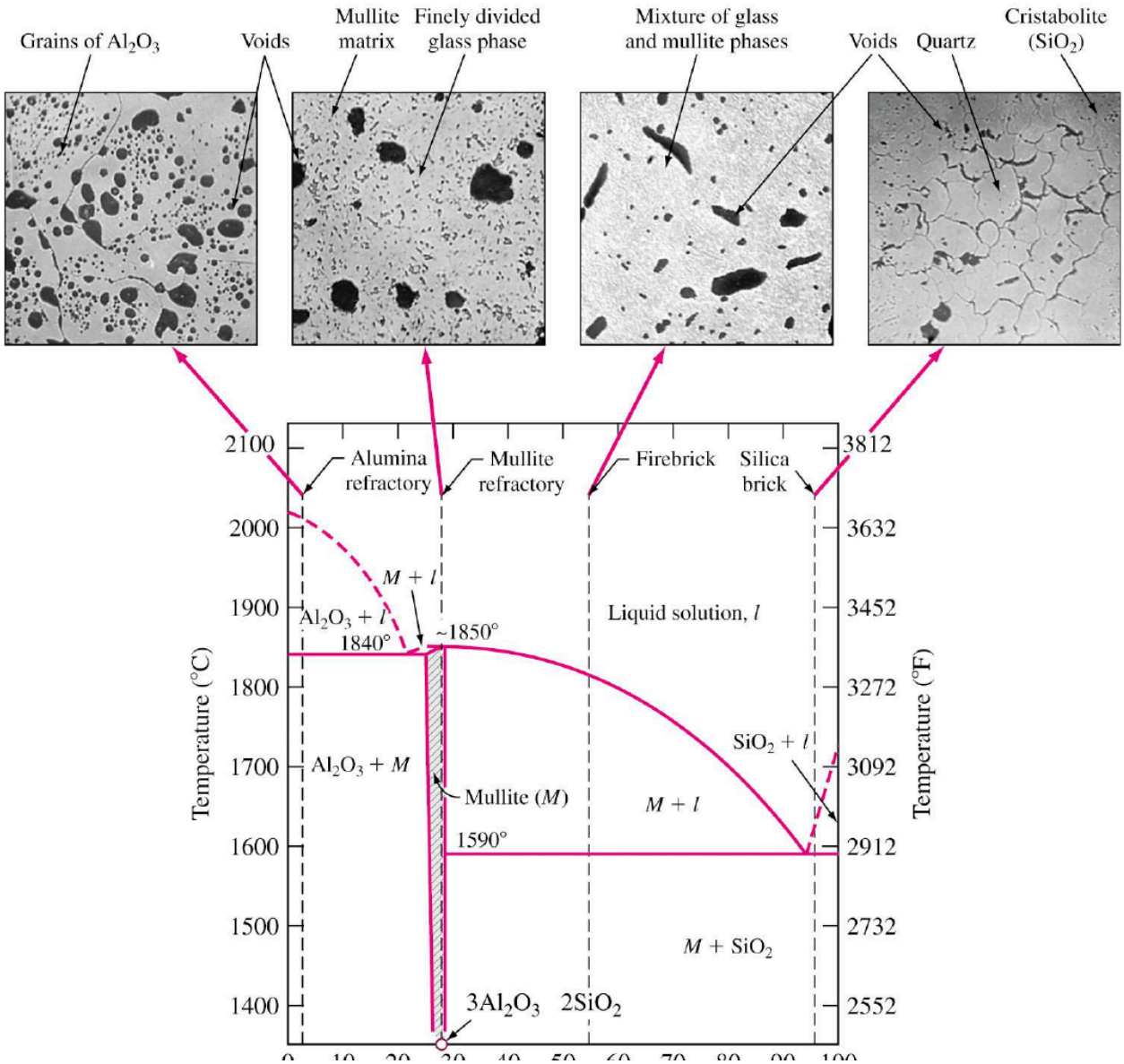


Figure 80. Al_2O_3-SiO_2 (composition in wt% SiO_2) with the microstructures of alumina and mullite refractories, firebrick and silica brick; mullite, $3Al_2O_3{\cdot}2SiO_2$, is the intermediate phase. From Session 10 Slide 4.

WORKED EXAMPLE 18 — THE EUTECTIC PRODUCT OF AL_2O_3-SIO_2 [SESSION 10 SLIDE 3]

Question. A liquid of eutectic composition (7.7 wt% Al2O3) solidifies at 1587 °C. How much mullite and how much cristobalite does the product contain?

Tie line just below 1587 °C, from cristobalite (0 wt% Al2O3) to mullite (ss), whose Al2O3-poor limit is at about **72 wt% Al2O3** (a reading of the figure):

$$W_{\text{mullite}} = \frac{C_0 - C_{SiO_2}}{C_{\text{mullite}} - C_{SiO_2}} = \frac{7.7 - 0}{72 - 0} = 0.107 \qquad W_{SiO_2} = \frac{72 - 7.7}{72} = \frac{64.3}{72} = 0.893$$

Check: $0.107 \times 72 + 0.893 \times 0 = 7.7$ wt% Al_2O_3. **Answer:** the eutectic is about 89 % cristobalite and 11 % mullite: the liquid is very rich in SiO_2, so the product is mostly silica.

§ 18 Ternary phase diagrams [Session 10 Slides 5-8]

18.1 The composition triangle [Session 10 Slide 5]

A **ternary** diagram has **3 components** (A, B, C). The compositions are drawn in an equilateral triangle:

- the **corners** are the pure components;
- the **sides** represent the **binary compositions** A-B, A-C and B-C;
- each point inside is one composition, and its three percentages add up to 100 %.

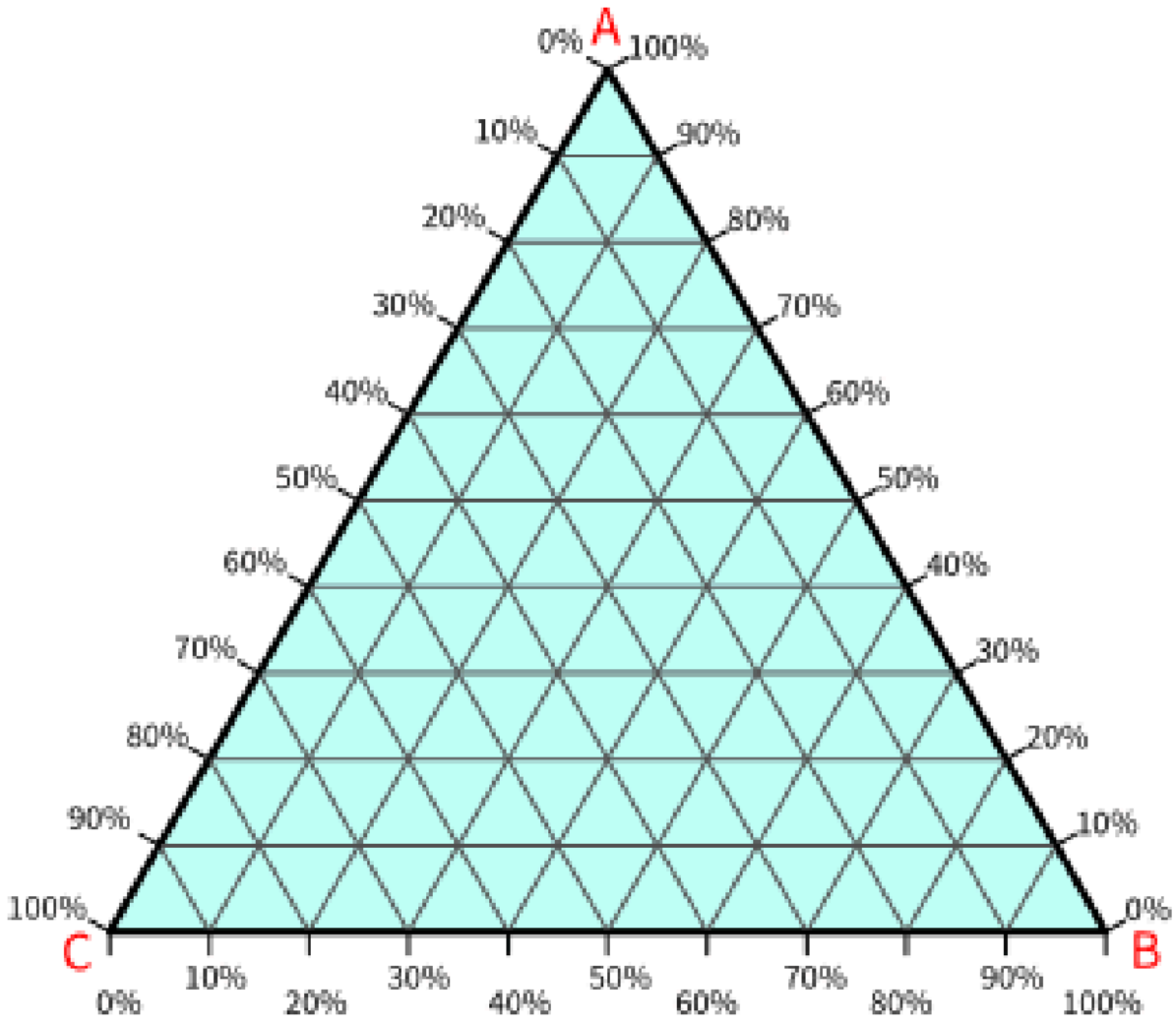


Figure 81. Composition triangle of a ternary system A-B-C (A at the top, C bottom left, B bottom right) with a 10 % grid on each side. Copied from Session 10 Slide 5.

With temperature added as a vertical axis the diagram becomes a solid: the binary diagrams are its side faces, and the liquidus and solidus become surfaces.

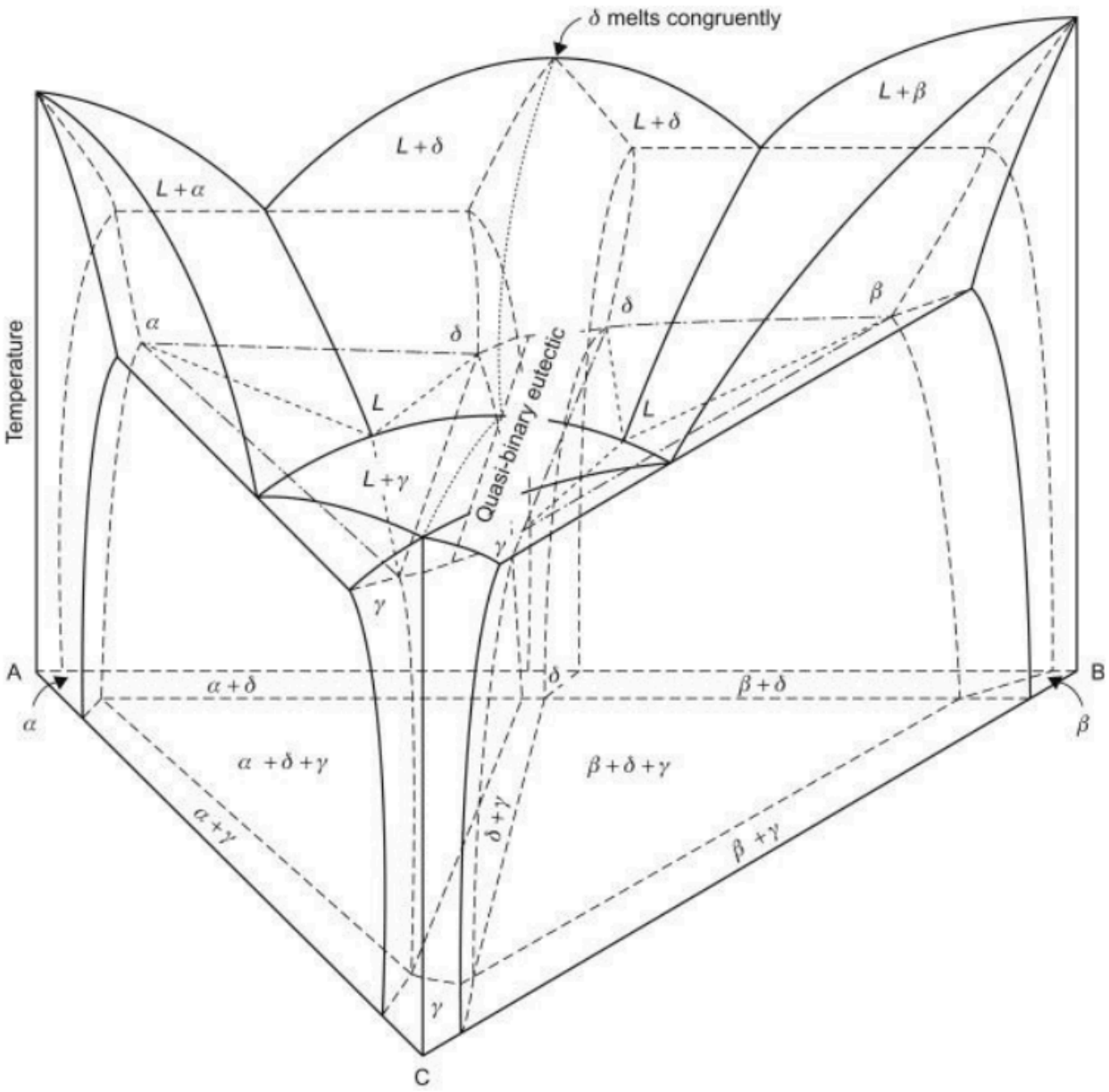


Figure 82. A ternary phase diagram as a solid: composition in the triangle base, temperature upwards. From Session 10 Slide 5.

How to read a point. Each percentage is read along the lines parallel to the side opposite the corresponding component: the lines of constant A are parallel to the side B-C, and A grows from 0 % on that side to 100 % at the corner A. Drawing through the point the three lines parallel to the sides gives the three percentages on the three axes.

18.2 Ternary diagrams of metallic alloys [Session 10 Slides 6-7]

Because the solid has three dimensions, ternary diagrams are shown as projections or sections: the **liquidus surface** and the **solidus surface** projected onto the triangle (with their temperature contours), and **isotherms**, horizontal sections at one temperature. The Ag-Cu-Zn system is shown in the three ways, with an isotherm at 350 °C [Slide 6].

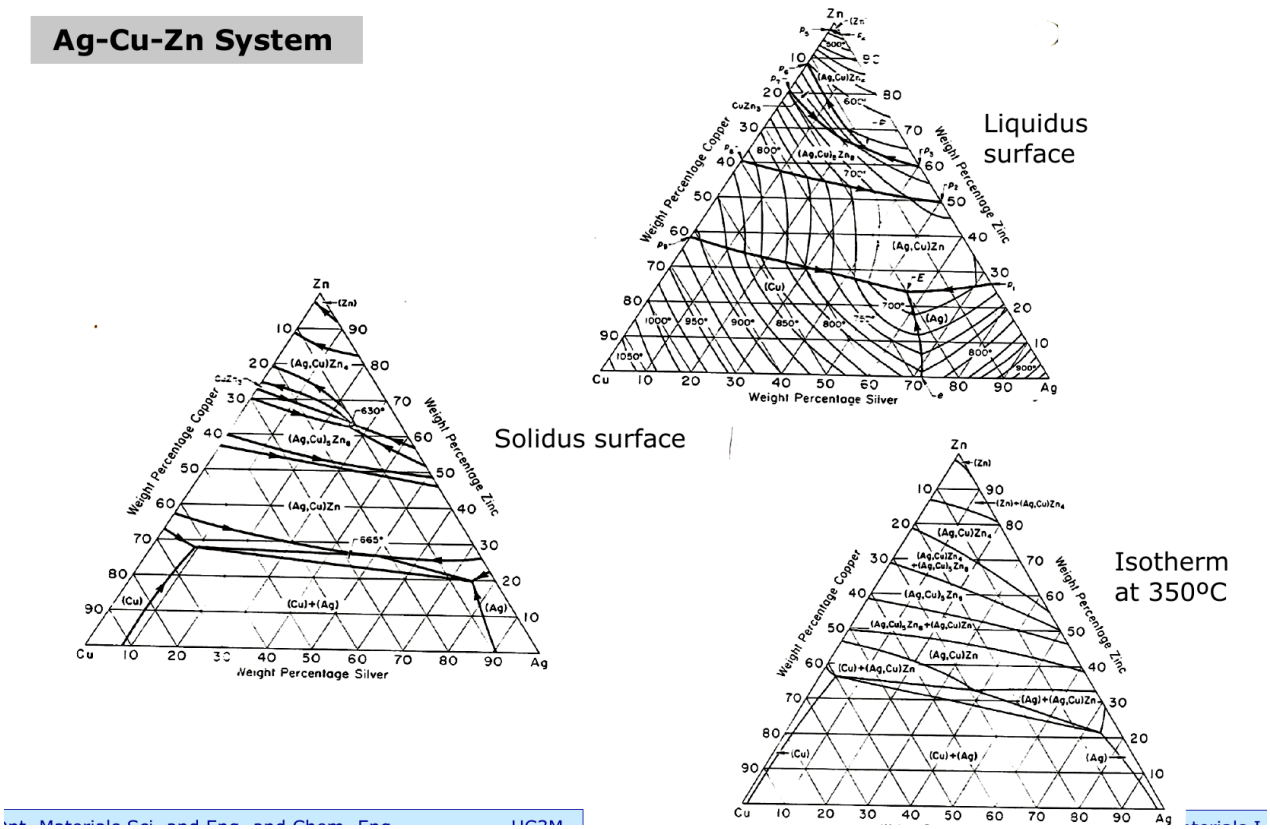


Figure 83. Ag-Cu-Zn: projections of the liquidus and solidus surfaces and the isotherm at 350 °C. From Session 10 Slide 6.

The **Fe-Cr-Ni** diagram (stainless steels) is shown as an isotherm at 650 °C, with the phase fields α, γ, σ and their two-phase regions [Slide 7]:

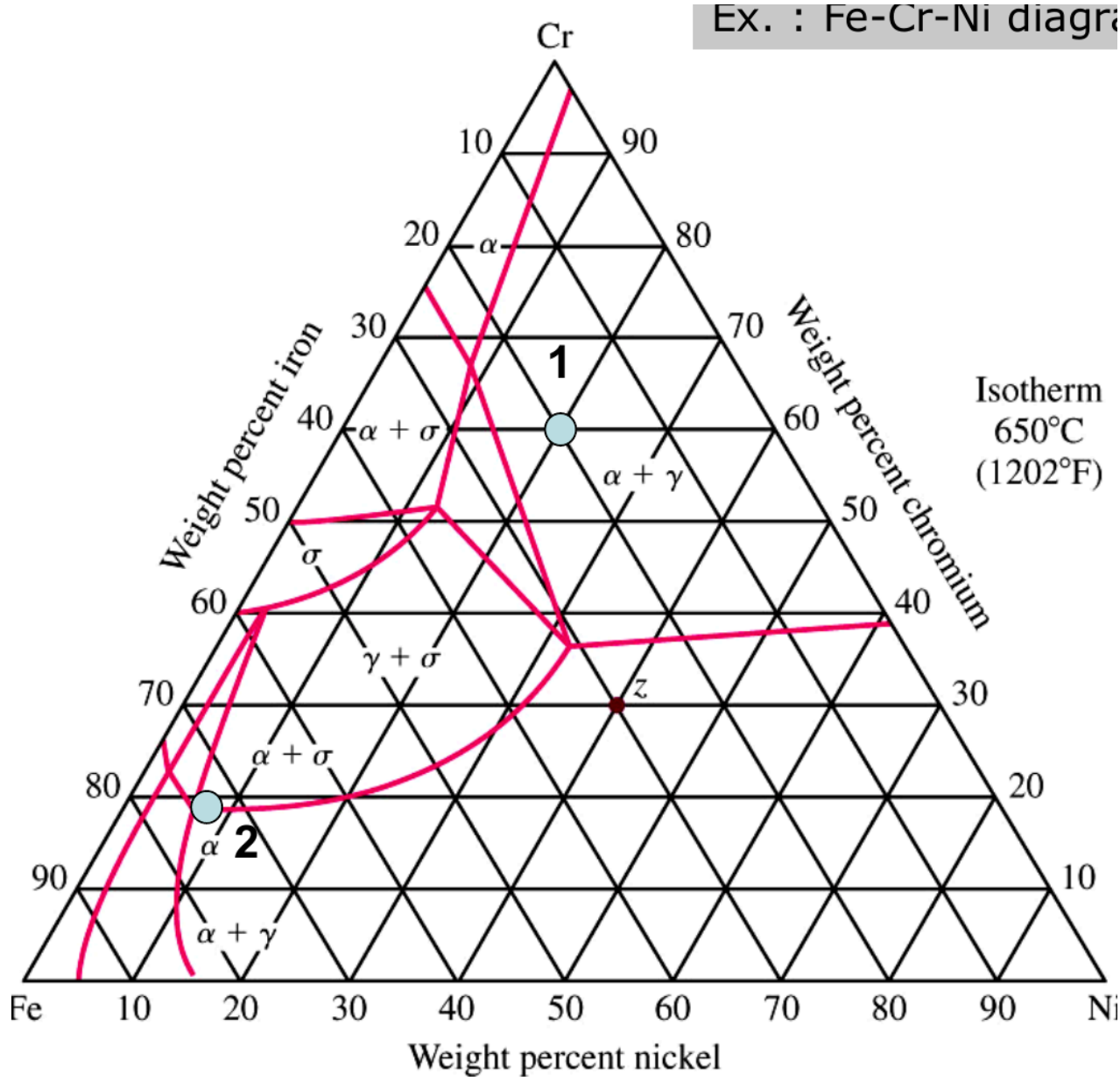


Figure 84. Fe-Cr-Ni isotherm at 650 °C (stainless steels) with points 1 and 2. From Session 10 Slide 7.

WORKED EXAMPLE 19 — READING POINTS 1 AND 2 OF THE FE-CR-NI ISOTHERM [SESSION 10 SLIDE 7]

Point 1. The line through the point parallel to the Fe-Ni side meets the chromium axis at **60 % Cr**; the line parallel to the Cr-Ni side meets the iron axis at **20 % Fe**; the line parallel to the Fe-Cr side meets the nickel axis at **20 % Ni**.

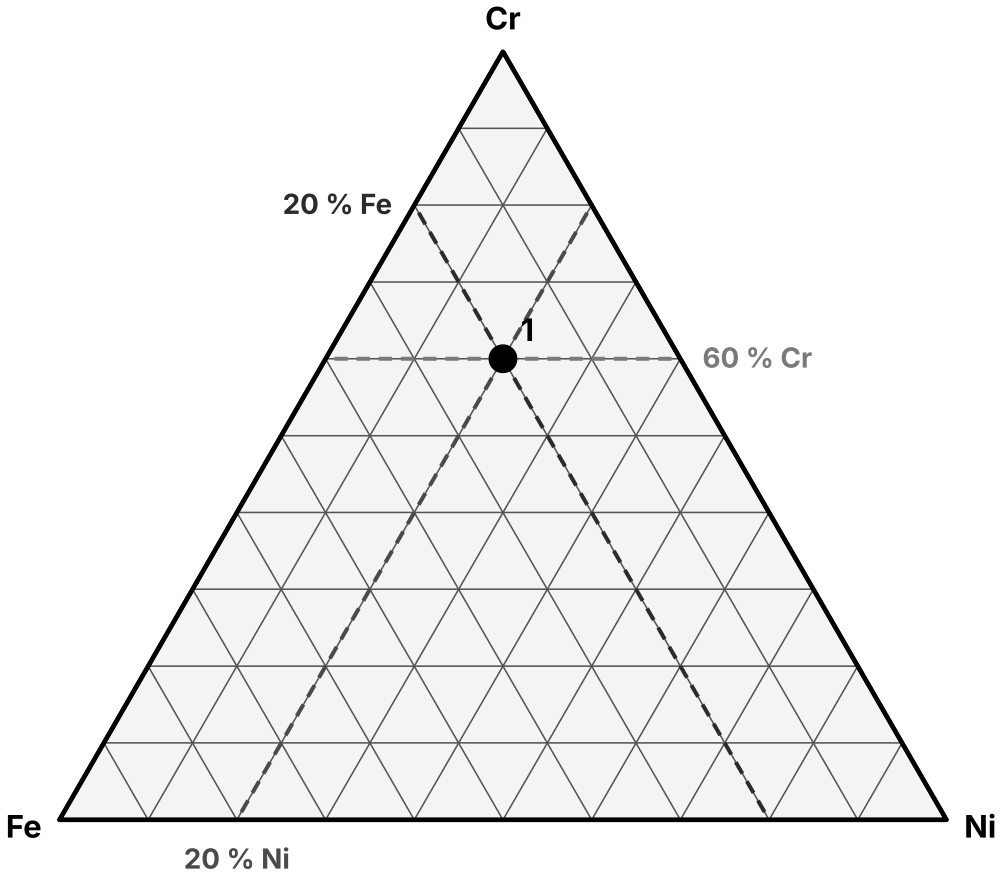


Figure 85. Reading of point 1 of the Fe-Cr-Ni isotherm at 650 °C: the three lines through it parallel to the sides meet the axes at 60 % Cr, 20 % Ni and 20 % Fe. Reading construction for point 1 of Session 10 Slide 7; the slide prints the values but does not draw these lines (EXTRA).

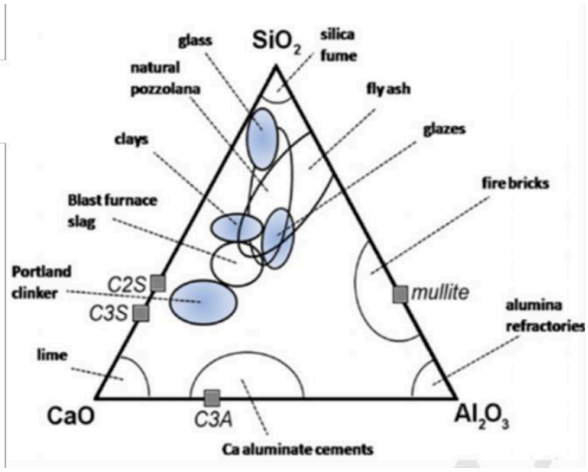
Check: $60 + 20 + 20 = 100$ %. At 650 °C the point lies in the field **α + γ**, so the alloy has two phases, as the slide says.

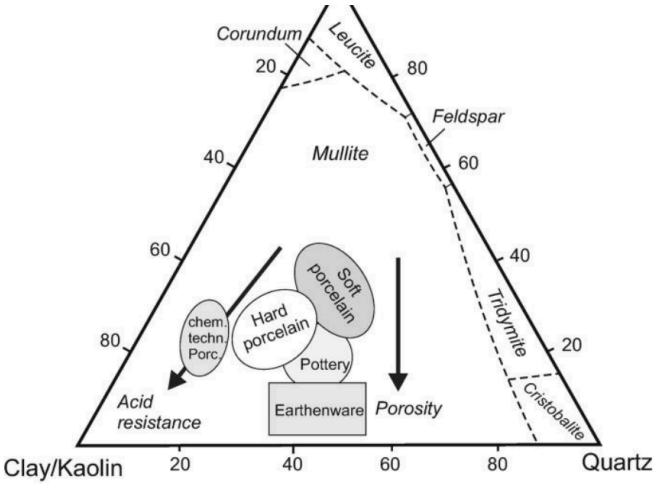
Point 2, the commercial stainless steel: **74 % Fe + 18 % Cr + 8 % Ni** ($74 + 18 + 8 = 100$ %). It lies near the Fe corner (it is mostly iron), on the low-Ni side of the diagram.

18.3 Ceramic ternary diagrams [Session 10 Slide 8]

Ternary diagrams are also used to describe ceramics: the CaO-Al_2O_3-SiO_2 system places cements (Portland clinker, calcium aluminate cements), glasses, slags and refractories; the **clay-feldspar-quartz** triangle gives the composition of **traditional ceramics** (porcelains, pottery, earthenware), with arrows for the changes in acid resistance and porosity.

Ex : Ceramic diagrams





Composition of traditional ceramics: clay-feldspar-quartz

Figure 86. Ceramic ternary diagrams: CaO-Al_2O_3-SiO_2 (cements, glasses, refractories) and clay-feldspar-quartz (traditional ceramics: porcelains, pottery, earthenware). From Session 10 Slide 8.